A / APPLICATIONS

Where did these faces come from?



Learning to generate · ES 667 · Nipun Batra

CONSIDER Where could these portraits have come from?

SAY Look across the faces. What varies, and what still makes each one look plausible?

Karras, Laine & Aila · StyleGAN · CC BY-NC 4.0

A / APPLICATIONS

Where did these faces come from?



Published generated samples · original StyleGAN

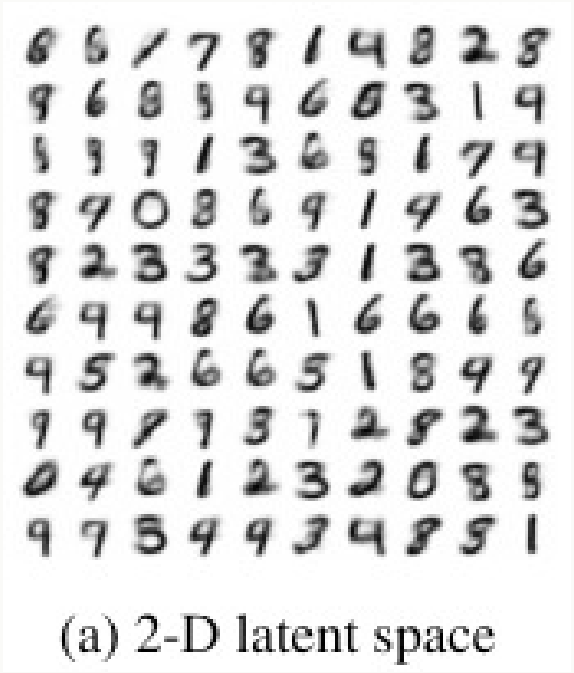
TAKEAWAY These are published generated faces. Change the random input and the face changes.

SAY Look across the faces. What varies, and what still makes each one look plausible?

Karras, Laine & Aila · StyleGAN · CC BY-NC 4.0

A / APPLICATIONS

How can a hidden space make digits?



Published prior samples · 2D latent

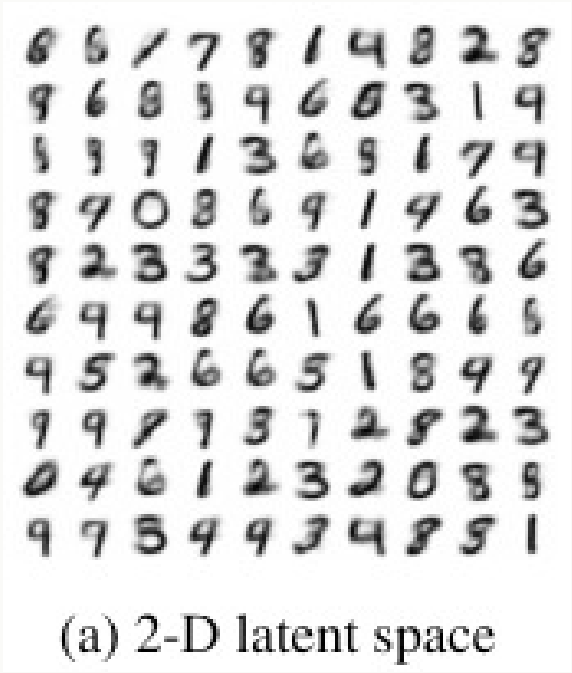
CONSIDER How could a model produce another digit after training?

SAY Watch how the digit changes as we move across this hidden space.

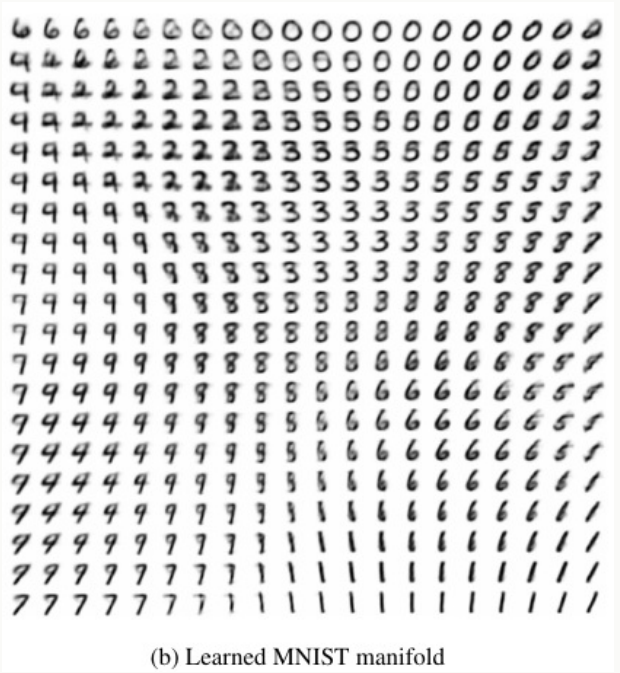
Kingma & Welling · AEVB

A / APPLICATIONS

How can a hidden space make digits?



Published prior samples · 2D latent



Decoded grid · inverse-CDF coordinates

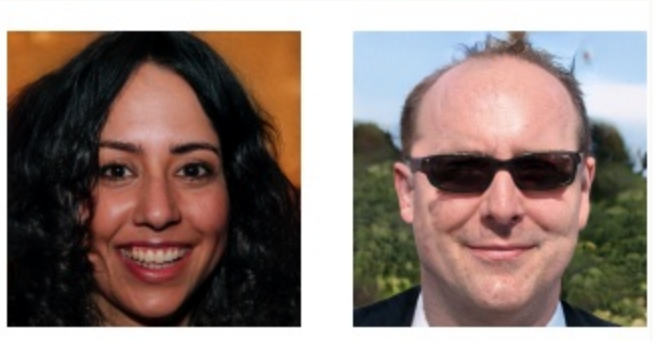
TAKEAWAY A decoded grid shows how the hidden coordinates change the digit.

SAY Watch how the digit changes as we move across this hidden space.

Kingma & Welling · AEVB

A / APPLICATIONS

Which properties can we mix?



Source A (left) · source B (right)

Read the source row and source column before inspecting their intersection.

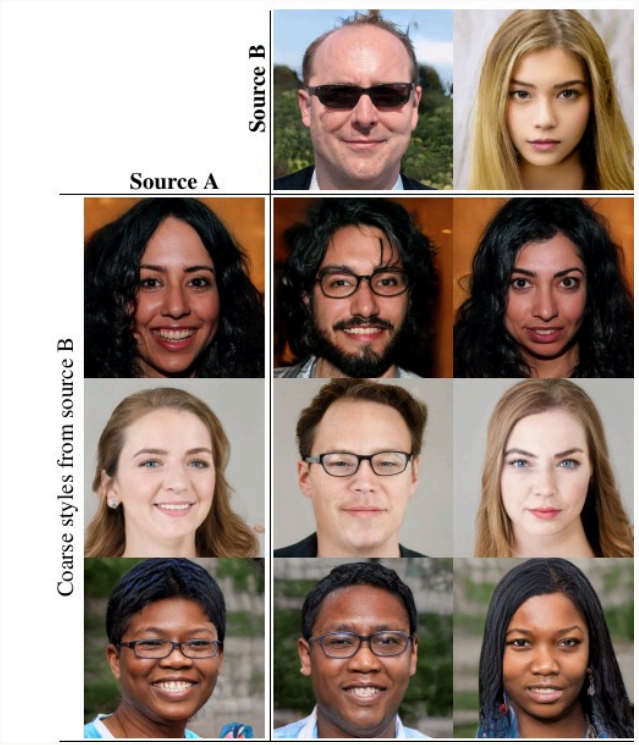
CONSIDER Which properties of these two sources could be combined?

SAY Try to identify which features came from each source.

Karras, Laine & Aila · StyleGAN · CC BY-NC 4.0

A / APPLICATIONS

Which properties can we mix?



Source A: rows · Source B: columns · coarse styles from B

Published style mixing changes selected layer groups.

TAKEAWAY Take coarse styles from B and the remaining styles from A: this is style mixing.

SAY Try to identify which features came from each source.

Karras, Laine & Aila · StyleGAN · CC BY-NC 4.0

A / APPLICATIONS

What does a shoe outline leave out?



Input: outline

CONSIDER An outline specifies shape. What does it leave unspecified?

SAY The outline tells us the shape. It leaves the colour and material open.

Isola et al. · pix2pix · row 100_AB

A / APPLICATIONS

What does a shoe outline leave out?



Input: outline



Output: generated



Ground truth: reference photo

TAKEAWAY The model fills in the appearance. The photograph is a separate reference.

SAY The outline tells us the shape. It leaves the colour and material open.

Isola et al. · pix2pix · row 100_AB

A / APPLICATIONS

Can a layout specify a landscape?



Published semantic layout · class legend retained

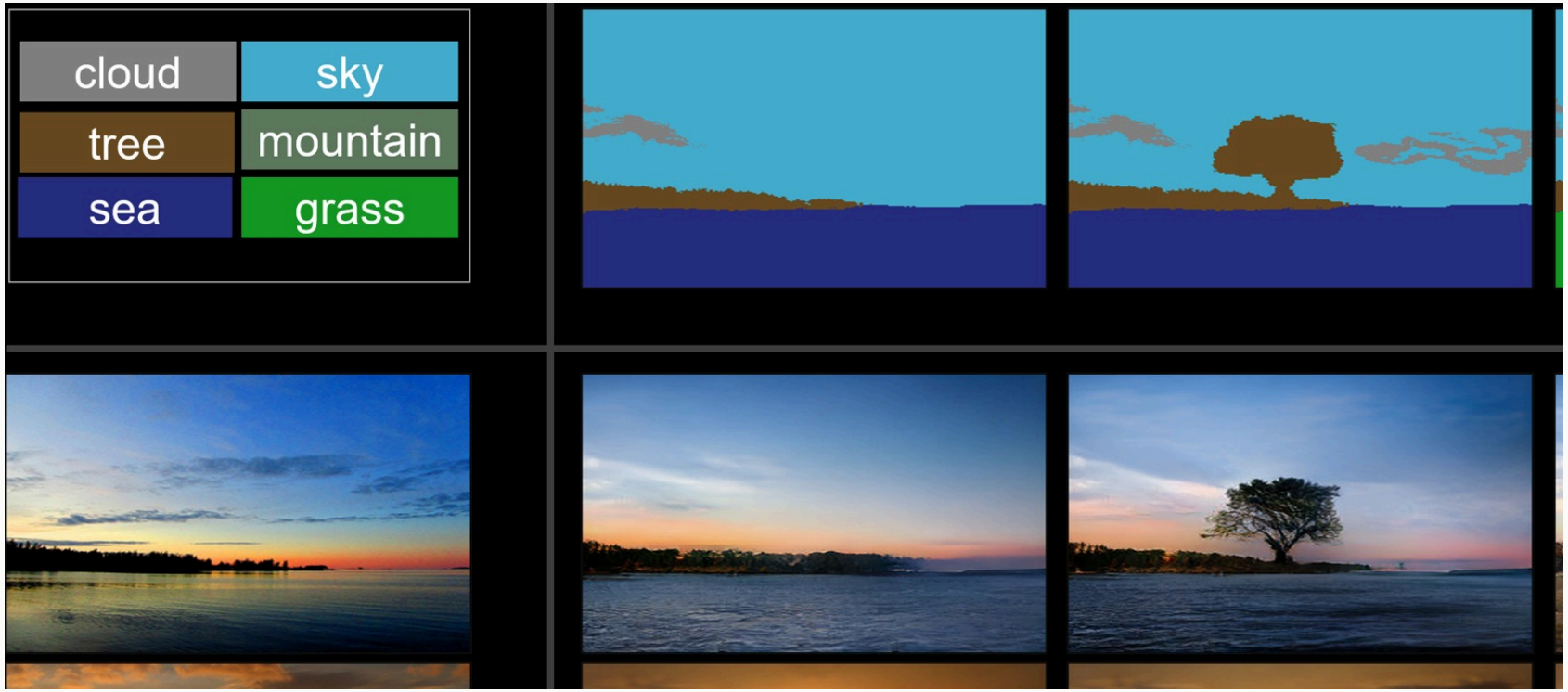
NOTICE The layout fixes where the regions belong.

SAY The layout says what goes where. What still has to be invented?

Park et al. · SPADE

A / APPLICATIONS

Can a layout specify a landscape?



Published semantic layout · class legend retained

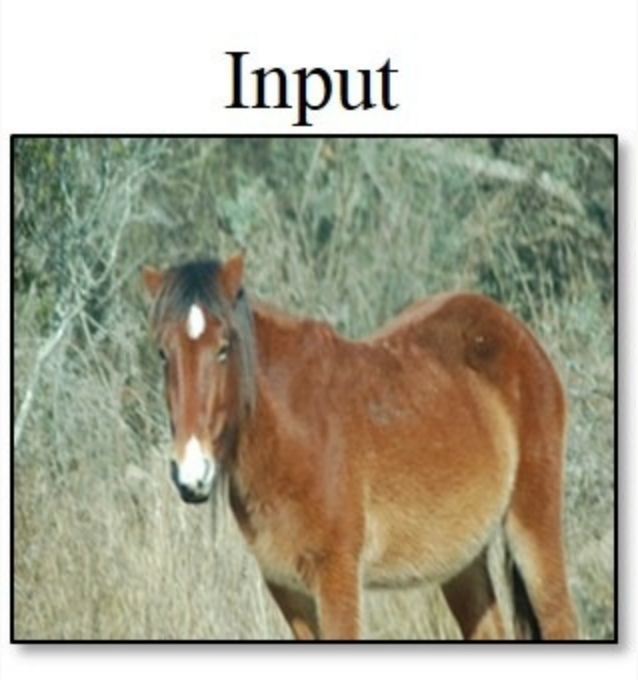
TAKEAWAY The layout constrains the scene; the generator supplies its appearance.

SAY The layout says what goes where. What still has to be invented?

Park et al. · SPADE

A / APPLICATIONS

What changes from horse to zebra?



Input horse photograph

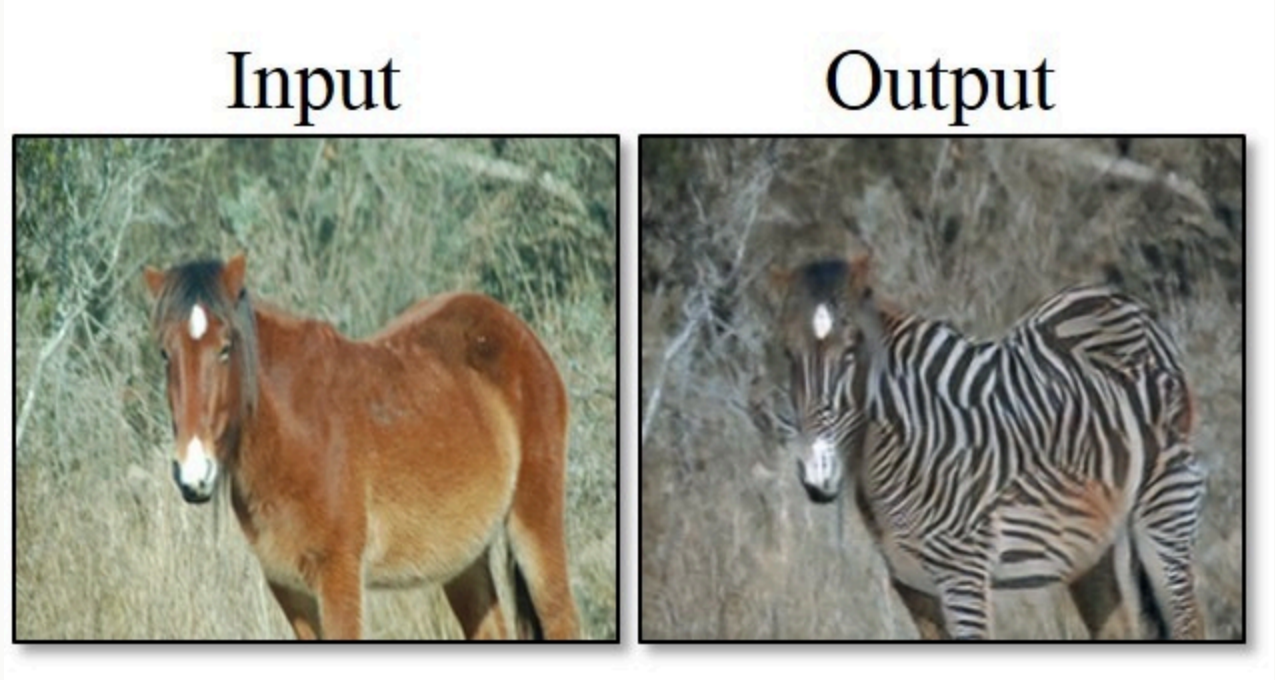
CONSIDER What should stay recognisable when the domain changes?

SAY We want the animal and scene to remain recognisable as the appearance changes.

Zhu et al. · CycleGAN

A / APPLICATIONS

What changes from horse to zebra?



Input photograph → generated zebra-domain image

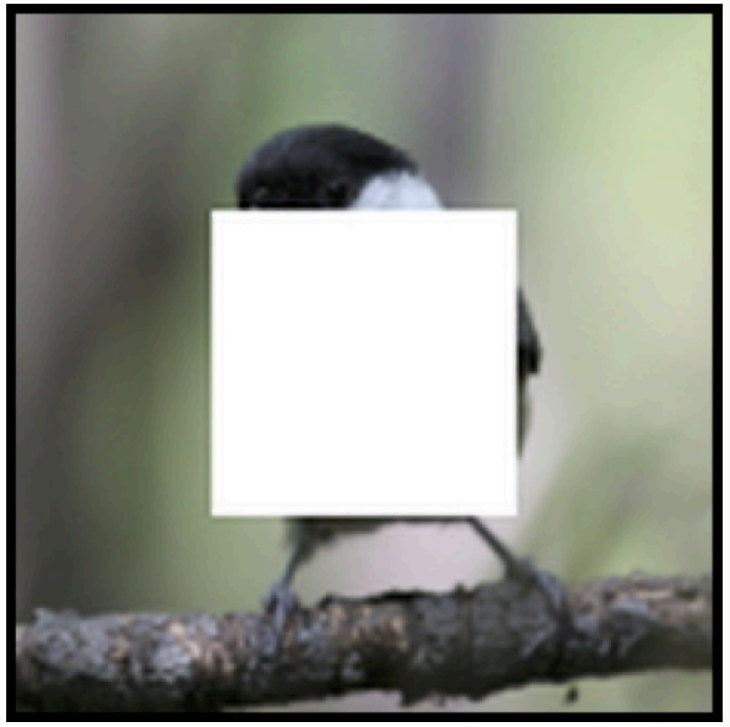
TAKEAWAY The horse is an observed input. The zebra-domain image is a generated translation.

SAY We want the animal and scene to remain recognisable as the appearance changes.

Zhu et al. · CycleGAN

A / APPLICATIONS

What belongs in the missing region?



Masked observed input

CONSIDER The visible context constrains the missing region. Does it determine every detail?

SAY More than one missing patch could fit the visible context.

Pathak et al. · Context Encoders · model result

A / APPLICATIONS

What belongs in the missing region?



Masked input (left) · model completion (right)

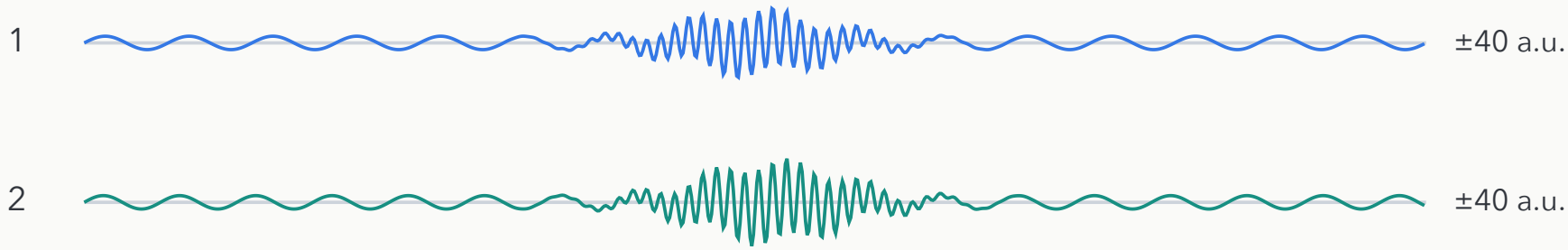
TAKEAWAY A plausible completion is not proof of the hidden original.

SAY More than one missing patch could fit the visible context.

Pathak et al. · Context Encoders · model result

A / APPLICATIONS

Could we generate a signal segment?



0 to 8 seconds · 100 samples / second · arbitrary amplitude units

Proposed lab application; illustrative signals. Deterministic equations, not recorded EEG.

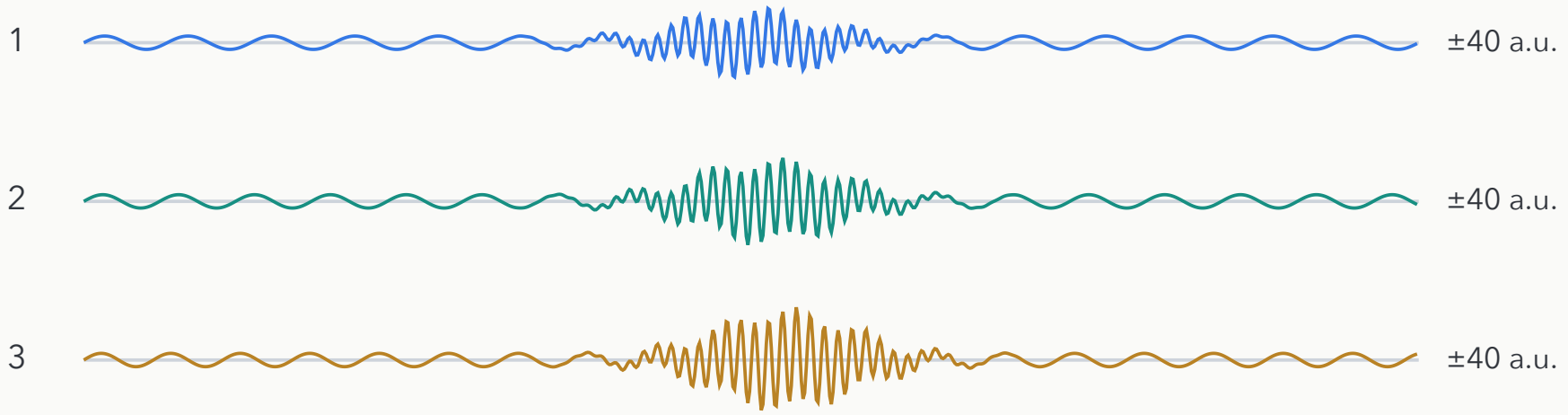
NOTICE Proposed lab application; illustrative signals. Condition: a spindle-like event.

SAY These traces illustrate the idea. They are not validated physiological recordings.

Authored deterministic signal illustration

A / APPLICATIONS

Could we generate a signal segment?



0 to 8 seconds · 100 samples / second · arbitrary amplitude units

Proposed lab application; illustrative signals. Deterministic equations, not recorded EEG.

TAKEAWAY Different traces can fit one event description. Scientific validity still needs testing.

SAY These traces illustrate the idea. They are not validated physiological recordings.

Authored deterministic signal illustration

A / APPLICATIONS

Which satellite images are possible?

Same caption: an electric substation in Finland



August

change
the month →

Published generated samples · fixed caption and other metadata · first method, first sample row

CONSIDER What does the metadata tell a satellite-image generator?

SAY Treat each output as a possible image under the metadata, not as a measurement.

Khanna et al. · DiffusionSat

A / APPLICATIONS

Which satellite images are possible?

Same caption: an electric substation in Finland



August

change the month →



January

Published generated samples · fixed caption and other metadata · first method, first sample row

> **TAKEAWAY** These are generated possibilities, not recovered measurements or forecasts.
>
> **SAY** Treat each output as a possible image under the metadata, not as a measurement.

Khanna et al. · DiffusionSat

A / APPLICATIONS

Have we generated something already?

Recreated character bigram model fitted to 14 teaching names · start ^ · end $

^	d	e	e	v	a	$

$$P(a \mid \hat{}) = 3/14 \qquad P(d \mid \hat{}) = 2/14 \qquad P(r \mid \hat{}) = 2/14$$

Seeds 667 and 668; categorical sampling, no temperature scaling.

NOTICE Choose a next character from its categorical distribution.

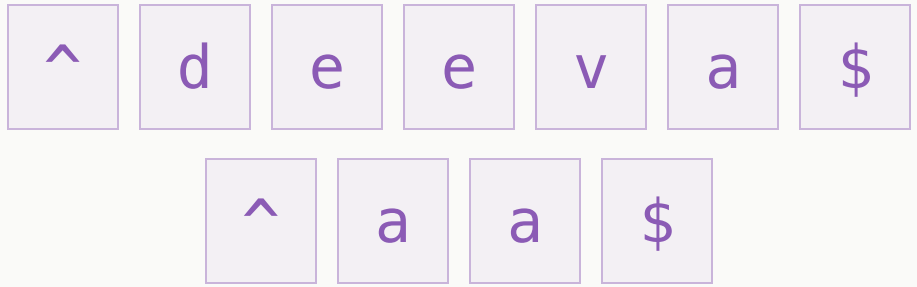
SAY We have already seen this pattern: predict probabilities, then draw one outcome.

Course name generator · reproduced categorical example

A / APPLICATIONS

Have we generated something already?

Recreated character bigram model fitted to 14 teaching names · start ^ · end $



$$P(a \mid \hat{}) = 3/14 \quad P(d \mid \hat{}) = 2/14 \quad P(r \mid \hat{}) = 2/14$$

Seeds 667 and 668; categorical sampling, no temperature scaling.

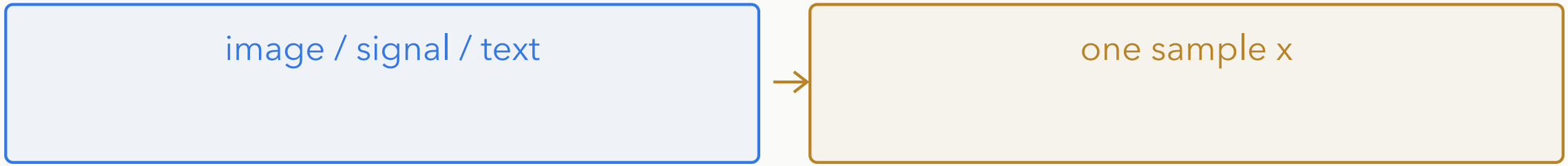
TAKEAWAY A new random seed gives another sequence from the same fitted model.

SAY We have already seen this pattern: predict probabilities, then draw one outcome.

Course name generator · reproduced categorical example

B / DISTRIBUTIONS

What did the gallery have in common?



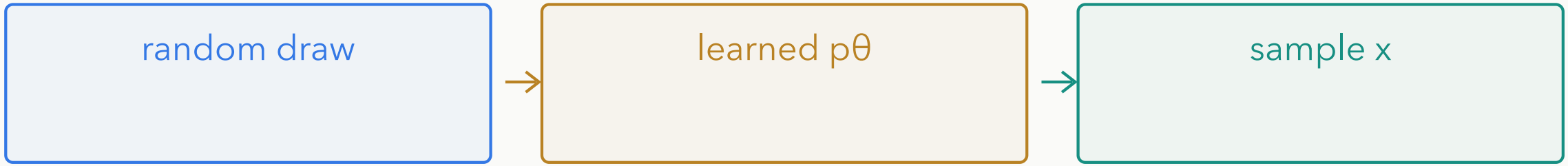
$$x$$

NOTICE A sample is one outcome. The distribution describes possible outcomes.

SAY A generator needs a way to assign probabilities, draw samples, or both.

B / DISTRIBUTIONS

What did the gallery have in common?



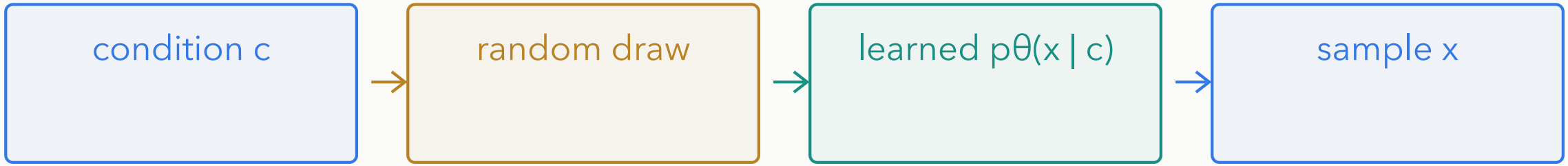
$$x \sim p_\theta(x)$$

NOTICE Learn a distribution from which we can draw new outputs.

SAY A generator needs a way to assign probabilities, draw samples, or both.

B / DISTRIBUTIONS

What did the gallery have in common?



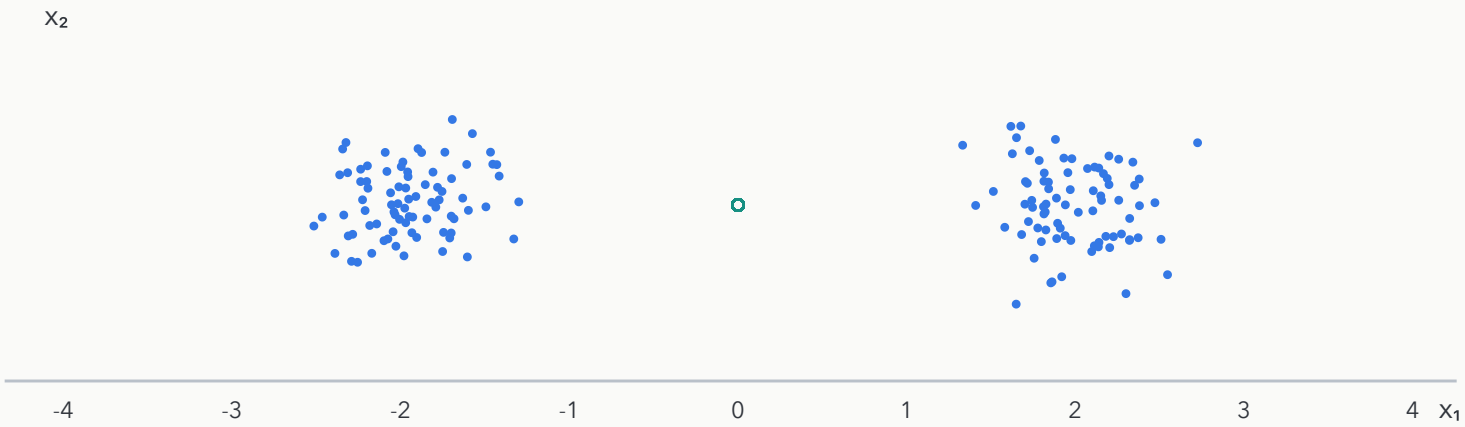
$$x \sim p_\theta(x \mid c)$$

TAKEAWAY A condition narrows which outputs are plausible.

SAY A generator needs a way to assign probabilities, draw samples, or both.

B / DISTRIBUTIONS

Can a good average be a bad sample?



within 0.65 of a true centre

0.0%

modes with ≥5 of 160 samples

0/2

Deterministic toy geometry, centres (−2,0), (2,0); these are not image-quality metrics.

$$\mathbb{E}[X] \neq X$$

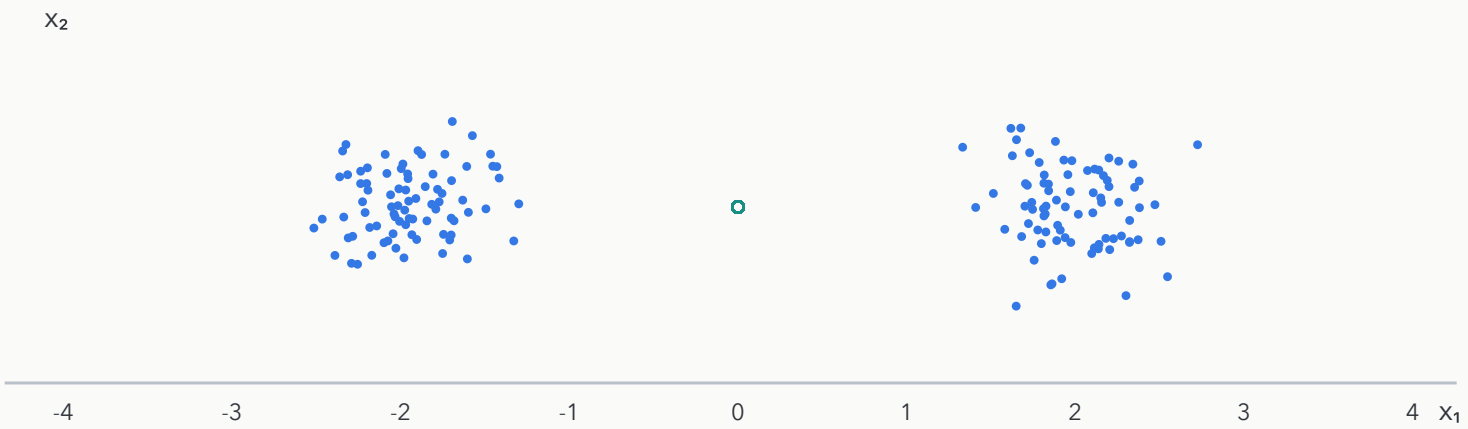
NOTICE The average falls between the two clusters.

SAY The average is in the middle. But almost none of our data live there.

ES 667 · original teaching diagrams and exact calculations

B / DISTRIBUTIONS

Can a good average be a bad sample?



within 0.65 of a true centre

0.0%

modes with ≥5 of 160 samples

0/2

Deterministic toy geometry, centres (−2,0), (2,0); these are not image-quality metrics.

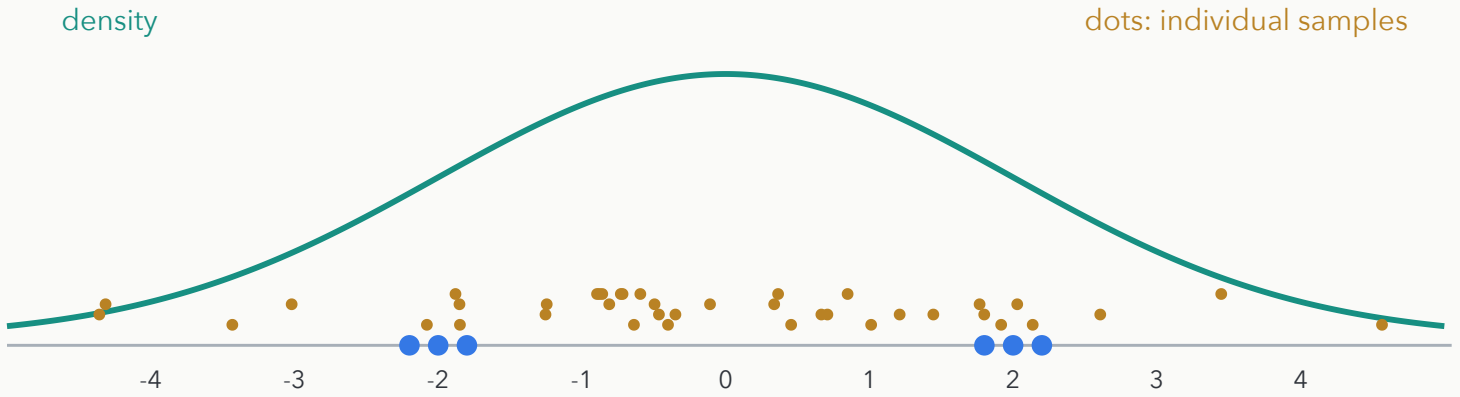
TAKEAWAY Good samples should look plausible and cover the data distribution.

SAY The average is in the middle. But almost none of our data live there.

ES 667 · original teaching diagrams and exact calculations

B / DISTRIBUTIONS

What does fitting a Gaussian learn?



Gaussian MLE μ **0.000000**

Gaussian MLE σ² **4.0266667**

$$\hat{\mu} = \frac{1}{N}\sum_{i} x_i = 0$$

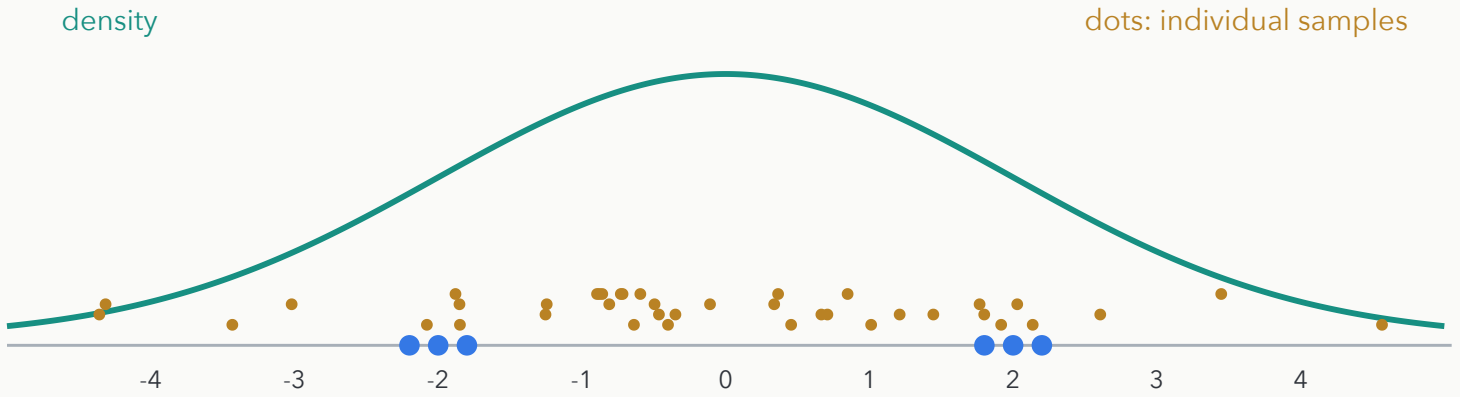
NOTICE Fit the centre to six observations.

SAY Fitting the parameters is straightforward. Choosing a suitable family is the harder question.

ES 667 · original teaching diagrams and exact calculations

B / DISTRIBUTIONS

What does fitting a Gaussian learn?



Gaussian MLE μ **0.000000**

Gaussian MLE σ² **4.0266667**

$$\hat{\sigma}^2 = \frac{1}{N}\sum_i (x_i - \hat{\mu})^2$$

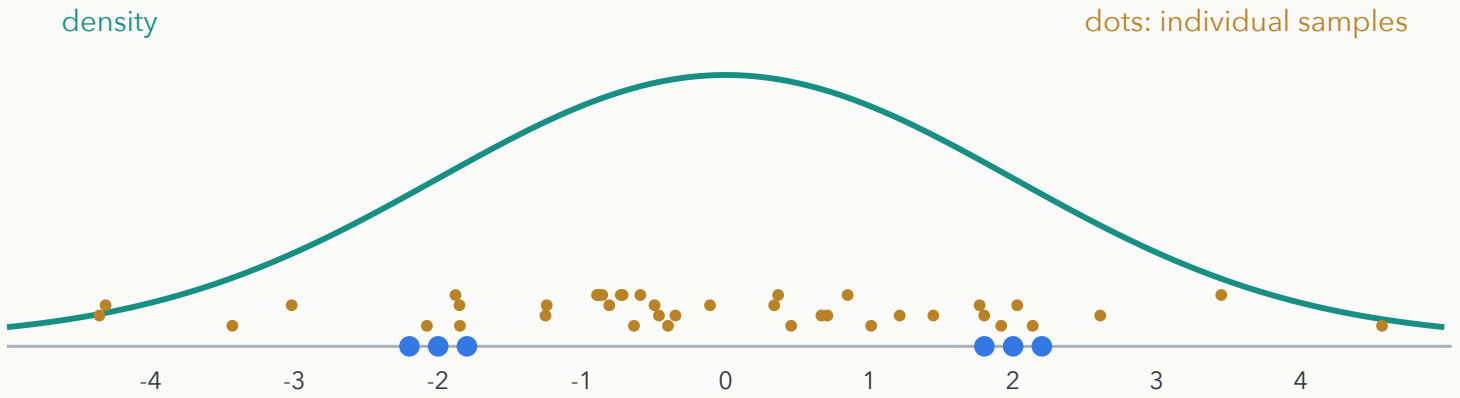
NOTICE For maximum likelihood, divide the squared deviations by N.

SAY Fitting the parameters is straightforward. Choosing a suitable family is the harder question.

ES 667 · original teaching diagrams and exact calculations

B / DISTRIBUTIONS

What does fitting a Gaussian learn?



Gaussian MLE μ **0.000000**

Gaussian MLE σ² **4.0266667**

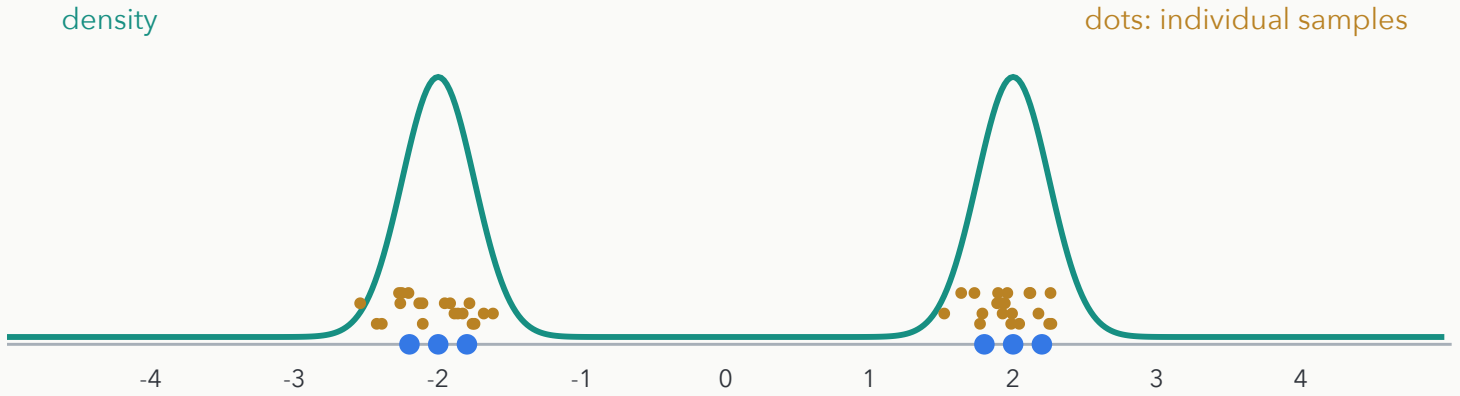
$$x = \hat{\mu} + \hat{\sigma}\epsilon$$

> **TAKEAWAY** Even the best fit within one family can miss the shape of the data.
>
> **SAY** Fitting the parameters is straightforward. Choosing a suitable family is the harder question.

ES 667 · original teaching diagrams and exact calculations

B / DISTRIBUTIONS

What if a hidden choice selects a cluster?



component means
$-2, +2$

each component σ
0.25

mixture weights
$\frac{1}{2}, \frac{1}{2}$

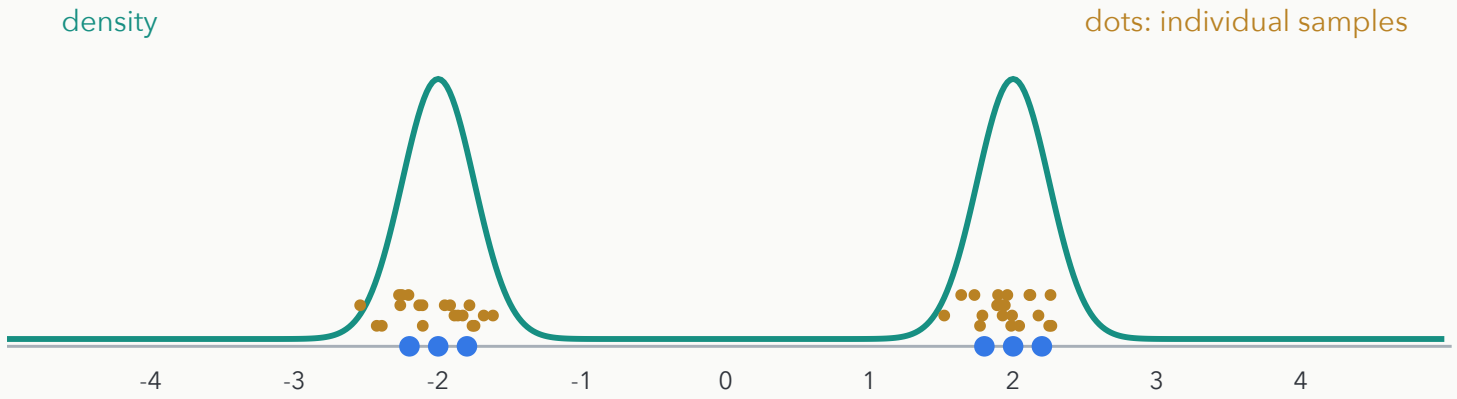
$$z \sim \text{Categorical}(1/2, 1/2)$$

NOTICE First choose a component.

SAY Think of two random choices: which cluster, then which point within it.

B / DISTRIBUTIONS

What if a hidden choice selects a cluster?



component means	each component σ	mixture weights
−2, +2	0.25	½, ½

$$p(x) = \sum_z p(z)p(x \mid z)$$

TAKEAWAY Then draw from its Gaussian. Sum over the hidden choice.

SAY Think of two random choices: which cluster, then which point within it.

ES 667 · original teaching diagrams and exact calculations

B / DISTRIBUTIONS

What distribution did our name model learn?

Recreated character bigram model fitted to 14 teaching names · start ^ · end $

^	d	e	e	v	a	$

```
P(a | ^) = 3/14    P(d | ^) = 2/14    P(r | ^) = 2/14
```

Seeds 667 and 668; categorical sampling, no temperature scaling.

$$p_\theta(x_t \mid x_{<t})$$

NOTICE Every token has a probability given its preceding context.

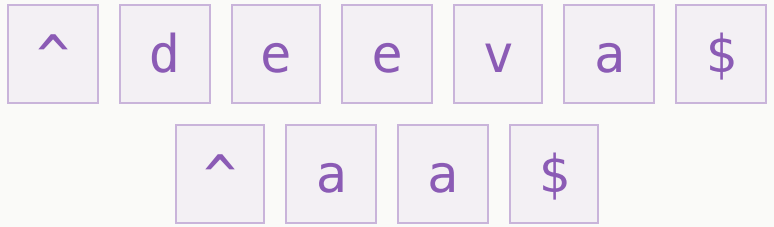
SAY We can score a whole name by following exactly the choices used to generate it.

Course name generator · reproduced categorical example

B / DISTRIBUTIONS

What distribution did our name model learn?

Recreated character bigram model fitted to 14 teaching names · start ^ · end $



`P(a | ^) = 3/14` `P(d | ^) = 2/14` `P(r | ^) = 2/14`

Seeds 667 and 668; categorical sampling, no temperature scaling.

$$p_\theta(x) = \prod_t p_\theta(x_t \mid x_{<t})$$

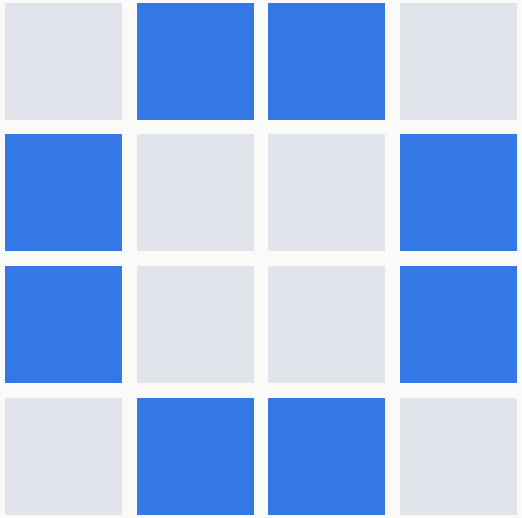
TAKEAWAY Multiply the next-token probabilities, including the probability of stopping.

SAY We can score a whole name by following exactly the choices used to generate it.

Course name generator · reproduced categorical example

B / DISTRIBUTIONS

Could we generate pixels in sequence?



Training: observed prefix

known pixels → next-pixel logits

One image: 16 binary values

$$x_1, x_2, \ldots, x_{16}$$

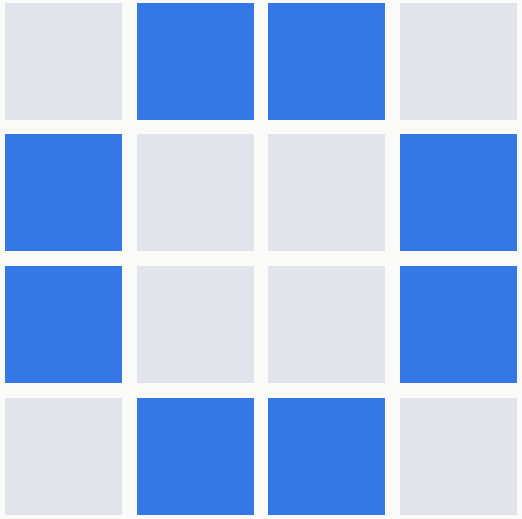
NOTICE An image can be ordered into a pixel sequence.

SAY During training we know the prefix. During generation we must build it ourselves.

Course name generator · reproduced categorical example

B / DISTRIBUTIONS

Could we generate pixels in sequence?



Training: observed prefix

known pixels → next-pixel logits

One image: 16 binary values

$$-\log p_\theta(x) = -\sum_i \log p_\theta(x_i \mid x_{<i})$$

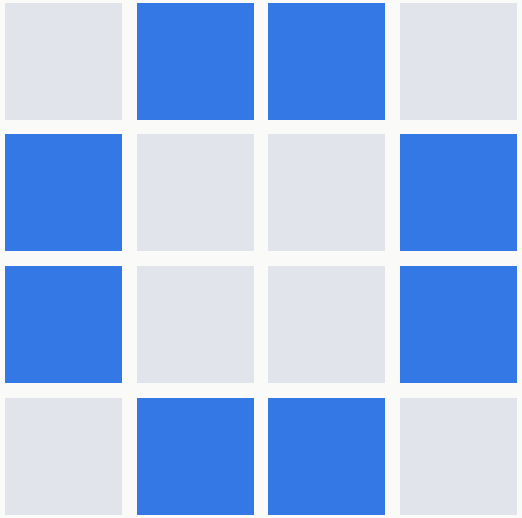
NOTICE Training uses the observed prefix. A causal mask lets us score positions in parallel.

SAY During training we know the prefix. During generation we must build it ourselves.

Course name generator · reproduced categorical example

B / DISTRIBUTIONS

Could we generate pixels in sequence?



Generation: sampled prefix

sample → append → repeat

One image: 16 binary values

$$x_i \sim p_\theta(x_i \mid x_{<i})$$

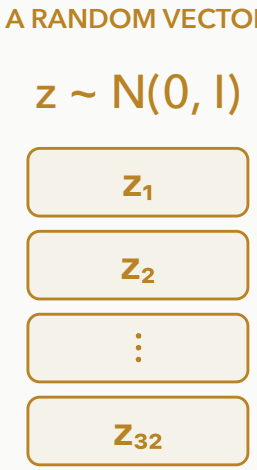
TAKEAWAY Generation feeds each sampled pixel into the next prediction.

SAY During training we know the prefix. During generation we must build it ourselves.

Course name generator · reproduced categorical example

B / DISTRIBUTIONS

Could one random vector make an image?



Local trained generator · one output from the short glyph run

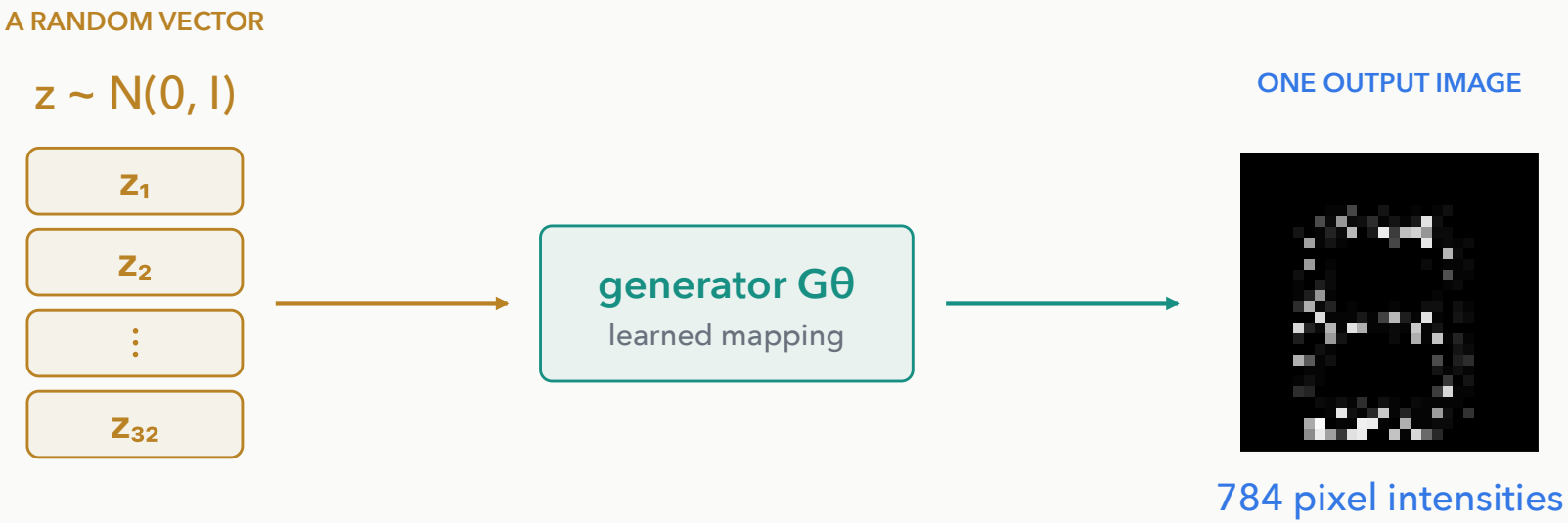
$$z \sim p(z)$$

CONSIDER Could a random vector select an entire output?

SAY Imagine choosing the whole output through one random vector.

B / DISTRIBUTIONS

Could one random vector make an image?



Local trained generator · one output from the short glyph run

$$x = G_\theta(z)$$

TAKEAWAY Learn a map that turns latent draws into plausible outputs.

SAY Imagine choosing the whole output through one random vector.

B / DISTRIBUTIONS

Does reconstruction tell us how to sample?



Reconstruction is supervised by the input.

$$x \to E_\phi(x) \to G_\theta(E_\phi(x))$$

NOTICE An autoencoder learns to reconstruct its inputs.

SAY Reconstructing the training images does not tell us where to draw a new code.

B / DISTRIBUTIONS

Does reconstruction tell us how to sample?



An irregular code cloud is possible: no Gaussian prior was specified.

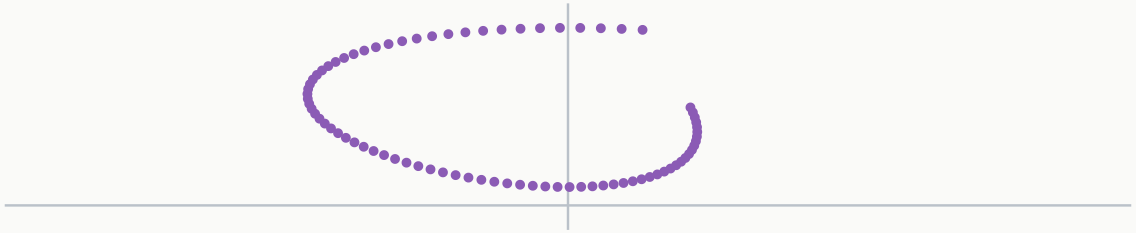


Illustration: irregular encoder codes (purple); standard-normal draws (amber)

NOTICE Reconstruction alone can leave irregularly arranged codes.

SAY Reconstructing the training images does not tell us where to draw a new code.

B / DISTRIBUTIONS

Does reconstruction tell us how to sample?



A density model over learned codes is one possible additional step.

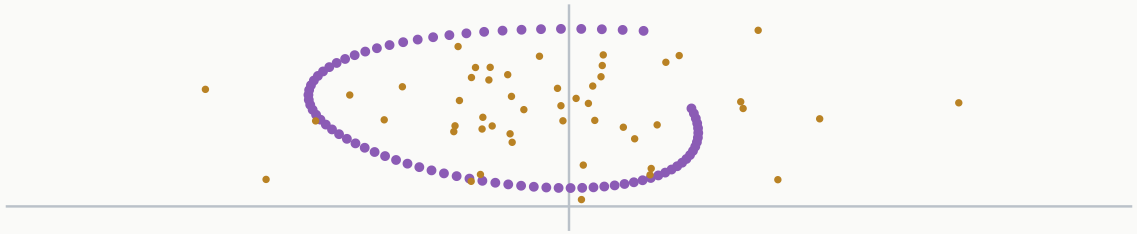


Illustration: irregular encoder codes (purple); standard-normal draws (amber)

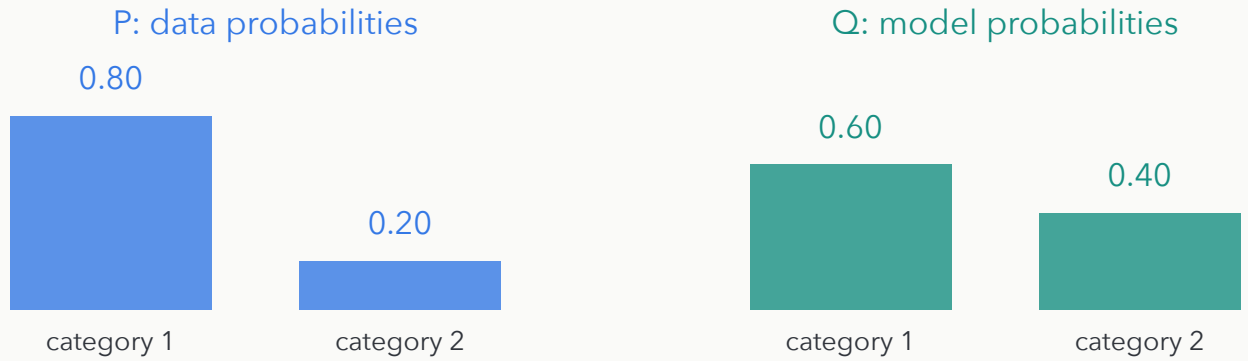
$$z \sim ?$$

TAKEAWAY Where should we draw a new code? Reconstruction alone has not answered that.

SAY Reconstructing the training images does not tell us where to draw a new code.

C / KL

How do two distributions disagree?



KL(P ‖ Q), nats

0.091516222

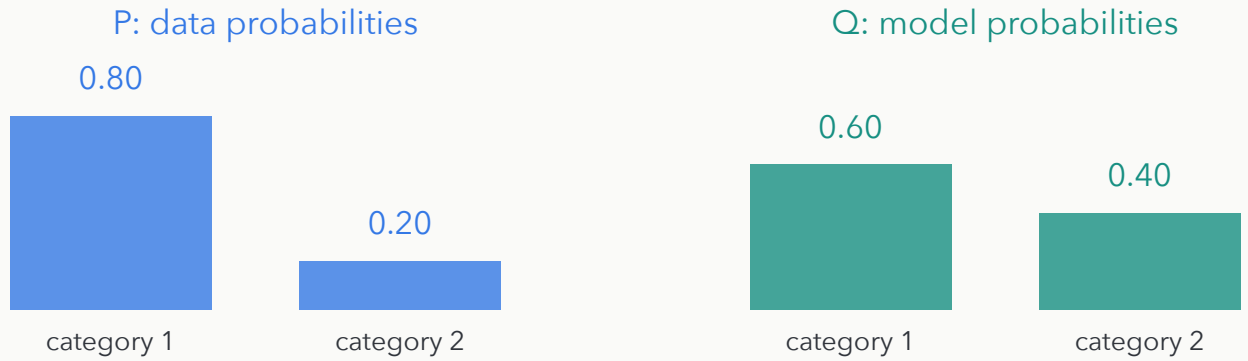
$$P = (0.8, 0.2), \quad Q = (0.6, 0.4)$$

NOTICE Two distributions assign different probabilities to the same categories.

SAY Read KL as the extra surprise we incur by using Q when the outcomes follow P.

C / KL

How do two distributions disagree?



KL(P ‖ Q), nats

0.091516222

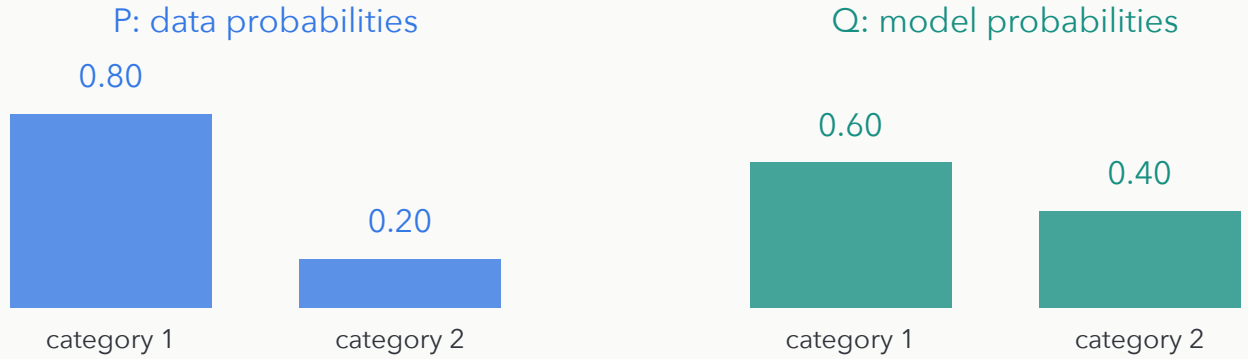
$$\text{surprisal}(i) = -\log Q_i$$

NOTICE An unlikely observation is more surprising.

SAY Read KL as the extra surprise we incur by using Q when the outcomes follow P.

C / KL

How do two distributions disagree?



KL(P ‖ Q), nats

0.091516222

$$D_{\mathrm{KL}}(P\|Q) = \sum_i P_i \log \frac{P_i}{Q_i}$$

TAKEAWAY Average the extra surprise, weighting each event by its probability under P.

SAY Read KL as the extra surprise we incur by using Q when the outcomes follow P.

C / KL

Can we compute KL by hand?

category	P_i	Q_i	$\log(P_i / Q_i)$	weighted contribution
1	0.80	0.60	0.287682072	0.230145658
2	0.20	0.40	−0.693147181	−0.138629436

KL(P ‖ Q), nats

0.091516222

NOTICE Category 1: weight the log ratio by its probability under P.

SAY Keep the negative term. It is the total, not each contribution, that must be nonnegative.

C / KL

Can we compute KL by hand?

category	P_i	Q_i	$\log(P_i / Q_i)$	weighted contribution
1	0.80	0.60	`0.287682072`	`0.230145658`
2	0.20	0.40	`−0.693147181`	`−0.138629436`

KL(P ‖ Q), nats

`0.091516222`

NOTICE Category 2 can make a negative contribution.

SAY Keep the negative term. It is the total, not each contribution, that must be nonnegative.

C / KL

Can we compute KL by hand?

category	P_i	Q_i	$\log(P_i / Q_i)$	weighted contribution
1	0.80	0.60	`0.287682072`	`0.230145658`
2	0.20	0.40	`−0.693147181`	`−0.138629436`

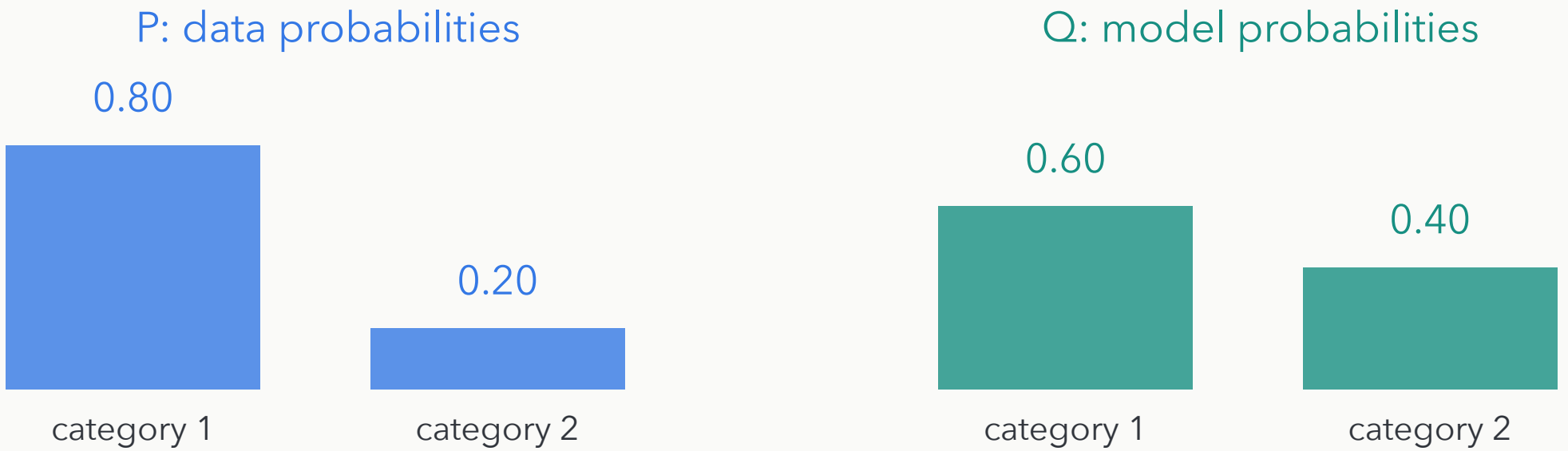
KL(P ‖ Q), nats

`0.091516222`

> **TAKEAWAY** Individual contributions can be negative. Their sum cannot.
>
> **SAY** Keep the negative term. It is the total, not each contribution, that must be nonnegative.

C / KL

Why do direction and support matter?



KL(P ‖ Q), nats
0.091516222

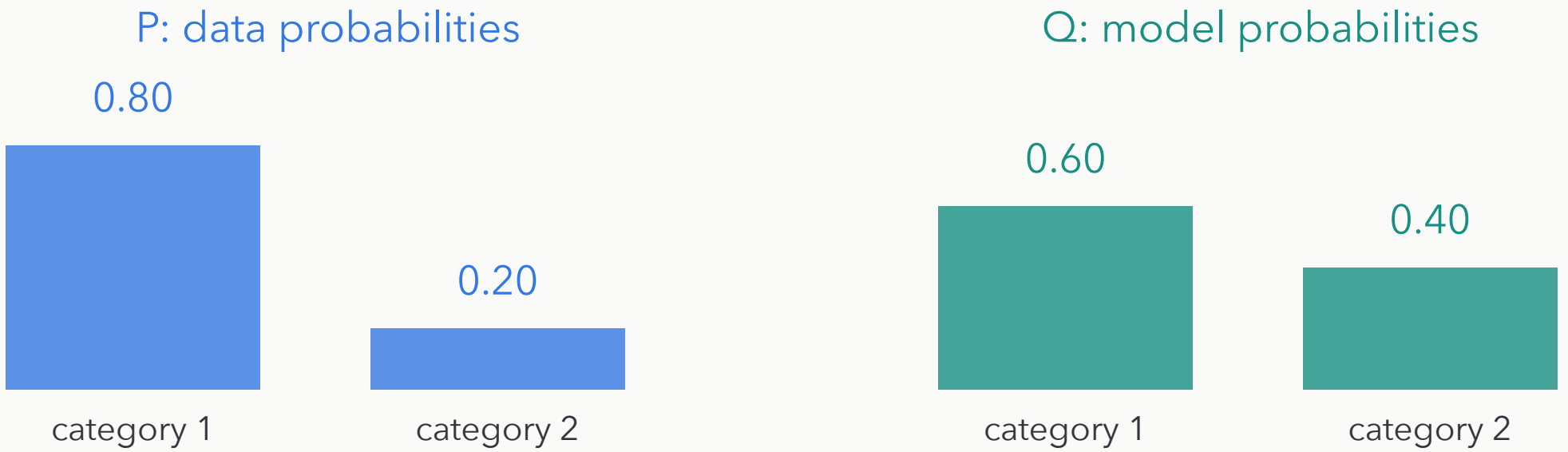
KL(Q ‖ P), nats
0.104649629

NOTICE Swap P and Q: the weights change as well as the log ratios.

SAY Swap the arguments first. Then try giving a possible event zero probability.

C / KL

Why do direction and support matter?



KL(P ‖ Q), nats
0.091516222

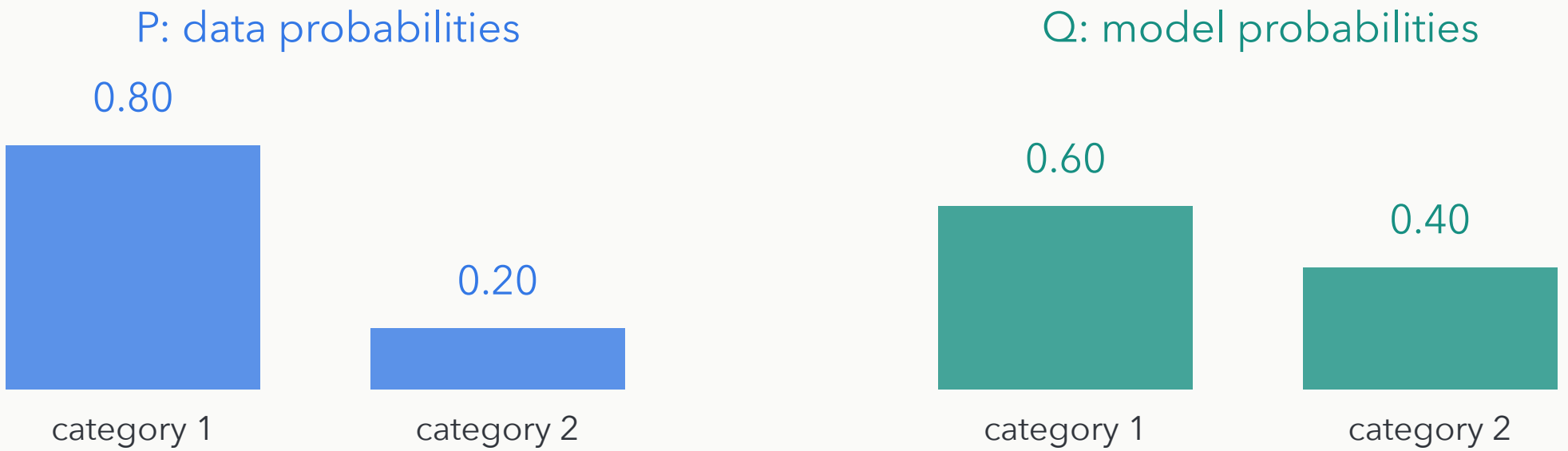
KL(Q ‖ P), nats
0.104649629

NOTICE Matching distributions incur no extra surprisal.

SAY Swap the arguments first. Then try giving a possible event zero probability.

C / KL

Why do direction and support matter?



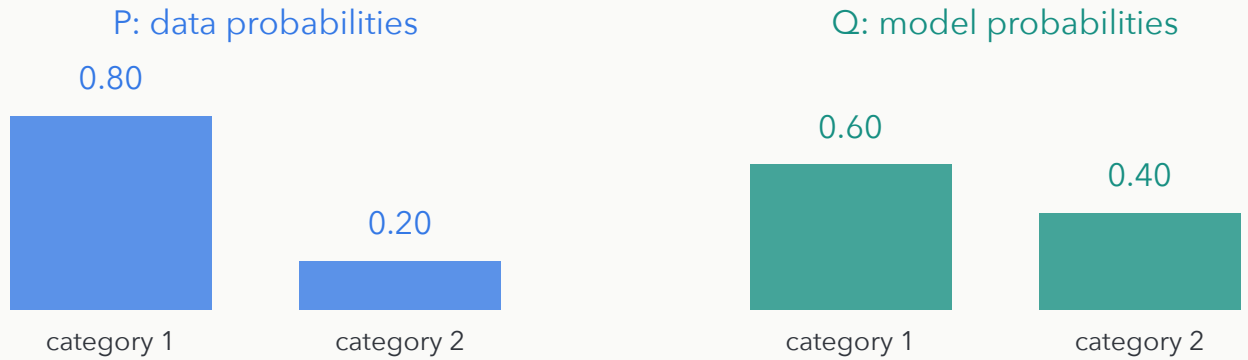
KL(P ‖ Q), nats
0.091516222

KL(Q ‖ P), nats
0.104649629

> **TAKEAWAY** A possible event given zero model probability has infinite cost.
>
> **SAY** Swap the arguments first. Then try giving a possible event zero probability.

C / KL

Where is KL hiding in cross-entropy?



KL(P ‖ Q), nats
`0.091516222`

H(P)
`0.500402424`

H(P,Q)
`0.591918645`

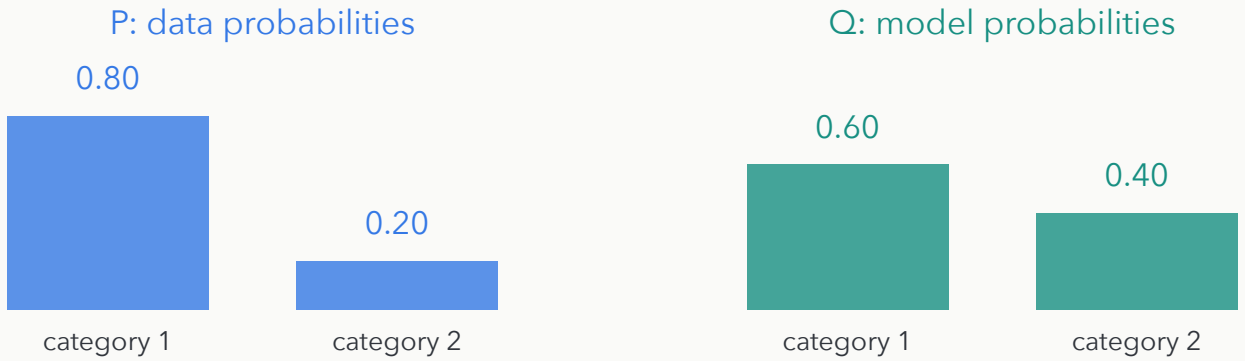
$$H(P) = -\sum_i P_i \log P_i$$

NOTICE Entropy is the uncertainty in P. Cross-entropy uses Q to describe it.

SAY With the data distribution fixed, entropy is a constant in the training objective.

C / KL

Where is KL hiding in cross-entropy?



KL(P ‖ Q), nats **0.091516222**

H(P) **0.500402424**

H(P,Q) **0.591918645**

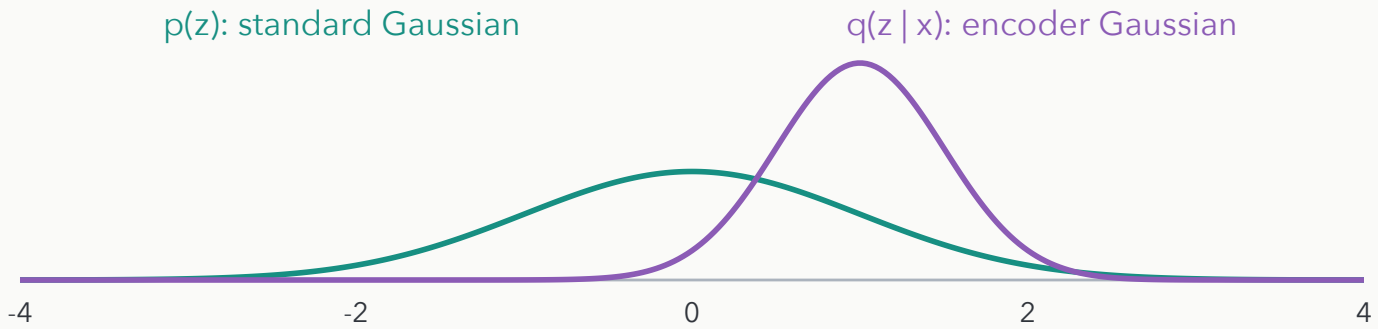
$$H(P,Q) = H(P) + D_{\mathrm{KL}}(P\|Q)$$

TAKEAWAY With P fixed, minimising cross-entropy also minimises KL.

SAY With the data distribution fixed, entropy is a constant in the training objective.

C / KL

What changes for a continuous latent?



Prior: N(0,1) · approximate posterior: N(μ, σ²)

μ	σ	KL(q ‖ prior)
1.00	0.50	0.818147

$$D_{\mathrm{KL}}(q\|p) = \int q(z) \log \frac{q(z)}{p(z)}\, dz$$

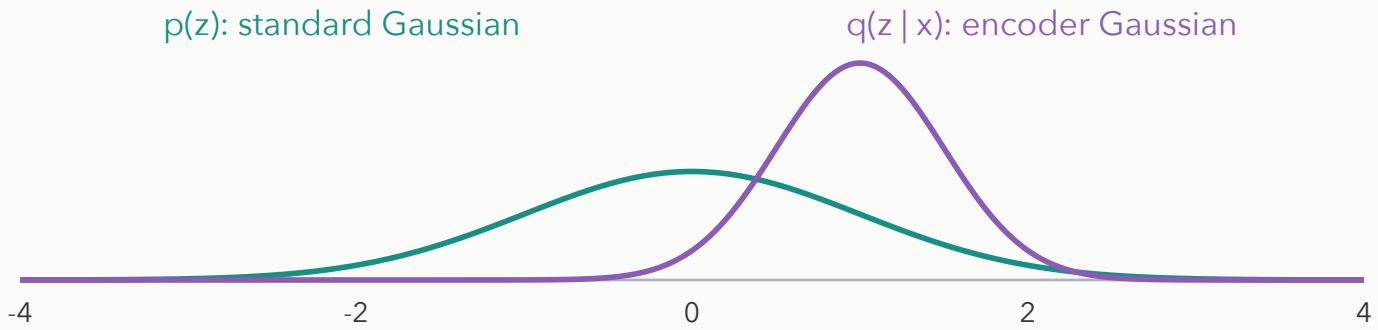
NOTICE For a continuous latent, sum becomes integral.

SAY Now the latent can take any real value, so the sum becomes an integral.

Kingma & Welling · AEVB · authored teaching diagram

C / KL

What changes for a continuous latent?



Prior: N(0,1) · approximate posterior: N(μ, σ²)

μ	σ	KL(q ‖ prior)
1.00	0.50	0.818147

$$D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z))$$

TAKEAWAY The encoder gives a distribution for this x. The prior does not depend on x.

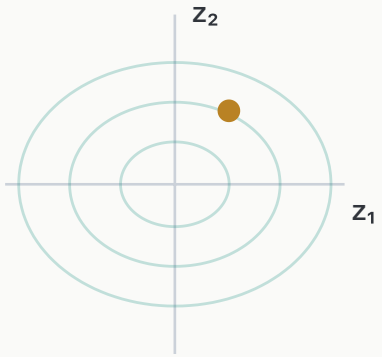
SAY Now the latent can take any real value, so the sum becomes an integral.

Kingma & Welling · AEVB · authored teaching diagram

D / ELBO

What is the generative story?

1 · DRAW A LATENT



$z \sim N(0, I)$

z = (0.96, 1.30)

$$z \sim \mathcal{N}(0, I)$$

NOTICE Begin with a prior. No observed image or encoder is needed.

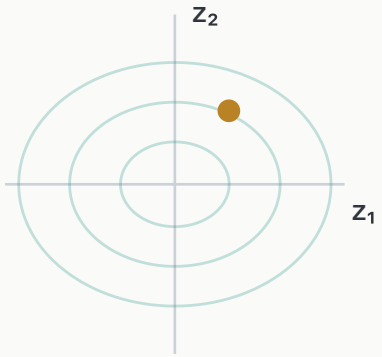
SAY Tell the sampling story in order: draw z, decode its parameters, then draw x.

Kingma & Welling · AEVB · authored teaching diagram

D / ELBO

What is the generative story?

1 · DRAW A LATENT



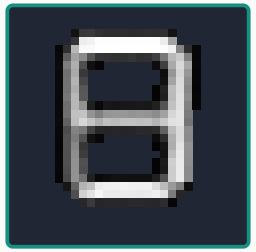
$z \sim N(0, I)$

z = (0.96, 1.30)

decoder θ

2 → 128 → 784

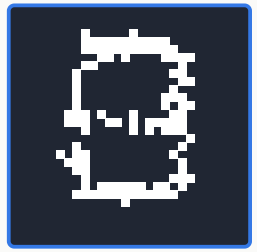
2 · DECODE PROBABILITIES



$\pi = \text{sigmoid}(\text{logits})$

one probability per pixel

3 · SAMPLE PIXELS



$x_i \sim \text{Bernoulli}(\pi_i)$

one binary image

$$p_\theta(x, z) = p(z)p_\theta(x \mid z)$$

TAKEAWAY Decode probabilities first, then sample the observation.

SAY Tell the sampling story in order: draw z, decode its parameters, then draw x.

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D / ELBO

What does maximum likelihood require?

MANY POSSIBLE LATENTS



Integrate over all z; these three paths only illustrate the calculation.

$$p_\theta(x) = \int p(z) p_\theta(x \mid z)\, dz$$

NOTICE Every latent that could explain this image contributes to its probability.

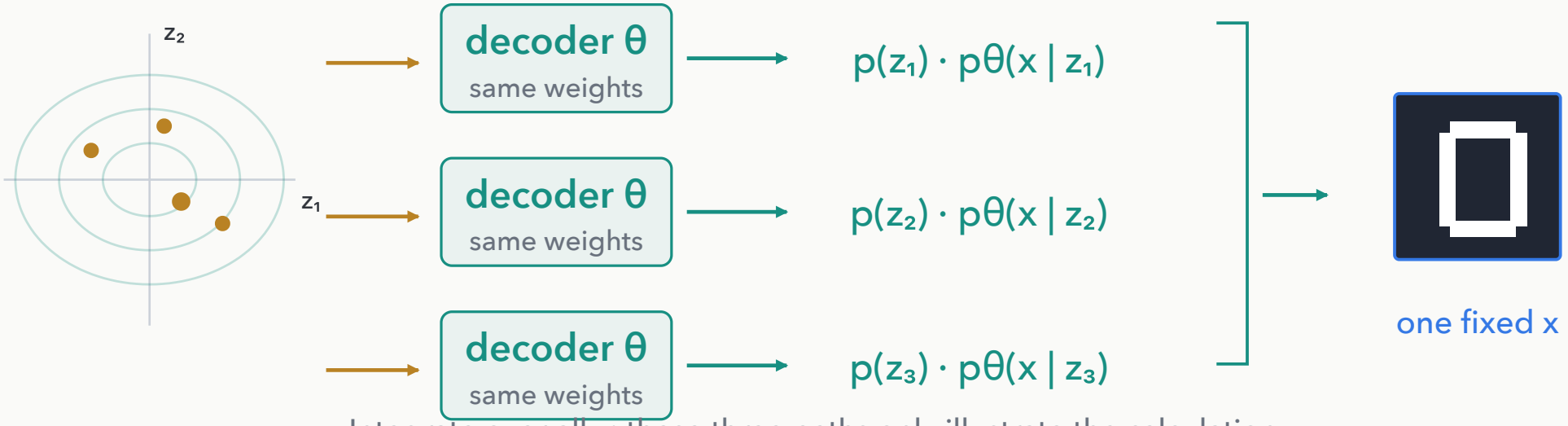
SAY To score one image, we must account for all the latent values that could explain it.

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D / ELBO

What does maximum likelihood require?

MANY POSSIBLE LATENTS



Integrate over all z; these three paths only illustrate the calculation.

$$\max_{\theta} \sum_{n} \log p_\theta(x_n)$$

TAKEAWAY With a nonlinear decoder, this integral is usually hard to compute exactly.

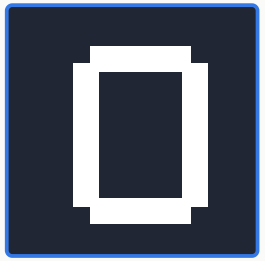
SAY To score one image, we must account for all the latent values that could explain it.

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D / ELBO

Which latent could explain this image?

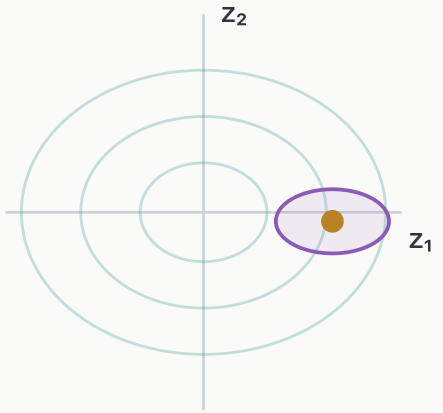
OBSERVED IMAGE



x is now fixed

encoder φ
approximate inference

qφ(z | x)



A distribution, not one code

$$p_\theta(z \mid x) = \frac{p_\theta(x, z)}{p_\theta(x)}$$

CONSIDER After observing x, which latent values explain it?

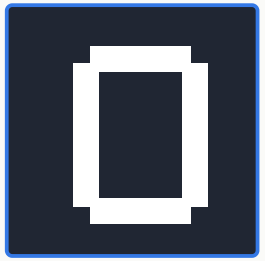
SAY The true posterior asks which latents could have produced the image we observed.

Kingma & Welling · AEVB · authored teaching diagram

D / ELBO

Which latent could explain this image?

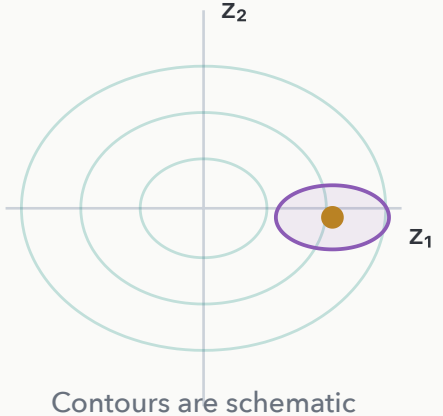
OBSERVED IMAGE



x is now fixed

encoder φ
approximate inference

qφ(z | x)



Contours are schematic

A distribution, not one code

Why approximate?

The true posterior needs pθ(x), the difficult integral.

$$q_\phi(z \mid x) \approx p_\theta(z \mid x)$$

TAKEAWAY Use a tractable encoder distribution to approximate the true posterior.

SAY The true posterior asks which latents could have produced the image we observed.

Kingma & Welling · AEVB · authored teaching diagram

How does log become a lower bound?

ELBO DERIVATION · LINE 1 OF 7

$$\log p_\theta(x) = \log \int p_\theta(x, z)\, dz$$

NOTICE Start with the marginal probability of the observation.

SAY Follow the equality signs. Jensen's inequality is the one step that introduces a bound.

D / ELBO

How does log become a lower bound?

ELBO DERIVATION · LINE 2 OF 7

$$\log p_\theta(x) = \log \int p_\theta(x, z)\, dz$$

$$= \log \int q_\phi(z \mid x) \frac{p_\theta(x, z)}{q_\phi(z \mid x)}\, dz$$

NOTICE Multiply and divide by q. It must be normalised and cover the required support.

SAY Follow the equality signs. Jensen's inequality is the one step that introduces a bound.

Kingma & Welling · AEVB · authored teaching diagram

D / ELBO

How does log become a lower bound?

ELBO DERIVATION · LINE 3 OF 7

$$= \log \int q_\phi(z \mid x) \frac{p_\theta(x, z)}{q_\phi(z \mid x)} \, dz$$

$$= \log \mathbb{E}_q \left[\frac{p_\theta(x, z)}{q_\phi(z \mid x)} \right]$$

NOTICE The integral is now an expectation under q.

SAY Follow the equality signs. Jensen's inequality is the one step that introduces a bound.

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How does log become a lower bound?

ELBO DERIVATION · LINE 4 OF 7

$$= \log \mathbb{E}_q \left[\frac{p_\theta(x, z)}{q_\phi(z \mid x)} \right]$$

$$\geq \mathbb{E}_q \left[\log \frac{p_\theta(x, z)}{q_\phi(z \mid x)} \right]$$

Concave log: log E[R] ≥ E[log R]. This is the first inequality.

TAKEAWAY Move log inside the expectation using Jensen. This is the first inequality.

SAY Follow the equality signs. Jensen's inequality is the one step that introduces a bound.

Kingma & Welling · AEVB · authored teaching diagram

D / ELBO

What are the two terms in the bound?

ELBO DERIVATION · LINE 5 OF 7

$$\mathbb{E}_q[\log p_\theta(x \mid z) + \log p(z) - \log q_\phi(z \mid x)]$$

NOTICE Factor the joint and separate the logarithms.

SAY The bound rewards explaining the image and subtracts a KL to the prior.

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D / ELBO

What are the two terms in the bound?

ELBO DERIVATION · LINE 6 OF 7

$$\mathbb{E}_q[\log p_\theta(x \mid z) + \log p(z) - \log q_\phi(z \mid x)]$$

$$\mathbb{E}_q[\log p_\theta(x \mid z)] - D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z))$$

NOTICE The bound is expected reconstruction log likelihood minus a KL to the prior.

SAY The bound rewards explaining the image and subtracts a KL to the prior.

Kingma & Welling · AEVB · authored teaching diagram

D / ELBO

What are the two terms in the bound?

ELBO DERIVATION · LINE 7 OF 7

$$\mathbb{E}_q[\log p_\theta(x \mid z)] - D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z))$$

$$\log p_\theta(x) \geq \mathcal{E}(\theta, \phi; x)$$

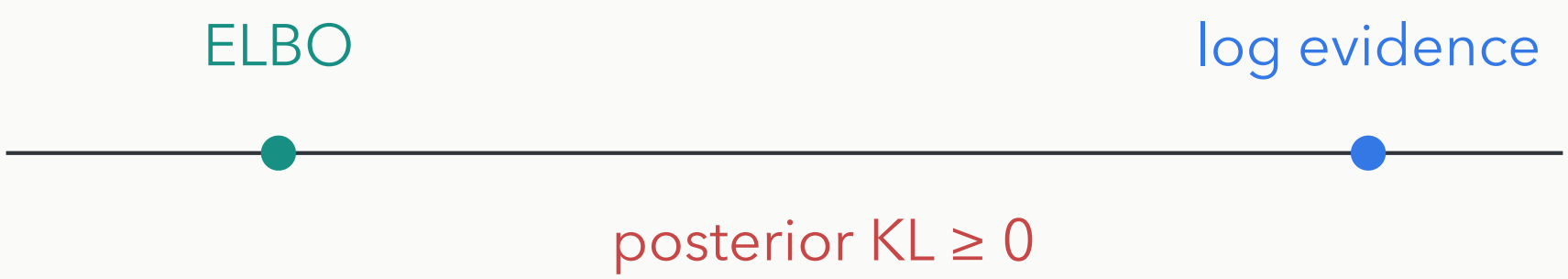
TAKEAWAY Maximise the evidence lower bound, or minimise its negative.

SAY The bound rewards explaining the image and subtracts a KL to the prior.

Kingma & Welling · AEVB · authored teaching diagram

D / ELBO

What is the gap to log evidence?



The KL in the training loss compares q with the prior.

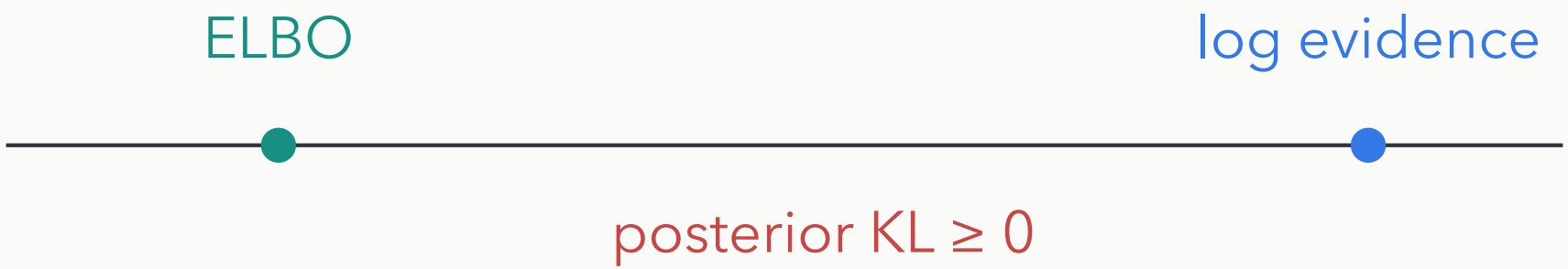
$$D_{\mathrm{KL}}(q \| p_\theta(z \mid x)) = \mathbb{E}_q[\log q - \log p_\theta(x, z)] + \log p_\theta(x)$$

NOTICE Expand the KL to the true posterior using Bayes' rule.

SAY There are two KLs in this story. Check which distribution is on the right each time.

Kingma & Welling · AEVB · authored teaching diagram

What is the gap to log evidence?



The KL in the training loss compares q with the prior.

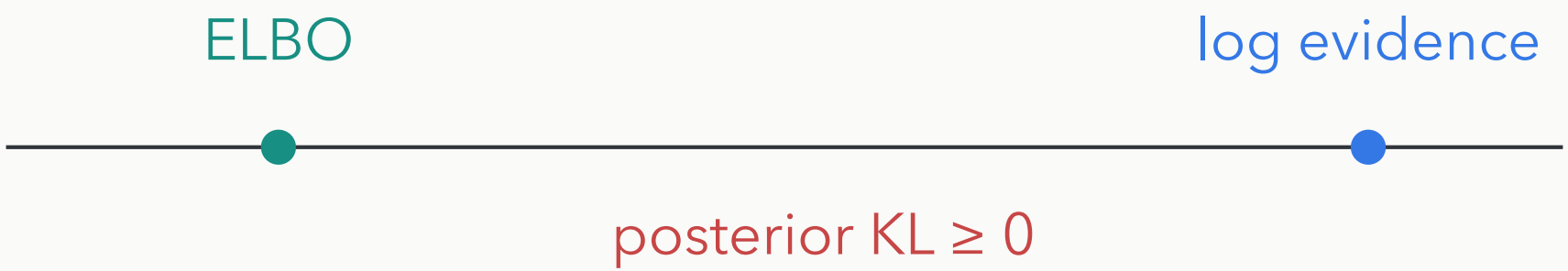
$$\log p_\theta(x) = \mathcal{E} + D_{\mathrm{KL}}(q \| p_\theta(z \mid x))$$

NOTICE The gap is a posterior KL, not the KL to the prior.

SAY There are two KLs in this story. Check which distribution is on the right each time.

D / ELBO

What is the gap to log evidence?



The KL in the training loss compares q with the prior.

$$q = p_\theta(z \mid x) \implies \mathcal{E} = \log p_\theta(x)$$

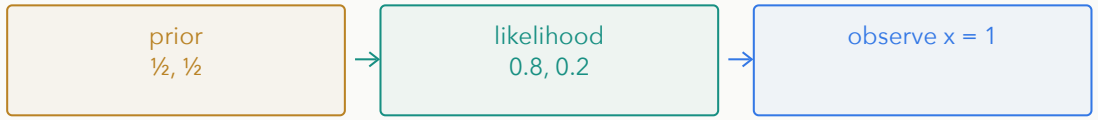
TAKEAWAY The gap closes when q matches the true posterior almost everywhere.

SAY There are two KLs in this story. Check which distribution is on the right each time.

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D / ELBO

Can we check the bound exactly?



evidence
0.500000

posterior
0.8, 0.2

E_q log p(x|z)	−0.777661
KL(q ‖ prior)	0.020136
ELBO	−0.797797
KL(q ‖ posterior)	0.104650
log evidence	−0.693147

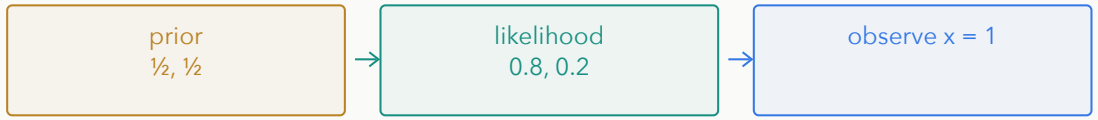
NOTICE Two latent states let us compute evidence and posterior exactly.

SAY Move q while watching the evidence stay fixed and the bound move below it.

Kingma & Welling · AEVB · authored teaching diagram

D / ELBO

Can we check the bound exactly?



evidence
`0.500000`

posterior
`0.8, 0.2`

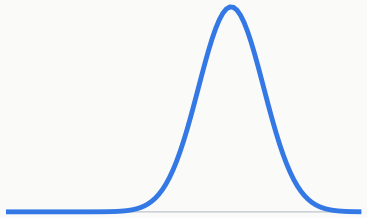
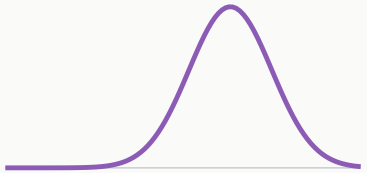
E_q log p(x|z)	`−0.777661`
KL(q ‖ prior)	`0.020136`
ELBO	`−0.797797`
KL(q ‖ posterior)	`0.104650`
log evidence	`−0.693147`

> **TAKEAWAY** Move q. The bound changes; the fixed model evidence does not.
>
> **SAY** Move q while watching the evidence stay fixed and the bound move below it.

Kingma & Welling · AEVB · authored teaching diagram

D / ELBO

Which distribution is which?

$p(z)$	$p_\theta(z \mid x)$	$q_\phi(z \mid x)$	$p_\theta(x \mid z)$
			
FIXED PRIOR	FOR THIS x	FOR THIS x	FOR THIS z
Before seeing x Fixed by the model	After seeing x Usually intractable	After seeing x Predicted by the encoder	After choosing z Pixel probabilities

SCHEMATIC LATENT DENSITIES · ACTUAL RECORDED DECODER IMAGE

$$p(z), \quad p_\theta(z \mid x), \quad q_\phi(z \mid x), \quad p_\theta(x \mid z)$$

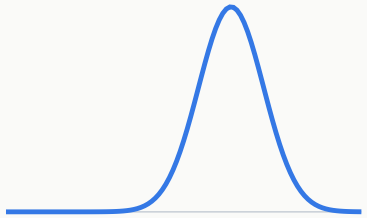
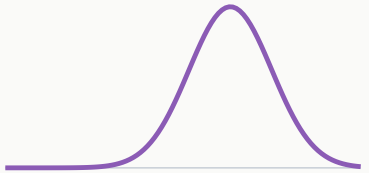
CONSIDER Which distribution belongs at each point in the model?

SAY Before using a symbol, say what is conditioned on and what it describes.

Kingma & Welling · AEVB · authored teaching diagram

D / ELBO

Which distribution is which?

$p(z)$	$p_\theta(z \mid x)$	$q_\varphi(z \mid x)$	$p_\theta(x \mid z)$
			
FIXED PRIOR	FOR THIS x	FOR THIS x	FOR THIS z
Before seeing x Fixed by the model	After seeing x Usually intractable	After seeing x Predicted by the encoder	After choosing z Pixel probabilities

SCHEMATIC LATENT DENSITIES · ACTUAL RECORDED DECODER IMAGE

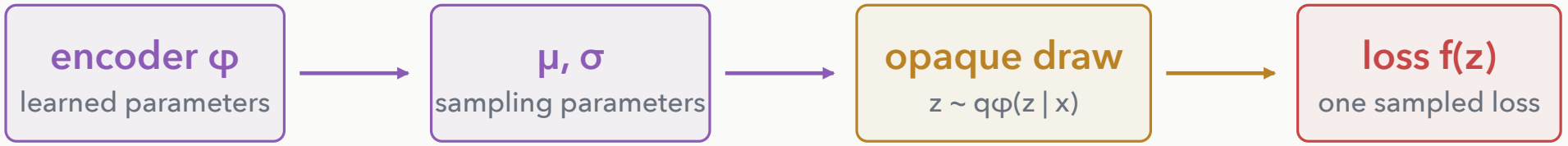
$$\hat{\mathcal{E}} \neq \mathcal{E} \text{ in general}$$

> TAKEAWAY The exact ELBO is a bound. One Monte Carlo estimate can sit above log evidence.
>
> SAY Before using a symbol, say what is conditioned on and what it describes.

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E / GRADIENTS

Where is the missing gradient path?



$$\nabla_\phi \mathbb{E}_{z\sim q_\phi}[f(z)]$$

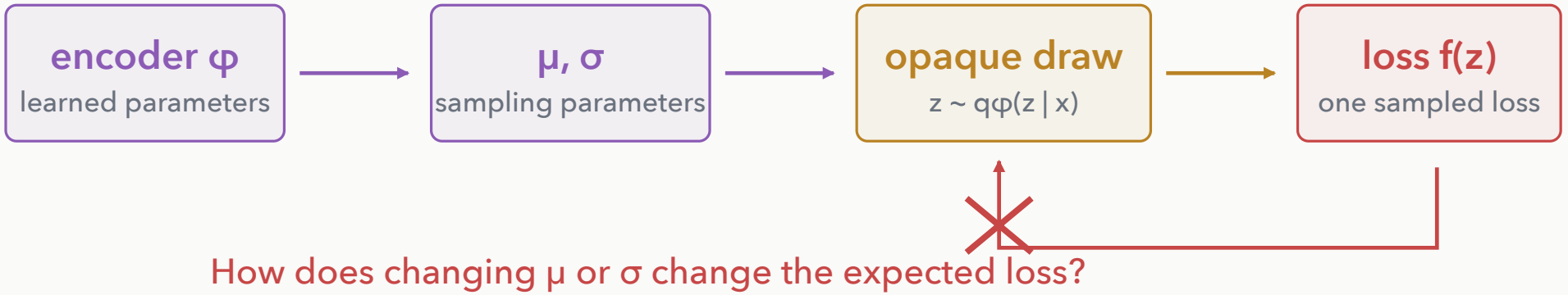
NOTICE The encoder controls the sampling distribution. We need its expected-loss gradient.

SAY We need the loss to tell the encoder how to move its sampling distribution.

Kingma & Welling · AEVB · authored teaching diagram

E / GRADIENTS

Where is the missing gradient path?



$$\mu, \sigma \rightarrow \text{sampler} \rightarrow z \rightarrow f(z)$$

TAKEAWAY An opaque random draw hides the path needed for ordinary backpropagation.

SAY We need the loss to tell the encoder how to move its sampling distribution.

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E / GRADIENTS

Can we move the randomness?

FIXED RANDOM INPUT

ε
-0.40

The distribution of ε stays fixed while μ and σ change.

ε stays fixed when parameters change.

$$\epsilon \sim \mathcal{N}(0, 1)$$

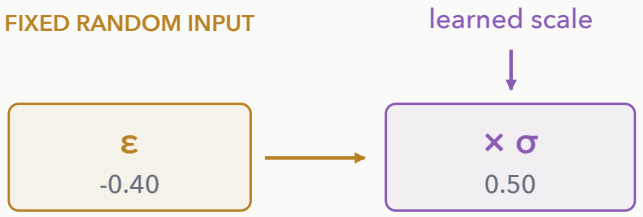
NOTICE Draw noise from a fixed, parameter-independent distribution.

SAY Keep the noise draw fixed. Stretch it, then shift it.

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E / GRADIENTS

Can we move the randomness?



The distribution of ε stays fixed while μ and σ change.

ε stays fixed when parameters change.

$$\sigma\epsilon$$

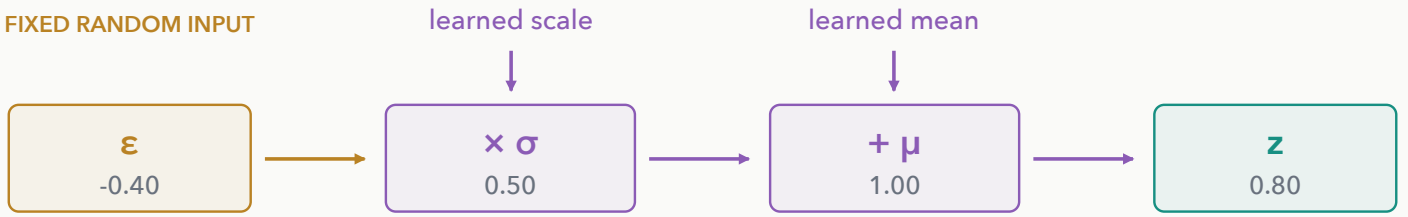
NOTICE Stretch the same noise draw.

SAY Keep the noise draw fixed. Stretch it, then shift it.

Kingma & Welling · AEVB · authored teaching diagram

E / GRADIENTS

Can we move the randomness?



The distribution of ε stays fixed while μ and σ change.

ε stays fixed when parameters change.

$$z = \mu + \sigma\epsilon \sim \mathcal{N}(\mu, \sigma^2)$$

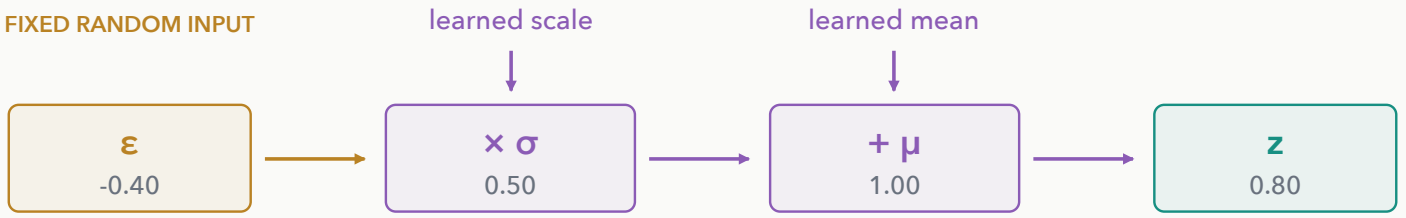
TAKEAWAY Shift the scaled noise. The result has exactly the Gaussian distribution we wanted.

SAY Keep the noise draw fixed. Stretch it, then shift it.

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E / GRADIENTS

Which derivatives follow the sample?



f(z) = z²

0.640

ε stays fixed when parameters change. Readouts use f(z) = z² and refer to this one draw.

$$\mathbb{E}_q f(z) = \mathbb{E}_\epsilon f(\mu + \sigma\epsilon)$$

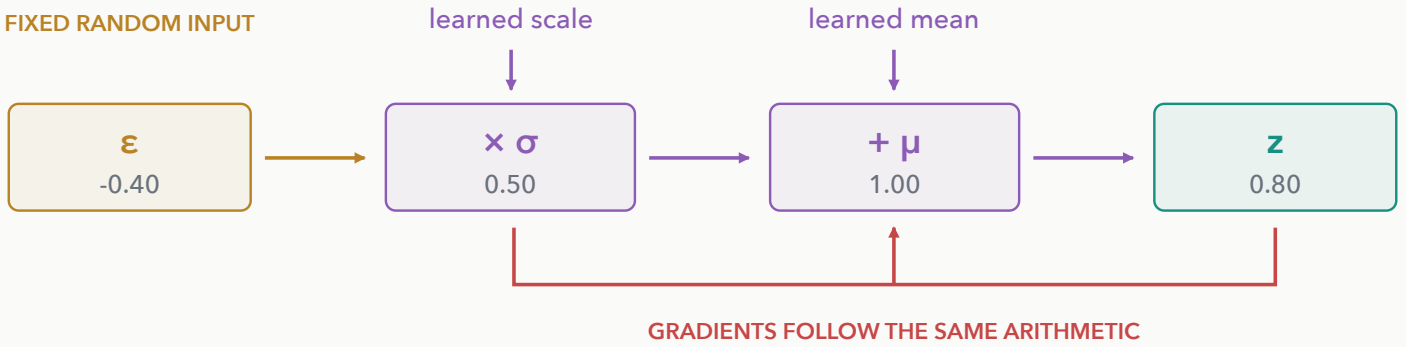
NOTICE Move the expectation to parameter-independent noise.

SAY For this derivative, epsilon is a fixed input, just like any other number in the graph.

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E / GRADIENTS

Which derivatives follow the sample?



f(z) = z²	∂f/∂μ	∂f/∂σ
0.640	1.600	−0.640

ε stays fixed when parameters change. Readouts use f(z) = z² and refer to this one draw.

$$\frac{\partial z}{\partial \mu} = 1, \qquad \frac{\partial z}{\partial \sigma} = \epsilon$$

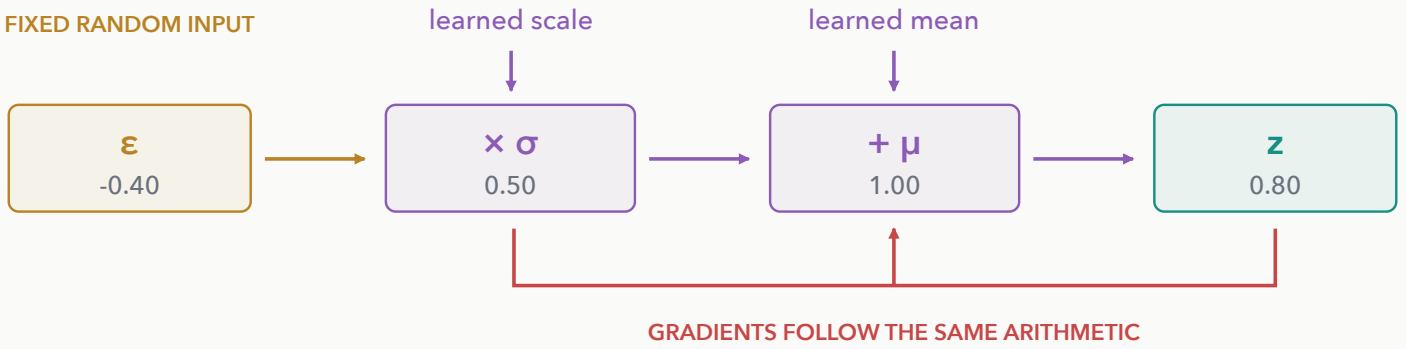
NOTICE For this draw, hold epsilon fixed and differentiate.

SAY For this derivative, epsilon is a fixed input, just like any other number in the graph.

Kingma & Welling · AEVB · authored teaching diagram

E / GRADIENTS

Which derivatives follow the sample?



f(z) = z²	∂f/∂μ	∂f/∂σ
0.640	1.600	−0.640

ε stays fixed when parameters change. Readouts use f(z) = z² and refer to this one draw.

$$\frac{\partial f}{\partial \mu} = f\prime(z), \quad \frac{\partial f}{\partial \sigma} = f\prime(z)\epsilon$$

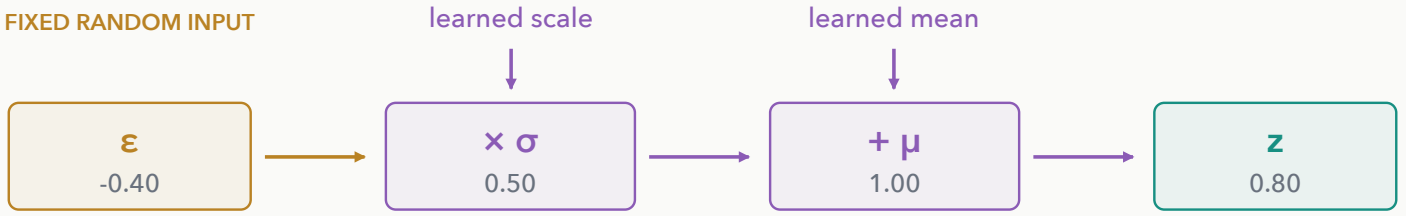
TAKEAWAY Apply the chain rule to each draw; average draws to estimate the expected gradient.

SAY For this derivative, epsilon is a fixed input, just like any other number in the graph.

Kingma & Welling · AEVB · authored teaching diagram

E / GRADIENTS

Can we check one gradient by hand?



f(z) = z²

0.640

ε stays fixed when parameters change. Readouts use f(z) = z² and refer to this one draw.

$$\mu = 1, \quad \sigma = .5, \quad \epsilon = -.4, \quad a = 0$$

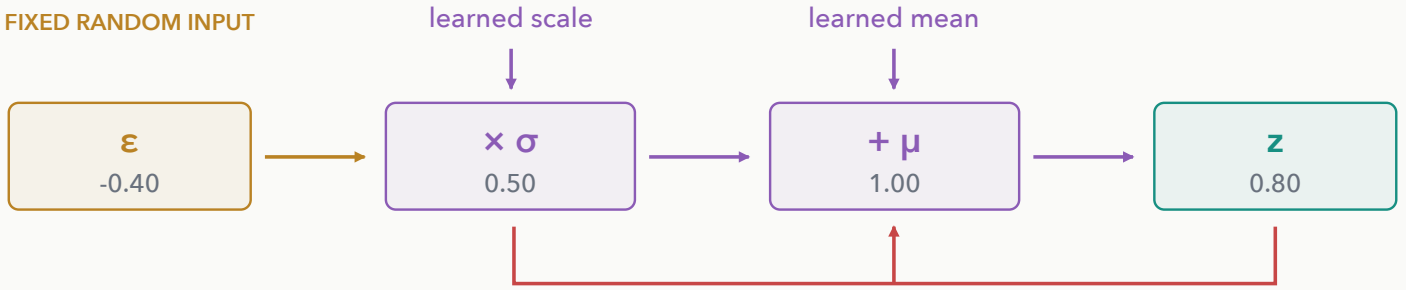
NOTICE A fixed draw gives a concrete sample and loss.

SAY Use the same epsilon on both sides of the finite-difference check.

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E / GRADIENTS

Can we check one gradient by hand?



f(z) = z²	∂f/∂μ	∂f/∂σ	∂f/∂logvar
0.640	1.600	−0.640	−0.160

ε stays fixed when parameters change. Readouts use f(z) = z² and refer to this one draw.

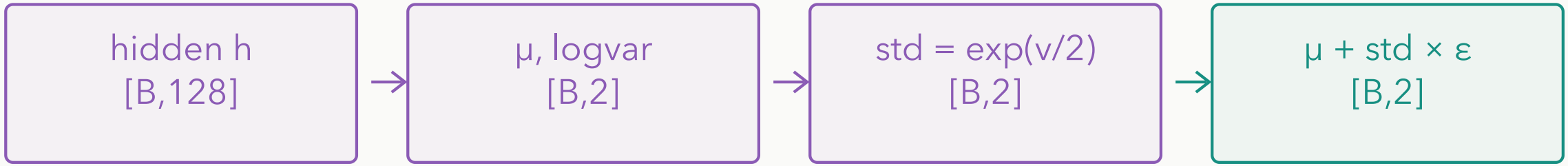
$$\frac{\partial f}{\partial v} = f\prime(z)\frac{\sigma\epsilon}{2}, \quad v = \log\sigma^2$$

> **TAKEAWAY** Use the same epsilon when checking the derivatives with finite differences.
>
> **SAY** Use the same epsilon on both sides of the finite-difference check.

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E / GRADIENTS

What should the encoder output?




```
std = torch.exp(0.5 * logvar)
eps = torch.randn_like(std)
z = mu + std * eps
```

$$\text{std} = \exp(\tfrac{1}{2}\,\text{logvar})$$

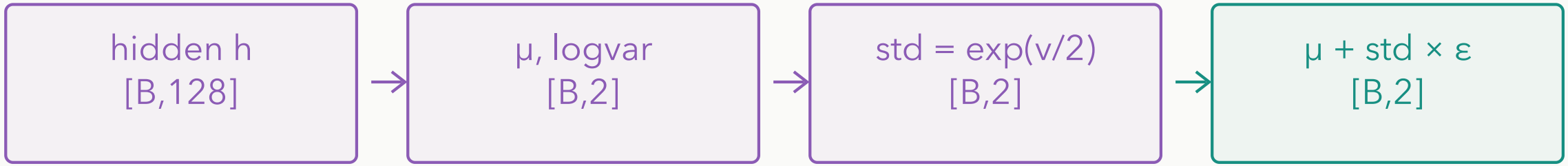
NOTICE The encoder predicts log variance. Convert it to standard deviation before sampling.

SAY Log variance is a convenient network output. It is not the scale used to multiply epsilon.

Kingma & Welling · AEVB · authored teaching diagram

E / GRADIENTS

What should the encoder output?




```
std = torch.exp(0.5 * logvar)
eps = torch.randn_like(std)
z = mu + std * eps
```

$$z = \mu + \text{std} \odot \epsilon$$

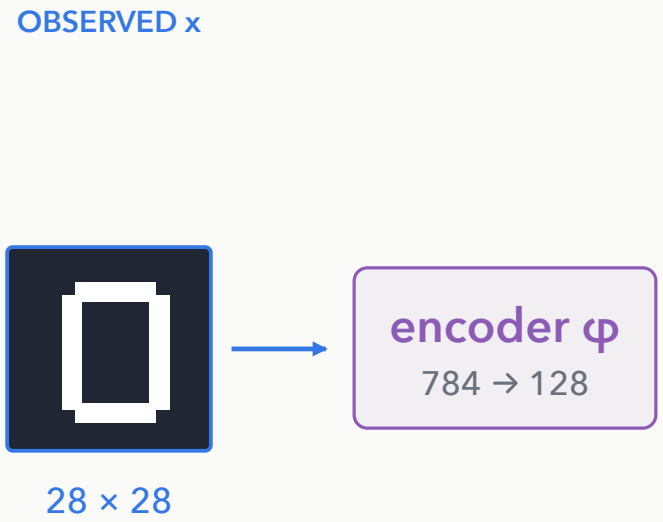
TAKEAWAY All five tensors have the same shape.

SAY Log variance is a convenient network output. It is not the scale used to multiply epsilon.

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F / VAE

How do the VAE tensors fit together?



$$[B, 1, 28, 28] \rightarrow [B, 784] \rightarrow [B, 128]$$

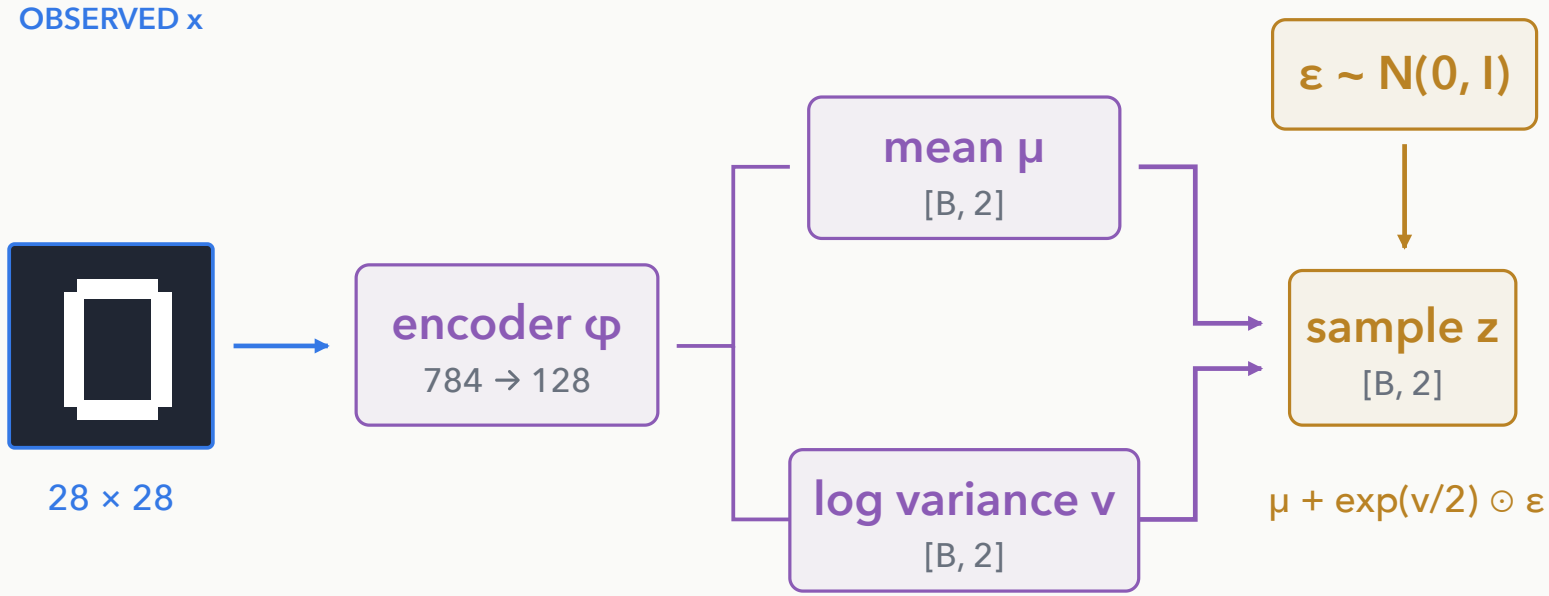
NOTICE Flatten each image, then compute a hidden representation.

SAY Trace one batch through the diagram and check the width at each arrow.

Kingma & Welling · AEVB · authored teaching diagram

F / VAE

How do the VAE tensors fit together?



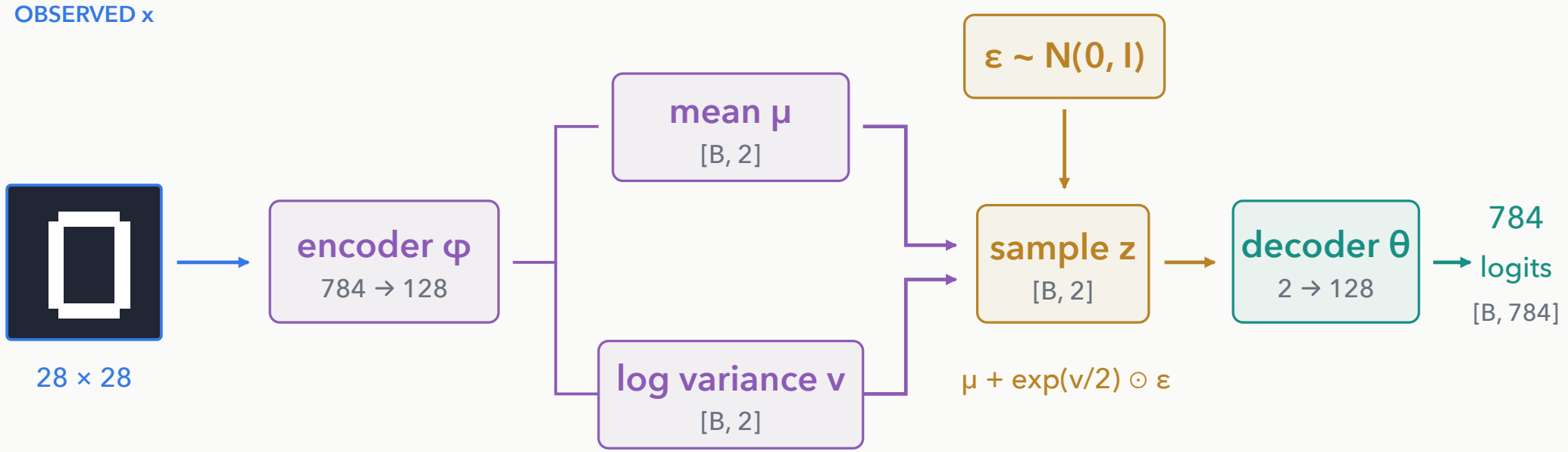
$$\mu, \text{logvar}, z : \ [B, 2]$$

NOTICE Two heads give a mean and a log variance for each image.

SAY Trace one batch through the diagram and check the width at each arrow.

Kingma & Welling · AEVB · authored teaching diagram

How do the VAE tensors fit together?



$$[B, 2] \rightarrow [B, 128] \rightarrow [B, 784]$$

TAKEAWAY The decoder returns logits. Sigmoid turns them into pixel probabilities.

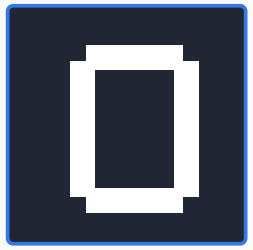
SAY Trace one batch through the diagram and check the width at each arrow.

Kingma & Welling · AEVB · authored teaching diagram

F / VAE

Why is reconstruction a likelihood?

OBSERVATION x



Binary pixels

Each observed pixel contributes one likelihood factor



$x_1 = 1$ $x_2 = 0$ $x_3 = 1$ $x_4 = 0$

$\pi = 0.75$ $\pi = 0.25$ $\pi = 0.7$ $\pi = 0.2$

FOUR-PIXEL ILLUSTRATION · VALUES CHOSEN FOR ARITHMETIC

$$p_\theta(x \mid z) = \prod_i \pi_i^{x_i} (1 - \pi_i)^{1 - x_i}$$

NOTICE Once z is given, the model assigns each binary pixel a Bernoulli probability.

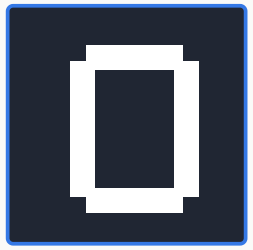
SAY Choose the observation model first. The reconstruction loss follows from that choice.

Kingma & Welling · AEVB · authored teaching diagram

F / VAE

Why is reconstruction a likelihood?

OBSERVATION x



Binary pixels

Each observed pixel contributes one likelihood factor



$x_1 = 1$	$x_2 = 0$	$x_3 = 1$	$x_4 = 0$
$\pi = 0.75$	$\pi = 0.25$	$\pi = 0.7$	$\pi = 0.2$
$-\log 0.75$	$-\log 0.75$	$-\log 0.7$	$-\log 0.8$

FOUR-PIXEL ILLUSTRATION · VALUES CHOSEN FOR ARITHMETIC

$$-\log p_\theta(x \mid z) = -\sum_i [x_i \log \pi_i + (1 - x_i) \log(1 - \pi_i)]$$

NOTICE Negative log likelihood is BCE summed over pixels.

SAY Choose the observation model first. The reconstruction loss follows from that choice.

Kingma & Welling · AEVB · authored teaching diagram

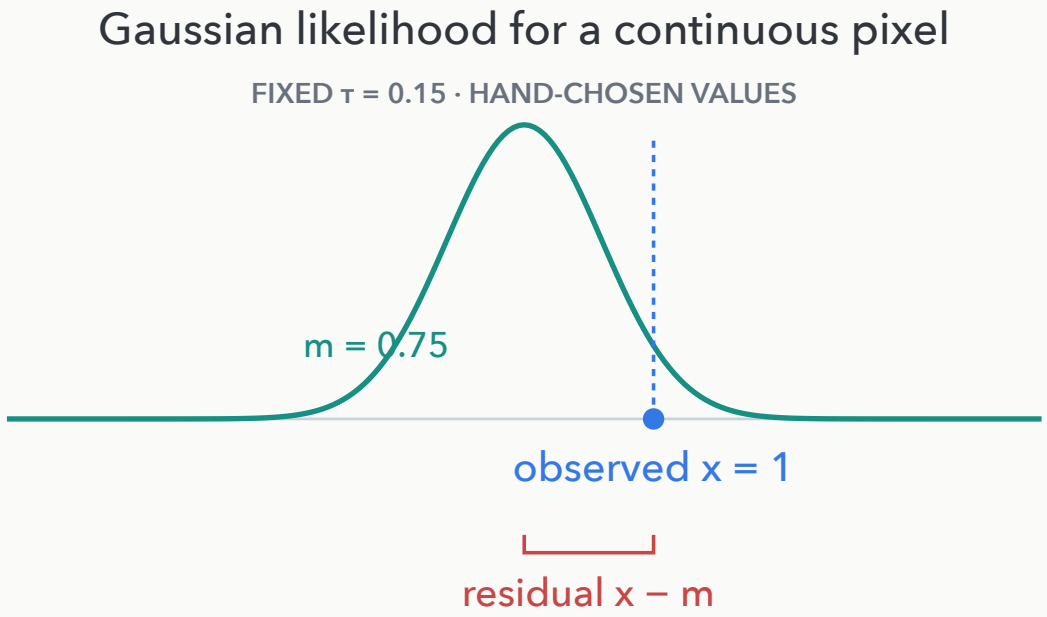
F / VAE

Why is reconstruction a likelihood?

CHANGE THE OBSERVATION MODEL

decoder mean m

illustration: m = 0.75



$$-\log p(x \mid z) = \frac{\|x - m_\theta(z)\|^2}{2\tau^2} + C$$

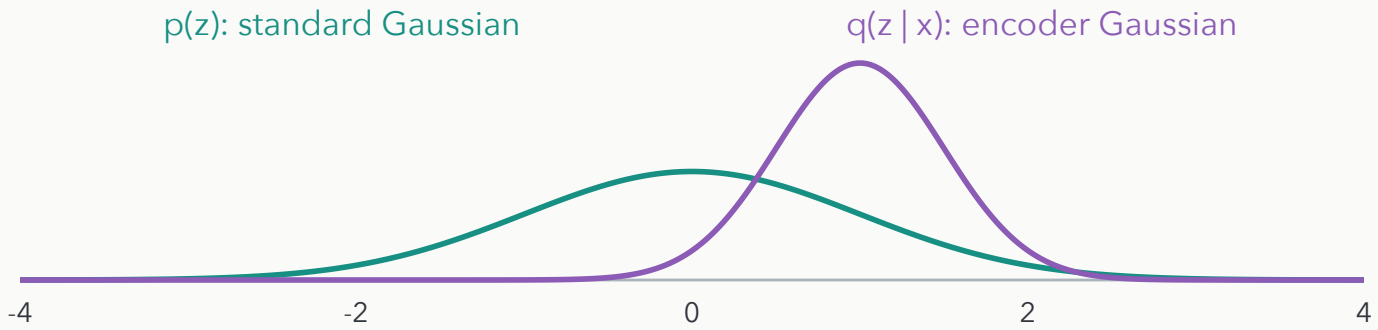
TAKEAWAY Fixed-variance Gaussian observations instead give squared error plus constants.

SAY Choose the observation model first. The reconstruction loss follows from that choice.

Kingma & Welling · AEVB · authored teaching diagram

F / VAE

Can we compute Gaussian KL exactly?



Prior: N(0,1) · approximate posterior: N(μ, σ²)

μ	σ	KL(q ‖ prior)
1.00	0.50	0.818147

$$\frac{1}{2}(\mu^2 + \sigma^2 - 1 - \log \sigma^2)$$

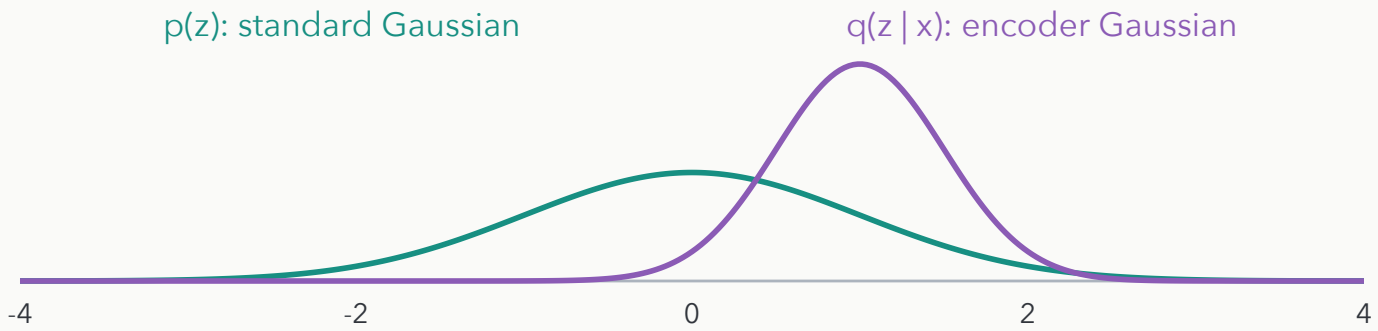
NOTICE For one Gaussian coordinate, the KL is analytic.

SAY Try mean zero and variance one. The posterior then matches the standard normal prior.

Kingma & Welling · AEVB · authored teaching diagram

F / VAE

Can we compute Gaussian KL exactly?



Prior: N(0,1) · approximate posterior: N(μ, σ²)

μ	σ	KL(q ‖ prior)
1.00	0.50	0.818147

$$\frac{1}{2}\sum_j(\mu_j^2+\sigma_j^2-1-\log\sigma_j^2)$$

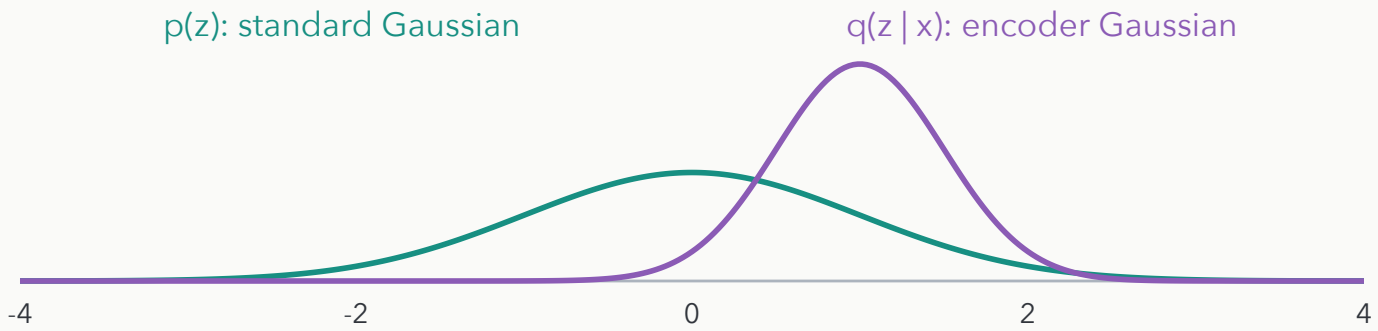
NOTICE Add the KL contributions from the independent latent coordinates.

SAY Try mean zero and variance one. The posterior then matches the standard normal prior.

Kingma & Welling · AEVB · authored teaching diagram

F / VAE

Can we compute Gaussian KL exactly?



Prior: N(0,1) · approximate posterior: N(μ, σ²)

μ	σ	KL(q ‖ prior)
1.00	0.50	0.818147

$$\frac{1}{2}\sum_j (\mu_j^2 + e^{v_j} - 1 - v_j)$$

TAKEAWAY At mean zero and variance one, each contribution is zero.

SAY Try mean zero and variance one. The posterior then matches the standard normal prior.

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F / VAE

What is a complete two-pixel VAE loss?

$x = [1, 0]$

$\mu = [1, -1]$

$\sigma = [0.5, 1]$

$\varepsilon = [0.2, -0.4]$

$z = [1.1, -1.4]$

Reconstruction NLL	0.575364145
KL coordinate 1	0.818147181
KL coordinate 2	0.500000000
Total Gaussian KL	1.318147181
One-sample −ELBO	1.893511325

Specified two-pixel decoder output $\pi = [0.75, 0.25]$. This is not the 784-pixel model.

NOTICE A temporary two-pixel example: sample the latent first.

SAY Keep the two calculations separate: sampled reconstruction cost, then exact latent KL.

Kingma & Welling · AEVB · authored teaching diagram

F / VAE

What is a complete two-pixel VAE loss?

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$\sigma = [0.5, 1]$

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KL coordinate 1	0.818147181
KL coordinate 2	0.500000000
Total Gaussian KL	1.318147181
One-sample −ELBO	1.893511325

Specified two-pixel decoder output $\pi = [0.75, 0.25]$. This is not the 784-pixel model.

NOTICE At this z, the specified decoder probabilities are (0.75, 0.25).

SAY Keep the two calculations separate: sampled reconstruction cost, then exact latent KL.

Kingma & Welling · AEVB · authored teaching diagram

F / VAE

What is a complete two-pixel VAE loss?

$x = [1, 0]$

$\mu = [1, -1]$

$\sigma = [0.5, 1]$

$\varepsilon = [0.2, -0.4]$

$z = [1.1, -1.4]$

Reconstruction NLL	0.575364145
KL coordinate 1	0.818147181
KL coordinate 2	0.500000000
Total Gaussian KL	1.318147181
One-sample −ELBO	1.893511325

Specified two-pixel decoder output $\pi = [0.75, 0.25]$. This is not the 784-pixel model.

> TAKEAWAY Add the exact Gaussian KL to this draw's reconstruction cost.
>
> SAY Keep the two calculations separate: sampled reconstruction cost, then exact latent KL.

Kingma & Welling · AEVB · authored teaching diagram

F / VAE

What happens in one VAE update?

```
logits, mu, logvar = model(x)
# encoder → sample → decoder
```

Update both φ and θ

For each example:
sum pixels → add latent KL

Across the batch:
mean → backpropagate → step

BCE_logits = binary_cross_entropy_with_logits · Complete runnable models

NOTICE Encode, sample, decode. The graph joins both networks.

SAY Build one loss per example before taking the batch mean.

Kingma & Welling · AEVB · authored teaching diagram

F / VAE

What happens in one VAE update?

```
logits, mu, logvar = model(x)
# encoder → sample → decoder
recon = BCE_logits(logits, x.flatten(1),
                   reduction="none").sum(1)
kl = 0.5 * (mu.square() + logvar.exp()
            - 1 - logvar).sum(1)
```

Update both φ and θ

For each example:
sum pixels → add latent KL

Across the batch:
mean → backpropagate → step

BCE_logits = binary_cross_entropy_with_logits · Complete runnable models

NOTICE For each image, sum over pixels and over latent coordinates separately.

SAY Build one loss per example before taking the batch mean.

Kingma & Welling · AEVB · authored teaching diagram

F / VAE

What happens in one VAE update?

```
logits, mu, logvar = model(x)
# encoder → sample → decoder
recon = BCE_logits(logits, x.flatten(1),
                   reduction="none").sum(1)
kl = 0.5 * (mu.square() + logvar.exp()
            - 1 - logvar).sum(1)
optimizer.zero_grad(set_to_none=True)
loss = (recon + kl).mean()
loss.backward()
optimizer.step()
```

Update both φ and θ

For each example:
sum pixels → add latent KL

Across the batch:
mean → backpropagate → step

BCE_logits = binary_cross_entropy_with_logits · Complete runnable models

TAKEAWAY Average per-example totals, then update encoder and decoder.

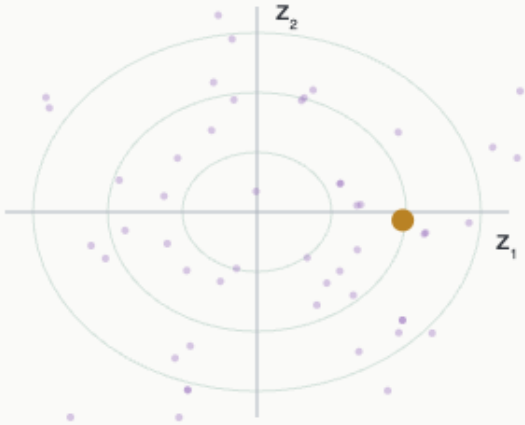
SAY Build one loss per example before taking the batch mean.

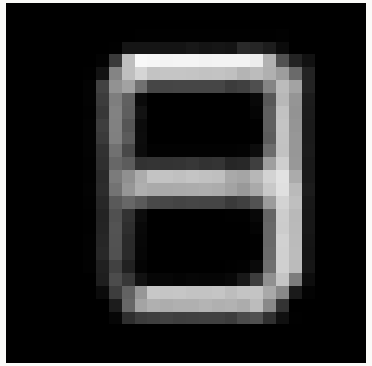
Kingma & Welling · AEVB · authored teaching diagram

F / VAE

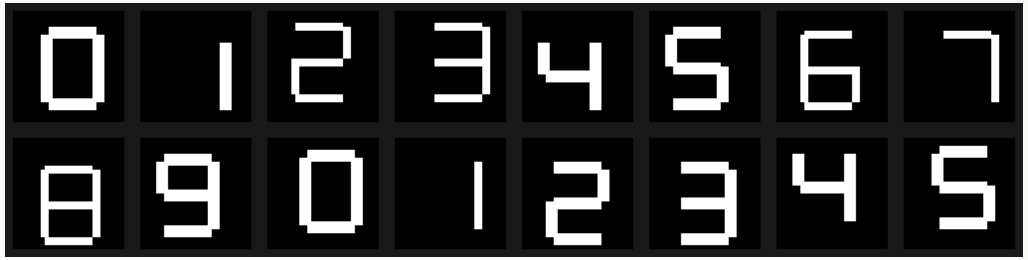
What is needed to generate a new image?

Observed x → encoder mean → decoder





Recorded training run · procedural binary glyphs



Held-out observations

$$z \sim q_\phi(z \mid x)$$

NOTICE Reconstruction starts with an observed image and its encoded posterior.

SAY Cover the encoder in the diagram. We can still generate from the prior.

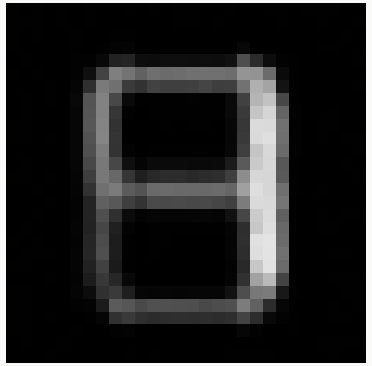
Kingma & Welling · AEVB · authored teaching diagram

F / VAE

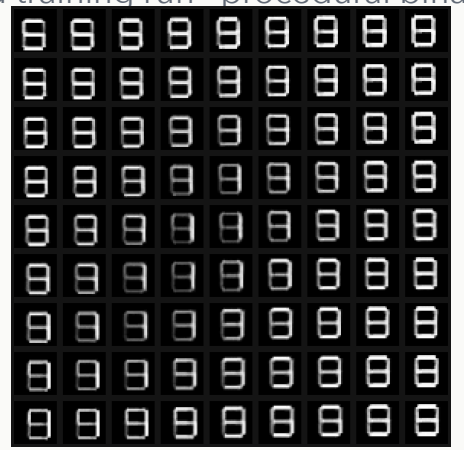
What is needed to generate a new image?

Prior draw → decoder only





Recorded training run · procedural binary glyphs



Uniform latent grid, −3 to 3

$$z \sim p(z), \quad x \sim p_\theta(x \mid z)$$

TAKEAWAY Fresh generation starts at the prior and skips the encoder.

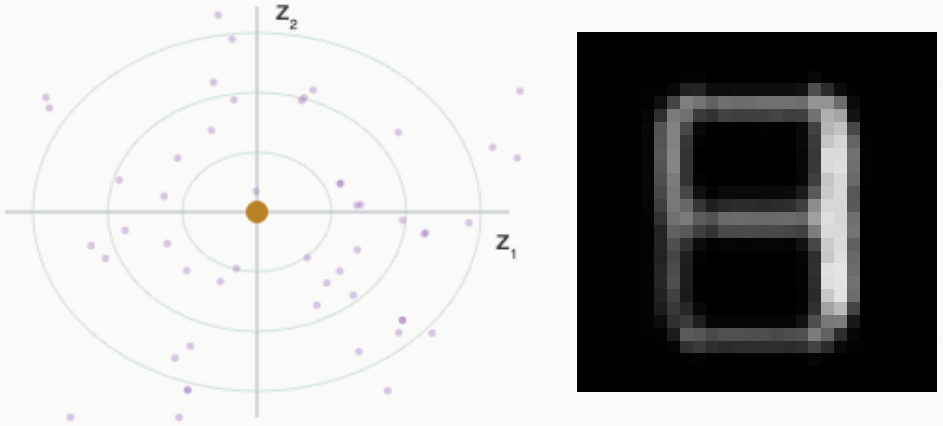
SAY Cover the encoder in the diagram. We can still generate from the prior.

Kingma & Welling · AEVB · authored teaching diagram

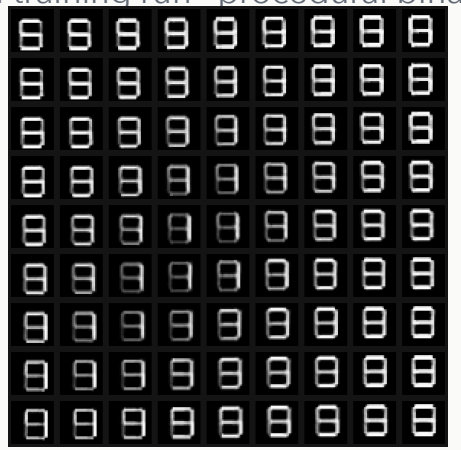
F / VAE

What does moving in latent space do?

Select z → fixed trained decoder



Recorded training run · procedural binary glyphs



Uniform latent grid, −3 to 3

NOTICE Move through z while keeping the trained decoder fixed.

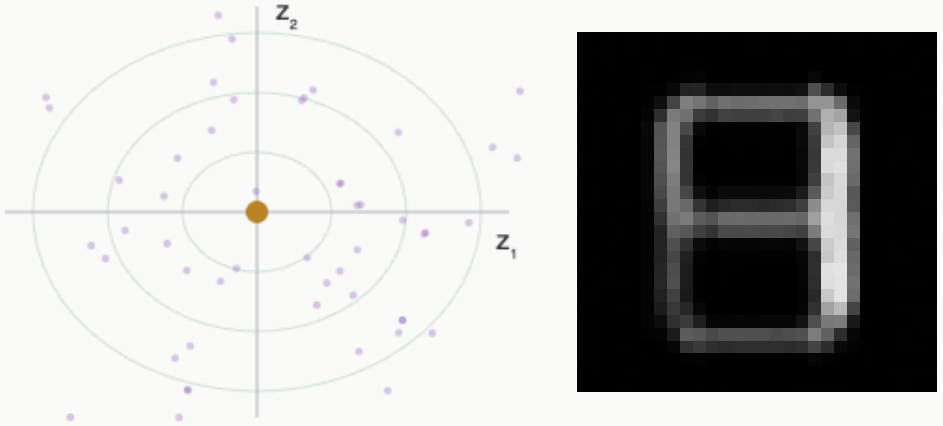
SAY The decoder stays fixed. Only the two latent coordinates move.

Kingma & Welling · AEVB · authored teaching diagram

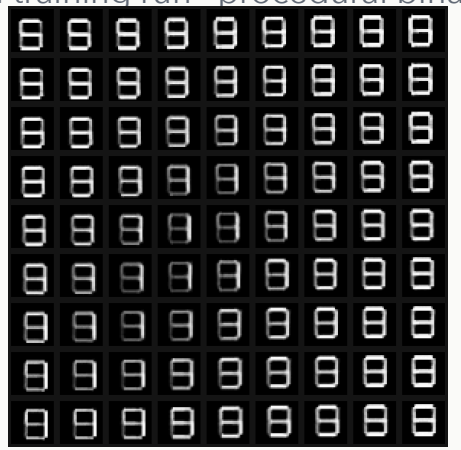
F / VAE

What does moving in latent space do?

Select z → fixed trained decoder



Recorded training run · procedural binary glyphs



Uniform latent grid, −3 to 3

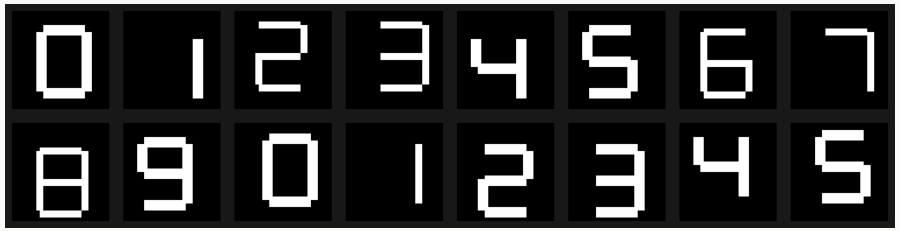
TAKEAWAY A regular grid helps us inspect the decoder. It is not a batch of prior draws.

SAY The decoder stays fixed. Only the two latent coordinates move.

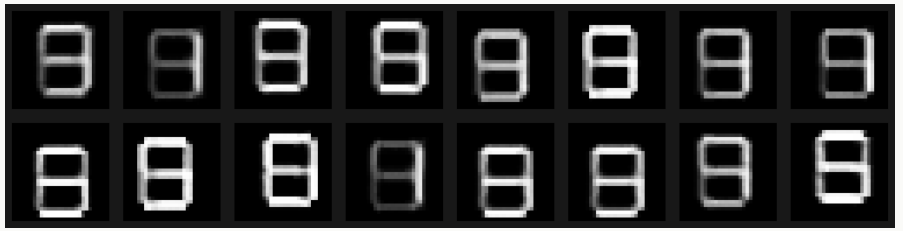
Kingma & Welling · AEVB · authored teaching diagram

F / VAE

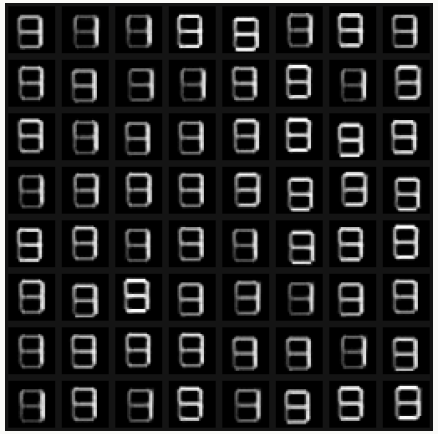
What does the VAE objective guarantee?



Held-out binary glyphs



Posterior-mean reconstructions



Prior probability images

One actual β = 1 run. Failure modes discussed here are possibilities, not fabricated retrained comparisons.

$$\mathcal{L} = \text{reconstruction NLL} + \text{KL}$$

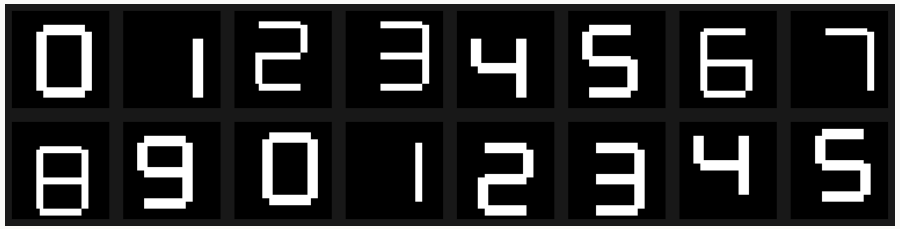
NOTICE The objective balances explaining x and keeping q near the prior.

SAY Look at what this run actually learned. The objective does not promise sharp or meaningful coordinates.

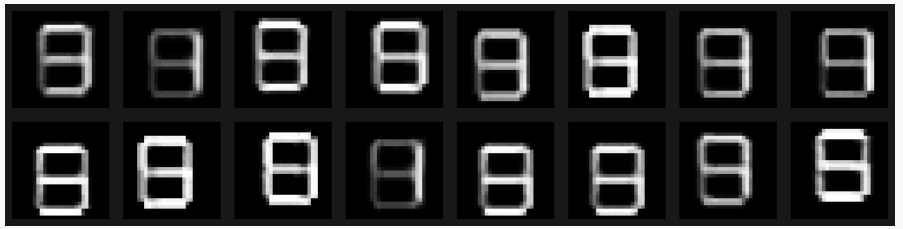
Kingma & Welling · AEVB · authored teaching diagram

F / VAE

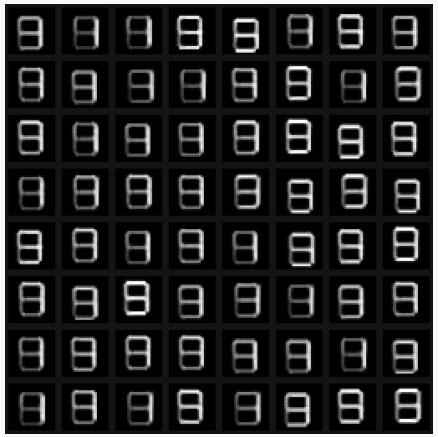
What does the VAE objective guarantee?



Held-out binary glyphs



Posterior-mean reconstructions



Prior probability images

One actual β = 1 run. Failure modes discussed here are possibilities, not fabricated retrained comparisons.

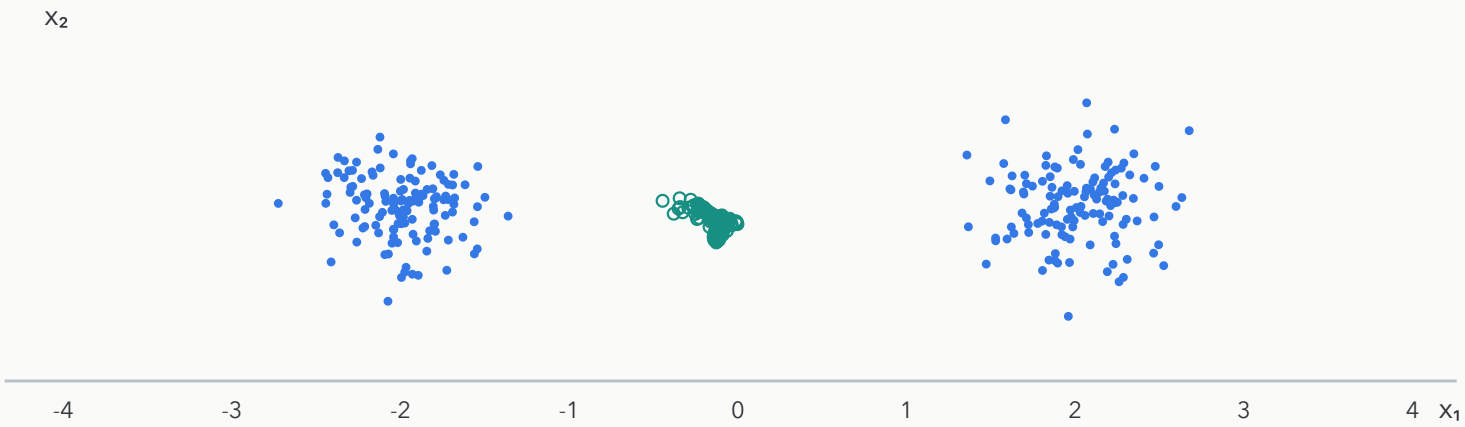
$$\beta = 1 : \ \text{vanilla negative ELBO}$$

> **TAKEAWAY** The objective does not guarantee sharp samples or interpretable latent coordinates.
>
> **SAY** Look at what this run actually learned. The objective does not promise sharp or meaningful coordinates.

Kingma & Welling · AEVB · authored teaching diagram

G / GAN

Can we learn the training signal?



within 0.65 of a true centre

0.0%

modes with ≥5 of 256 samples

0/2

Blue: real data · teal outlines: current G samples · recorded training run

$$x_{\text{fake}} = G_\theta(z)$$

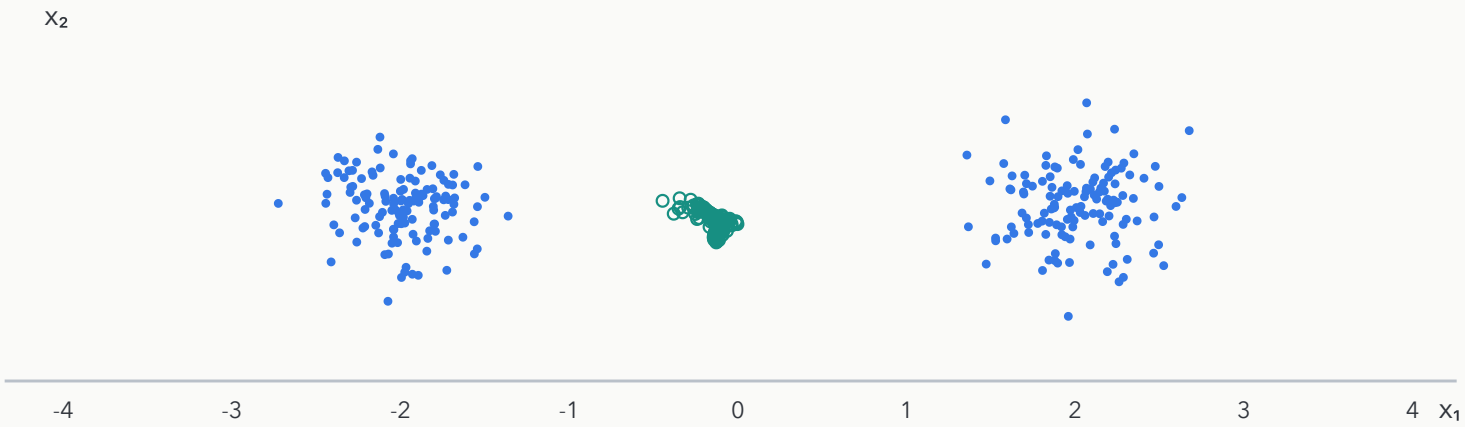
NOTICE A generator makes fake samples. Compare them with real data.

SAY This time we learn a critic of the samples as well as the generator.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Can we learn the training signal?



within 0.65 of a true centre

0.0%

modes with ≥5 of 256 samples

0/2

Blue: real data · teal outlines: current G samples · recorded training run

$$D_\psi(x) = \text{sigmoid}(a_\psi(x))$$

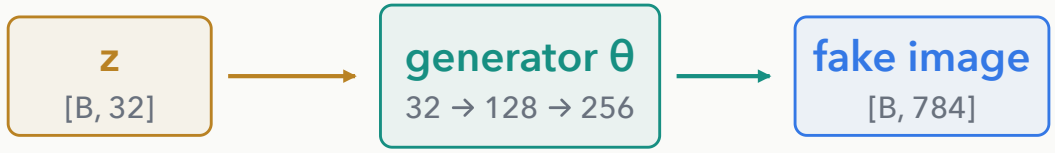
TAKEAWAY Train a classifier to recognise the source.

SAY This time we learn a critic of the samples as well as the generator.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

How do the two networks connect?



$$z : [B, 32] \rightarrow G_\theta \rightarrow [B, 784]$$

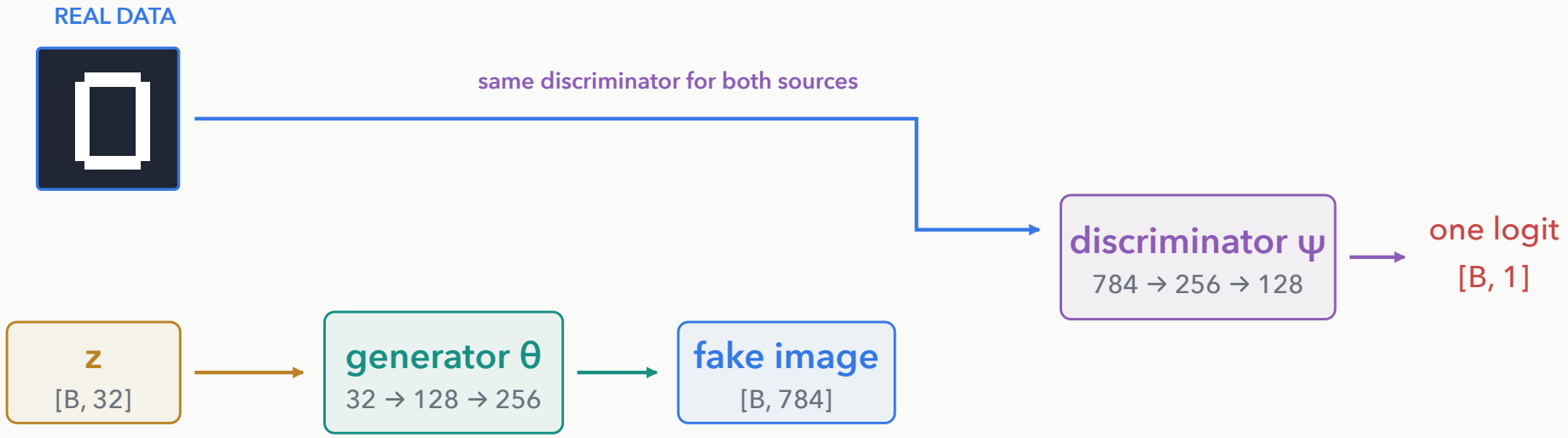
NOTICE The generator maps noise into the data space.

SAY Both paths end at the same discriminator with the same parameters.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

How do the two networks connect?



$$x : [B, 784] \rightarrow D_\psi \rightarrow [B, 1]$$

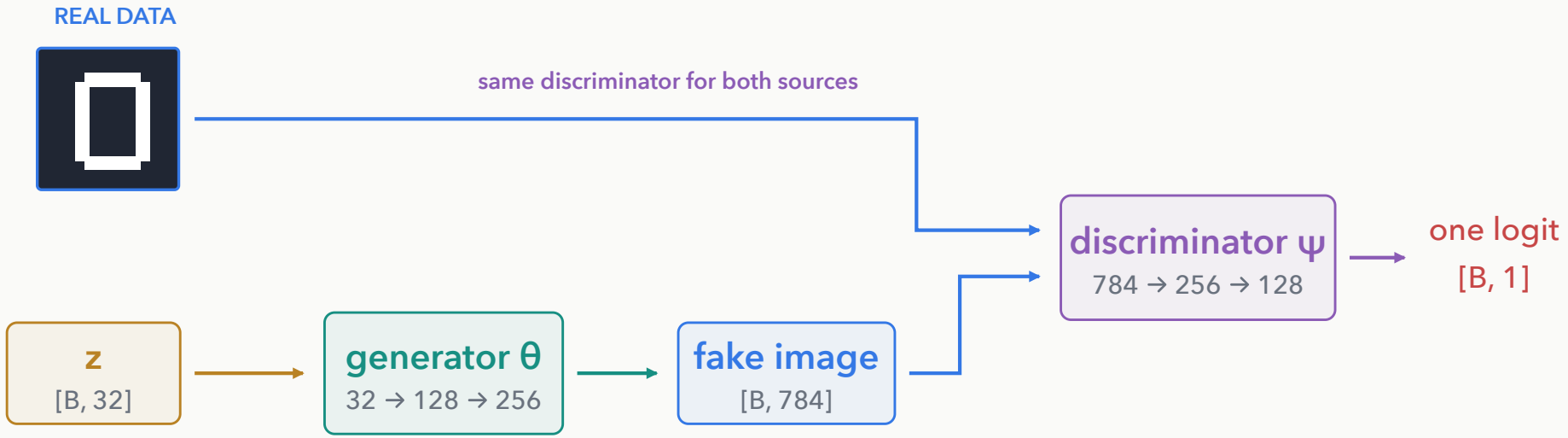
NOTICE Real examples enter a discriminator that outputs logits.

SAY Both paths end at the same discriminator with the same parameters.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

How do the two networks connect?



Two forward paths meet at one set of discriminator weights.

$$G_\theta(z) \to D_\psi \to [B, 1]$$

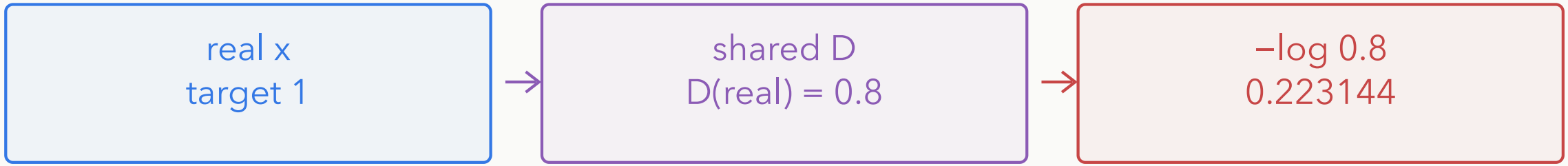
TAKEAWAY Both real and generated images go through the same discriminator.

SAY Both paths end at the same discriminator with the same parameters.

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G / GAN

Which labels train the discriminator?



fake target 0; D(fake)=0.3

`-log(0.7)`

L_D (sum of means)

`0.579818495`

L_G (non-saturating)

`1.203972804`

$$-\mathbb{E}_{\text{real}} \log D(x)$$

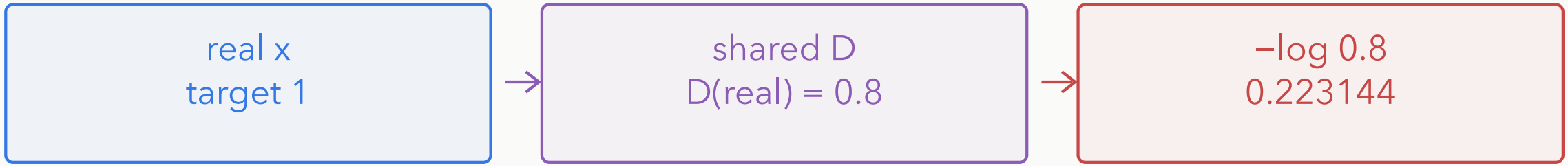
NOTICE Real examples have target 1.

SAY In the discriminator's task, real means one and generated means zero.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Which labels train the discriminator?



fake target 0; D(fake)=0.3
`-log(0.7)`

L_D (sum of means)
`0.579818495`

L_G (non-saturating)
`1.203972804`

$$-\mathbb{E}_z \log(1 - D(G(z)))$$

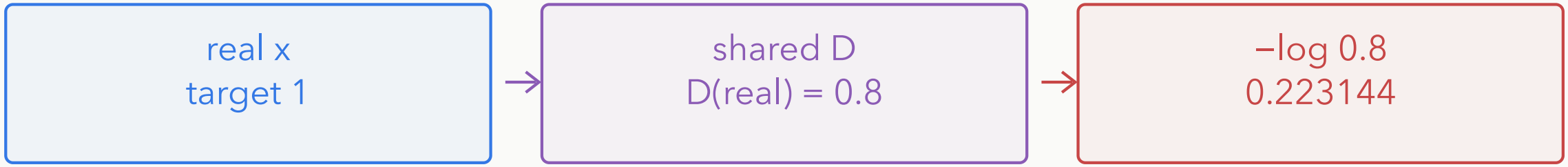
NOTICE Generated examples have target 0.

SAY In the discriminator's task, real means one and generated means zero.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Which labels train the discriminator?



fake target 0; D(fake)=0.3
`-log(0.7)`

L_D (sum of means)
`0.579818495`

L_G (non-saturating)
`1.203972804`

$$\mathcal{L}_D = -\mathbb{E}\log D(x) - \mathbb{E}\log(1 - D(G(z)))$$

TAKEAWAY The discriminator loss adds the two batch means.

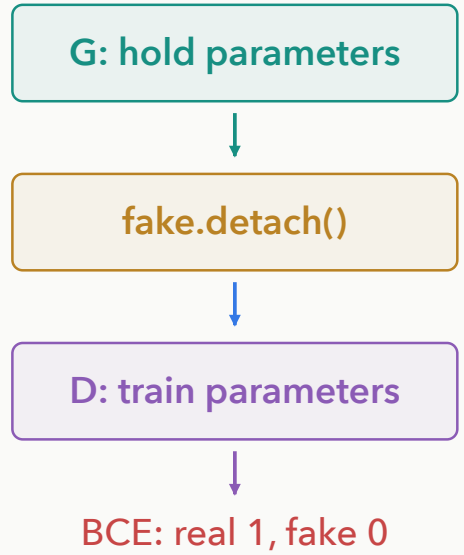
SAY In the discriminator's task, real means one and generated means zero.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Where should gradients stop for D?

```
opt_D.zero_grad(set_to_none=True)
opt_G.zero_grad(set_to_none=True)
fake = G(torch.randn(B, 32))
```



BCE_logits = binary_cross_entropy_with_logits · Complete runnable models

NOTICE Clear gradients and generate a fake batch.

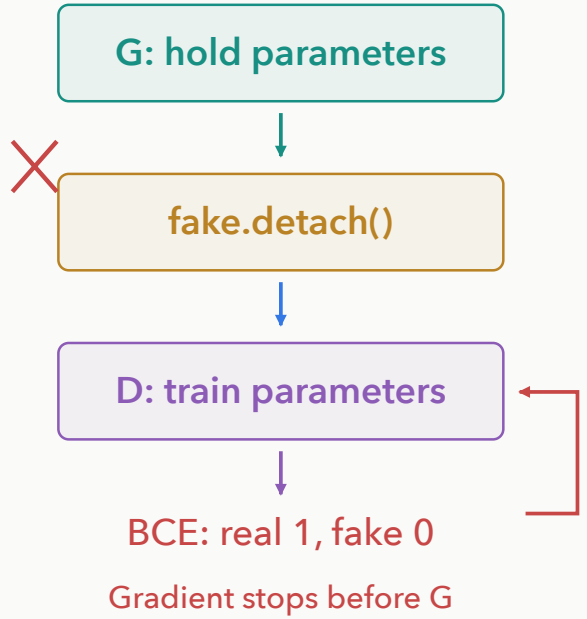
SAY Detaching keeps the fake values but cuts the path back to the generator.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Where should gradients stop for D?

```
opt_D.zero_grad(set_to_none=True)
opt_G.zero_grad(set_to_none=True)
fake = G(torch.randn(B, 32))
real_logits = D(real)
fake_logits = D(fake.detach())
loss_D = BCE_logits(real_logits, ones)
loss_D += BCE_logits(fake_logits, zeros)
```



BCE_logits = binary_cross_entropy_with_logits · Complete runnable models

NOTICE Detach the fake batch: keep its values, cut its connection to G.

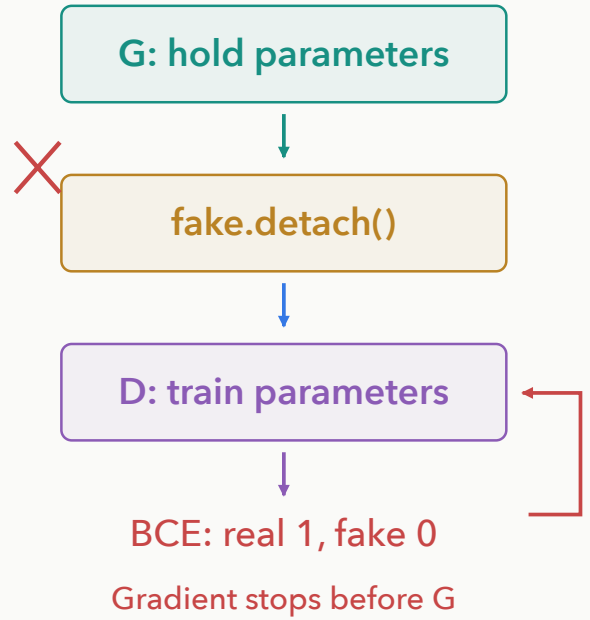
SAY Detaching keeps the fake values but cuts the path back to the generator.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Where should gradients stop for D?

```
opt_D.zero_grad(set_to_none=True)
opt_G.zero_grad(set_to_none=True)
fake = G(torch.randn(B, 32))
real_logits = D(real)
fake_logits = D(fake.detach())
loss_D = BCE_logits(real_logits, ones)
loss_D += BCE_logits(fake_logits, zeros)
loss_D.backward()
opt_D.step()
# G has no new gradients
```



BCE_logits = binary_cross_entropy_with_logits · Complete runnable models

TAKEAWAY Step D's optimiser. G receives no new gradient in this phase.

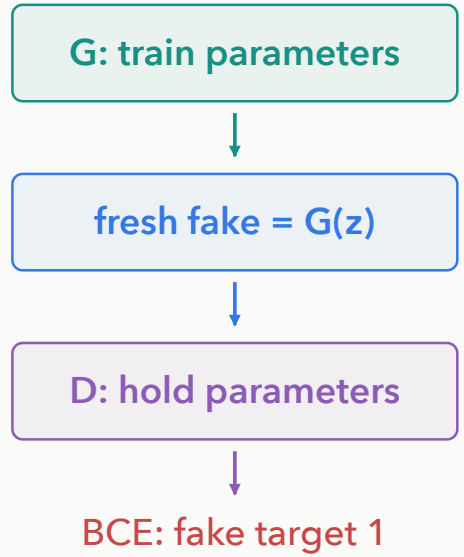
SAY Detaching keeps the fake values but cuts the path back to the generator.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

How can frozen D train G?

```
D.requires_grad_(False)
opt_G.zero_grad(set_to_none=True)
fake = G(torch.randn(B, 32))
```



BCE_logits = binary_cross_entropy_with_logits · Complete runnable models

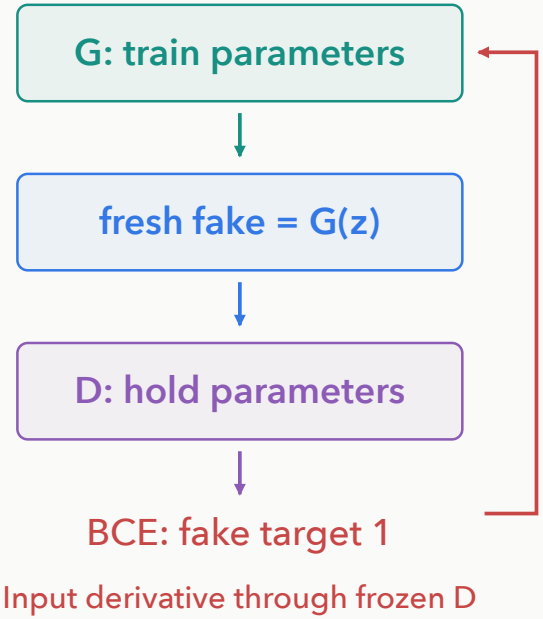
NOTICE Use a fresh forward pass through the updated discriminator.

SAY Frozen parameters do not mean a frozen function. We still differentiate D with respect to its input.

G / GAN

How can frozen D train G?

```
D.requires_grad_(False)
opt_G.zero_grad(set_to_none=True)
fake = G(torch.randn(B, 32))
fake_logits = D(fake)
loss_G = BCE_logits(fake_logits, ones)
# no detach; no no_grad around D
```



BCE_logits = binary_cross_entropy_with_logits · Complete runnable models

NOTICE Freeze D's parameters while keeping the derivative with respect to its input.

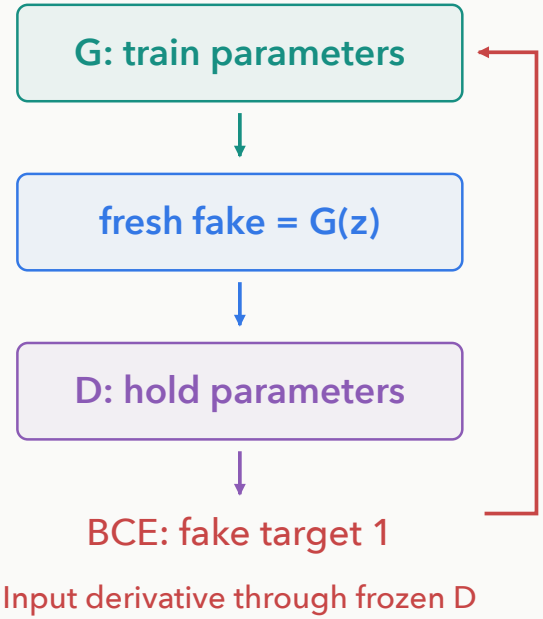
SAY Frozen parameters do not mean a frozen function. We still differentiate D with respect to its input.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

How can frozen D train G?

```
D.requires_grad_(False)
opt_G.zero_grad(set_to_none=True)
fake = G(torch.randn(B, 32))
fake_logits = D(fake)
loss_G = BCE_logits(fake_logits, ones)
# no detach; no no_grad around D
loss_G.backward()
opt_G.step()
D.requires_grad_(True)
```



BCE_logits = binary_cross_entropy_with_logits · Complete runnable models

TAKEAWAY Use target 1 to update G. D's parameter values stay fixed.

SAY Frozen parameters do not mean a frozen function. We still differentiate D with respect to its input.

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G / GAN

Which generator objective do we use?



$$\log(1 - D(G(z))) \quad \neq \quad -\log D(G(z))$$

$$\min_G \max_D \; \mathbb{E} \log D(x) + \mathbb{E} \log(1 - D(G(z)))$$

NOTICE The original game asks G to minimise what D maximises.

SAY The minimax game explains the idea. Our code uses the non-saturating generator loss.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Which generator objective do we use?



$$\log(1 - D(G(z))) \quad \neq \quad -\log D(G(z))$$

$$\mathcal{L}_G^{\text{NS}} = -\mathbb{E} \log D(G(z))$$

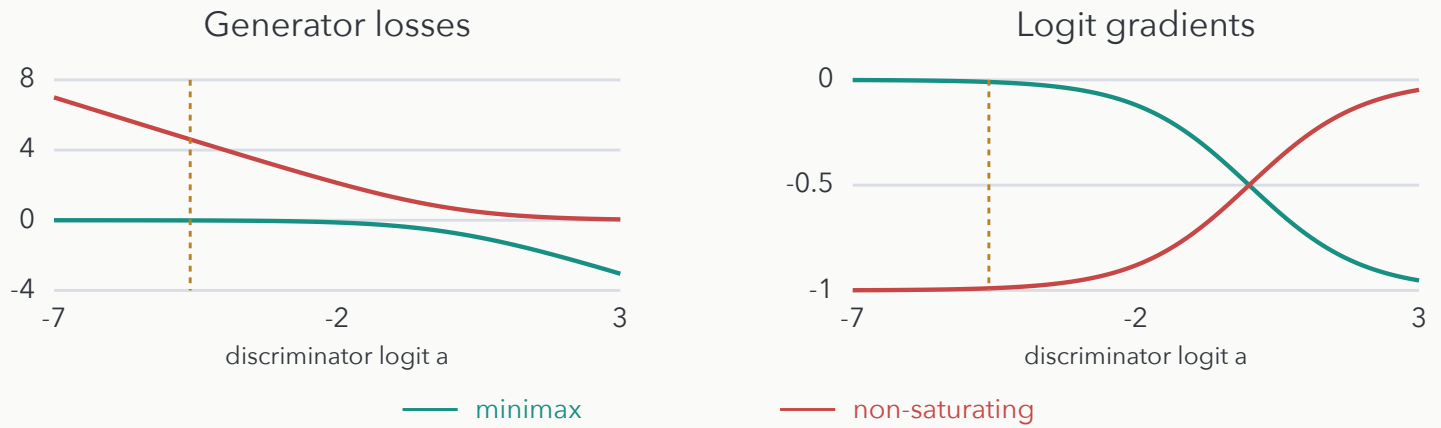
TAKEAWAY Our code uses the non-saturating generator loss, which gives different gradients.

SAY The minimax game explains the idea. Our code uses the non-saturating generator loss.

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G / GAN

Why use the non-saturating objective?



D = sigmoid(a)	minimax loss	non-saturating loss	minimax derivative	non-saturating derivative
0.010	−0.010	4.605	−0.010	−0.990

$$\frac{d \log(1 - \text{sigmoid}(a))}{da} = -D$$

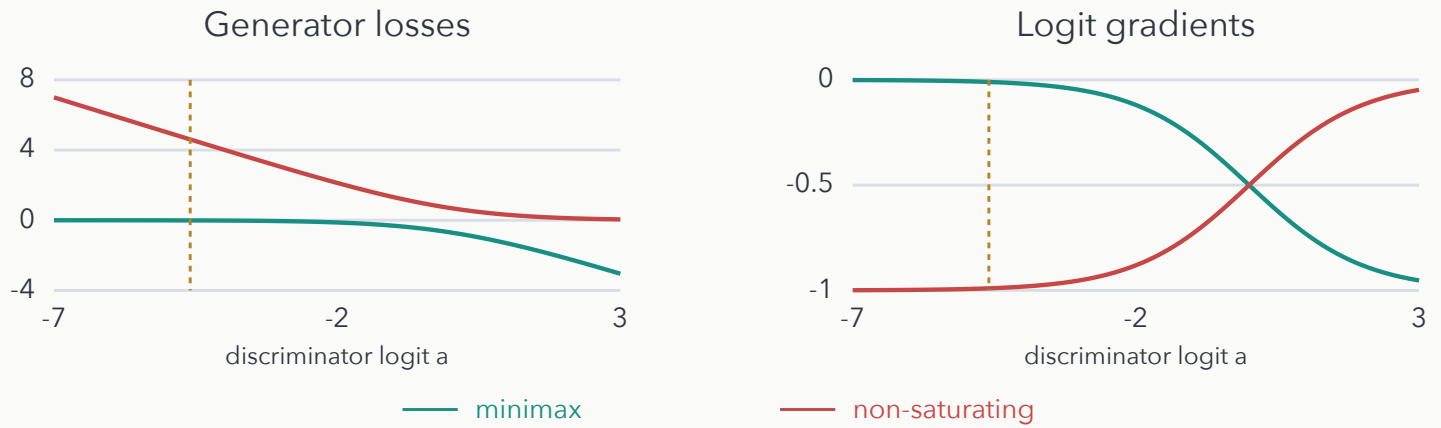
NOTICE When D confidently rejects a fake, the minimax logit gradient becomes small.

SAY At D equal to 0.01, compare the two slopes before following the rest of the graph.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Why use the non-saturating objective?



D = sigmoid(a)	minimax loss	non-saturating loss	minimax derivative	non-saturating derivative
0.010	−0.010	4.605	−0.010	−0.990

$$\frac{d[-\log \text{sigmoid}(a)]}{da} = D - 1$$

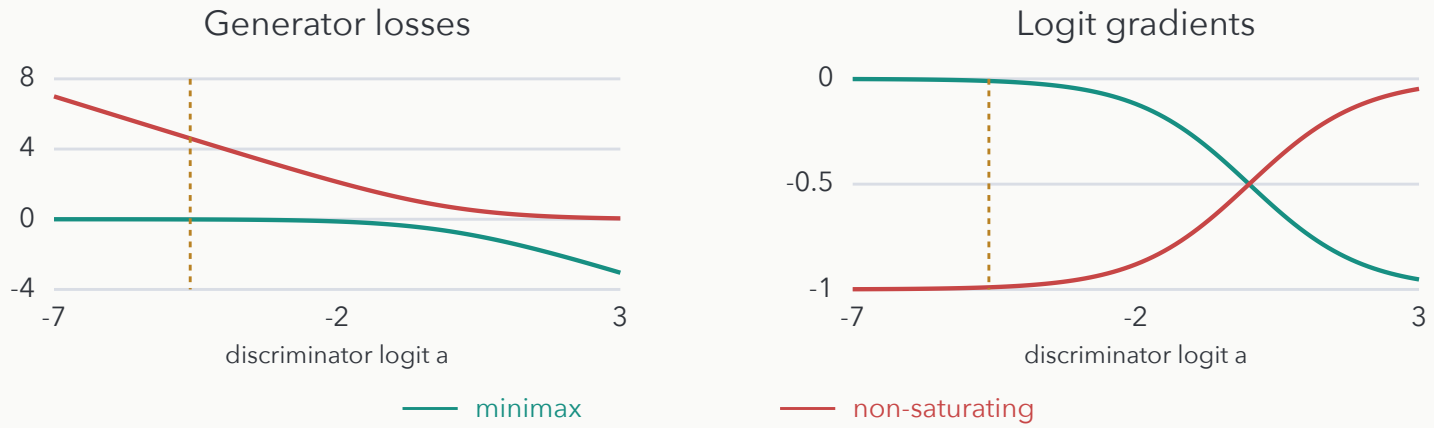
NOTICE At small D, the non-saturating loss still has a strong logit gradient.

SAY At D equal to 0.01, compare the two slopes before following the rest of the graph.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Why use the non-saturating objective?



D = sigmoid(a)	minimax loss	non-saturating loss	minimax derivative	non-saturating derivative
0.010	−0.010	4.605	−0.010	−0.990

$$\frac{d\mathcal{L}}{d\theta} = \frac{d\mathcal{L}}{da}\frac{da}{dx_{\text{fake}}}\frac{dx_{\text{fake}}}{d\theta}$$

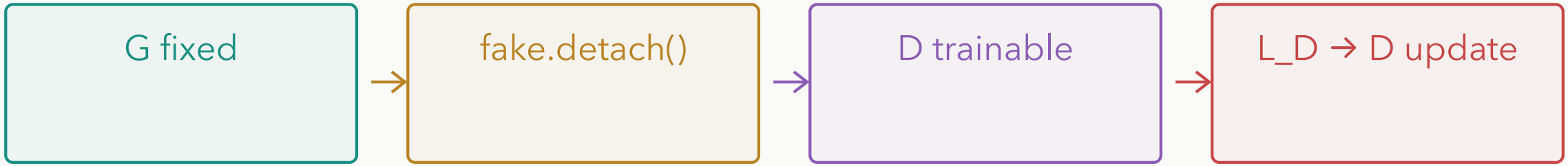
TAKEAWAY At D = 0.01, compare −0.01 with −0.99. Continue the chain rule through G.

SAY At D equal to 0.01, compare the two slopes before following the rest of the graph.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

What changes in one alternating iteration?



operation	max |ΔG|	max |ΔD|	G gradient norm
D · forward	0.00e+0	0.00e+0	0.00000

The detach boundary stops gradients entering G.

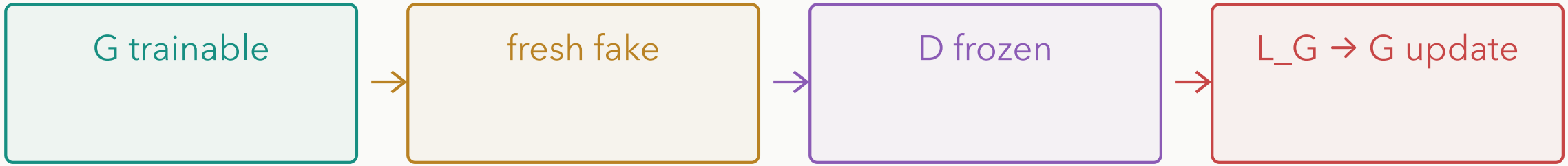
NOTICE D phase: detached fake, targets 1 and 0, update D.

SAY After each phase, check which parameter values changed and which stayed fixed.

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G / GAN

What changes in one alternating iteration?



operation	max |ΔG|	max |ΔD|	G gradient norm
`G · forward`	`0.00e+0`	`0.00e+0`	`0.00000`

D parameters stay fixed; its input derivative remains part of G's graph.

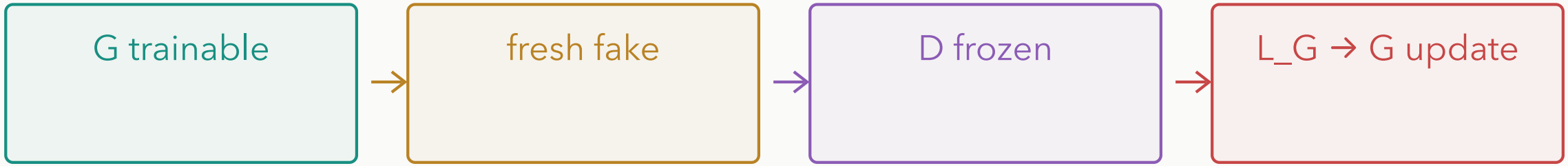
NOTICE G phase: fresh graph, D frozen, target 1, update G.

SAY After each phase, check which parameter values changed and which stayed fixed.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

What changes in one alternating iteration?



operation	max |ΔG|	max |ΔD|	G gradient norm
G · update	1.00e−3	0.00e+0	0.03680

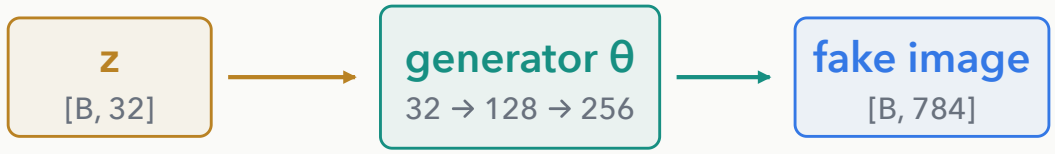
D parameters stay fixed; its input derivative remains part of G's graph.

> **TAKEAWAY** After G's update, enable D's parameter gradients for the next iteration.
>
> **SAY** After each phase, check which parameter values changed and which stayed fixed.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Where does each GAN tensor go?



$$[64, 32] \rightarrow [64, 128] \rightarrow [64, 256] \rightarrow [64, 784]$$

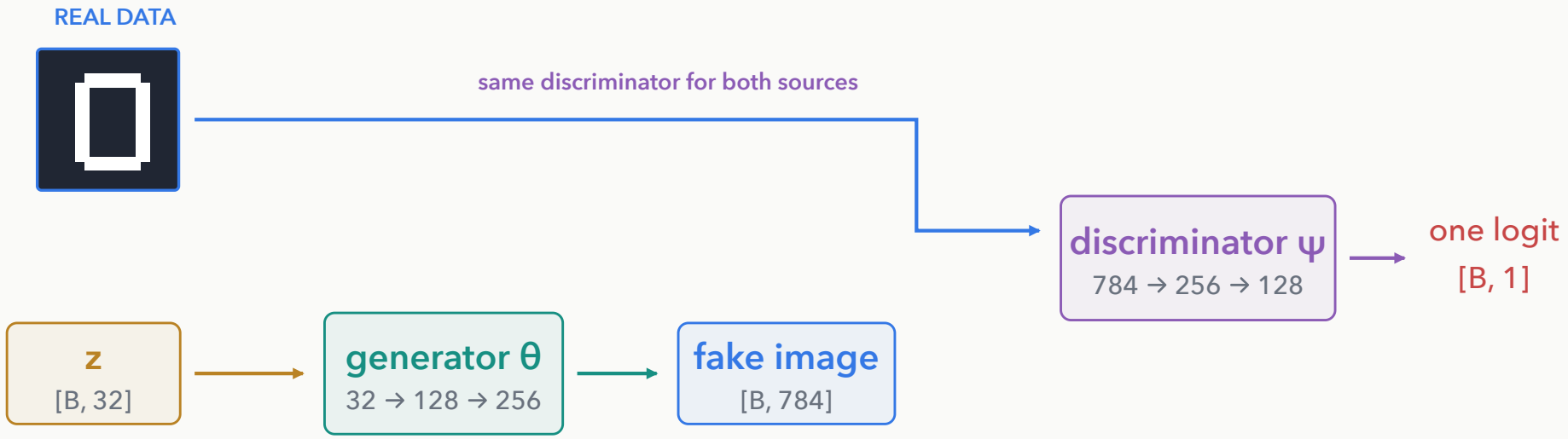
NOTICE Noise dimension, batch size and pixel dimension have different roles.

SAY The first dimension counts examples. It should survive every layer unchanged.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Where does each GAN tensor go?



$$[64, 784] \rightarrow [64, 256] \rightarrow [64, 128] \rightarrow [64, 1]$$

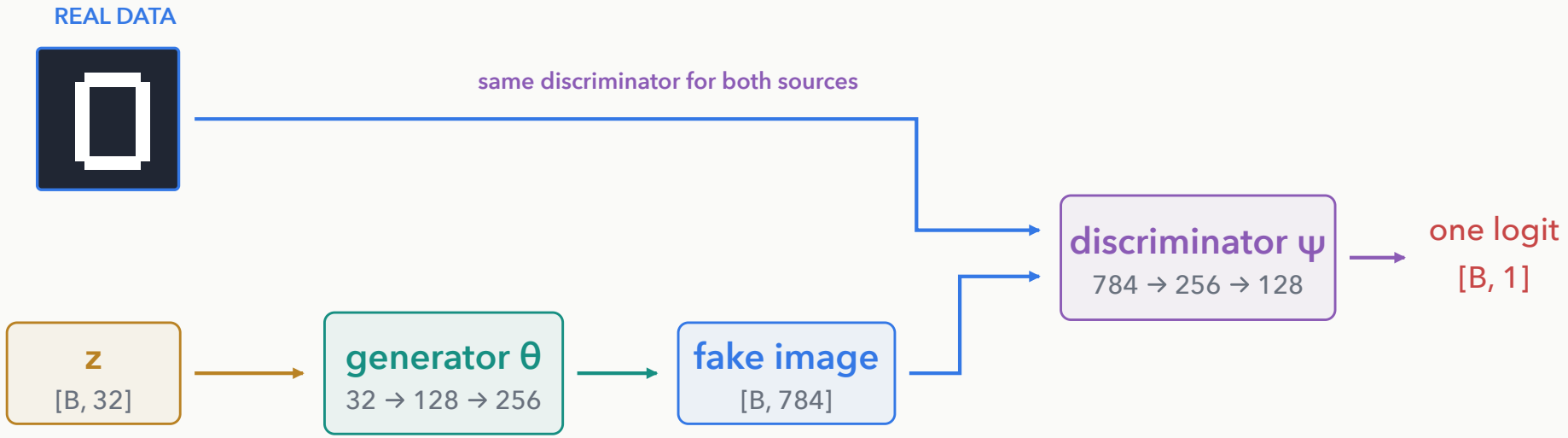
NOTICE Reshape for viewing. D still receives flattened pixels.

SAY The first dimension counts examples. It should survive every layer unchanged.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Where does each GAN tensor go?



Two forward paths meet at one set of discriminator weights.

$$\text{shape}(y) = \text{shape}(a) = [64, 1]$$

TAKEAWAY Targets must match logits exactly.

SAY The first dimension counts examples. It should survive every layer unchanged.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Which network remains at generation?



Observed real x [B,784] also enters the same D ψ → real logits [B,1].

observed data · random draw · generator / latent · encoder · loss / gradient

NOTICE Training needs real examples and the discriminator.

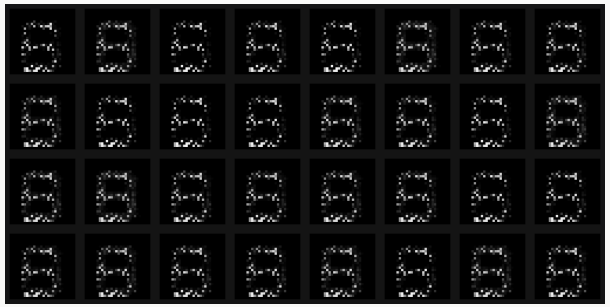
SAY Once training is over, keep the generator and feed it fresh noise.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Which network remains at generation?





Local glyph GAN · fixed noise · short run



Return to published StyleGAN · different architecture

$$z \sim p(z), \quad x = G_\theta(z)$$

> **TAKEAWAY** To sample after training, we only need noise and G.
>
> **SAY** Once training is over, keep the generator and feed it fresh noise.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Can realistic samples miss a mode?



within 0.65 of a true centre

97.5%

modes with ≥5 of 160 samples

1/2

Deterministic toy geometry, centres (−2,0), (2,0); these are not image-quality metrics.

NOTICE Plausible samples can all come from one mode.

SAY Every point may look plausible while an entire cluster is missing.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Can realistic samples miss a mode?



within 0.65 of a true centre

97.5%

modes with ≥5 of 160 samples

1/2

Deterministic toy geometry, centres (−2,0), (2,0); these are not image-quality metrics.

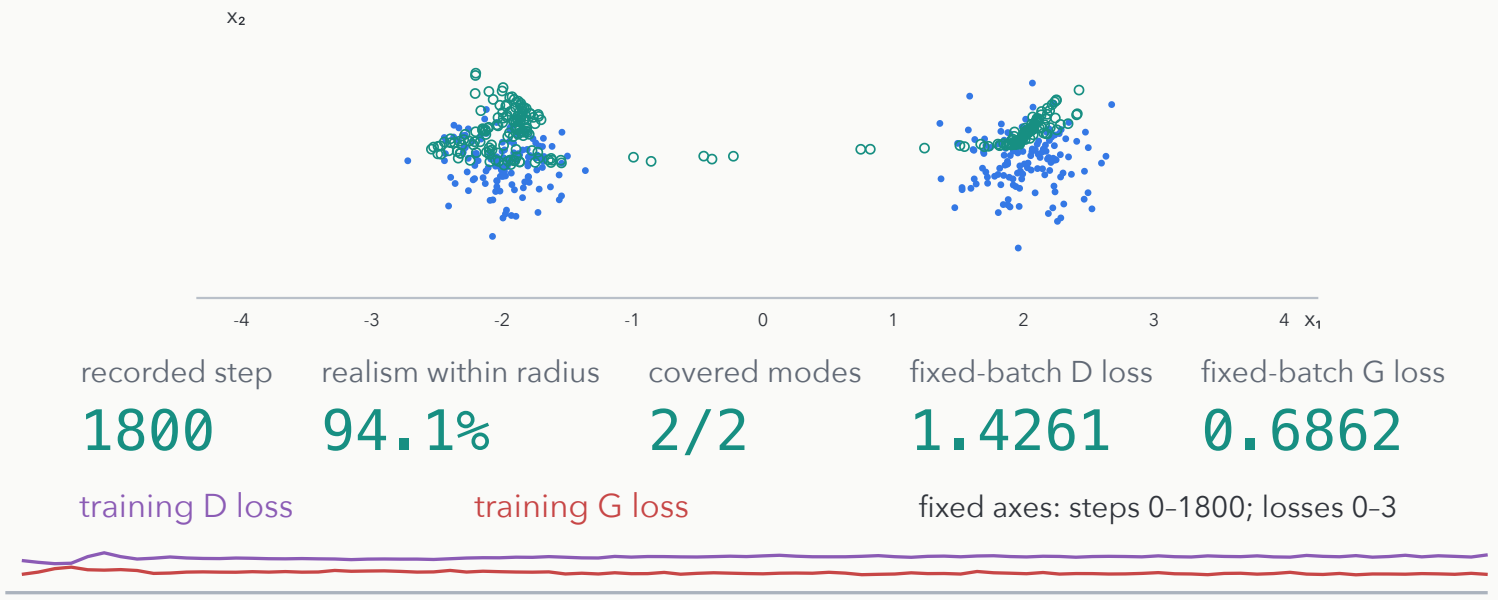
TAKEAWAY Check whether samples are plausible and whether all modes are represented.

SAY Every point may look plausible while an entire cluster is missing.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

What should we monitor during training?



Same 256 latent draws. Readout losses use exactly these real/fake points; curves show training-batch history.

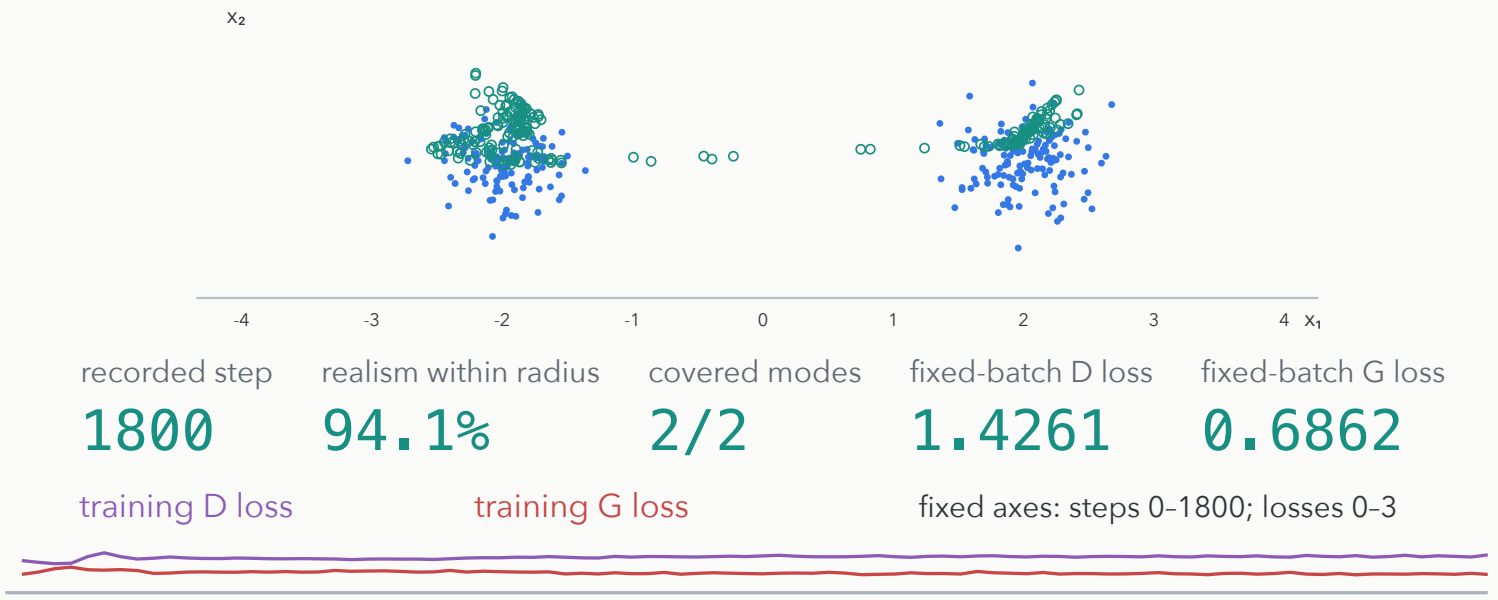
NOTICE Keep the noise fixed while replaying the recorded training checkpoints.

SAY Follow the same latent draws across checkpoints so movement reflects training.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

What should we monitor during training?



Same 256 latent draws. Readout losses use exactly these real/fake points; curves show training-batch history.

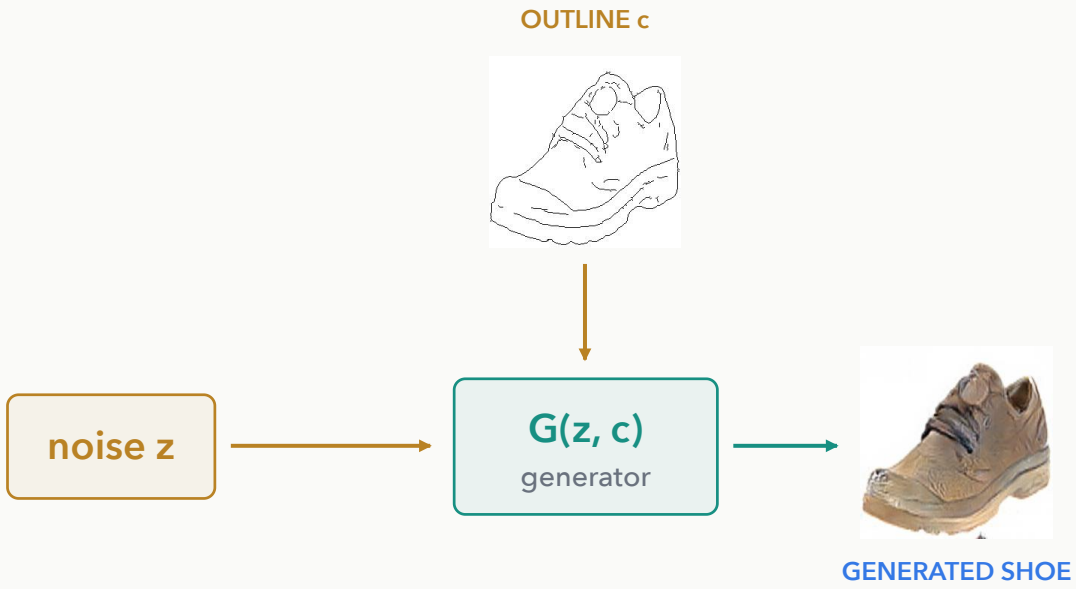
TAKEAWAY Read samples, coverage and losses together. D ≈ 1/2 is not proof of success.

SAY Follow the same latent draws across checkpoints so movement reflects training.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

How can both players use a condition?



The same conditioning idea applies to semantic layouts.

Published pix2pix shoe example · conceptual conditioning paths

$$x_{\text{fake}} = G(z, c)$$

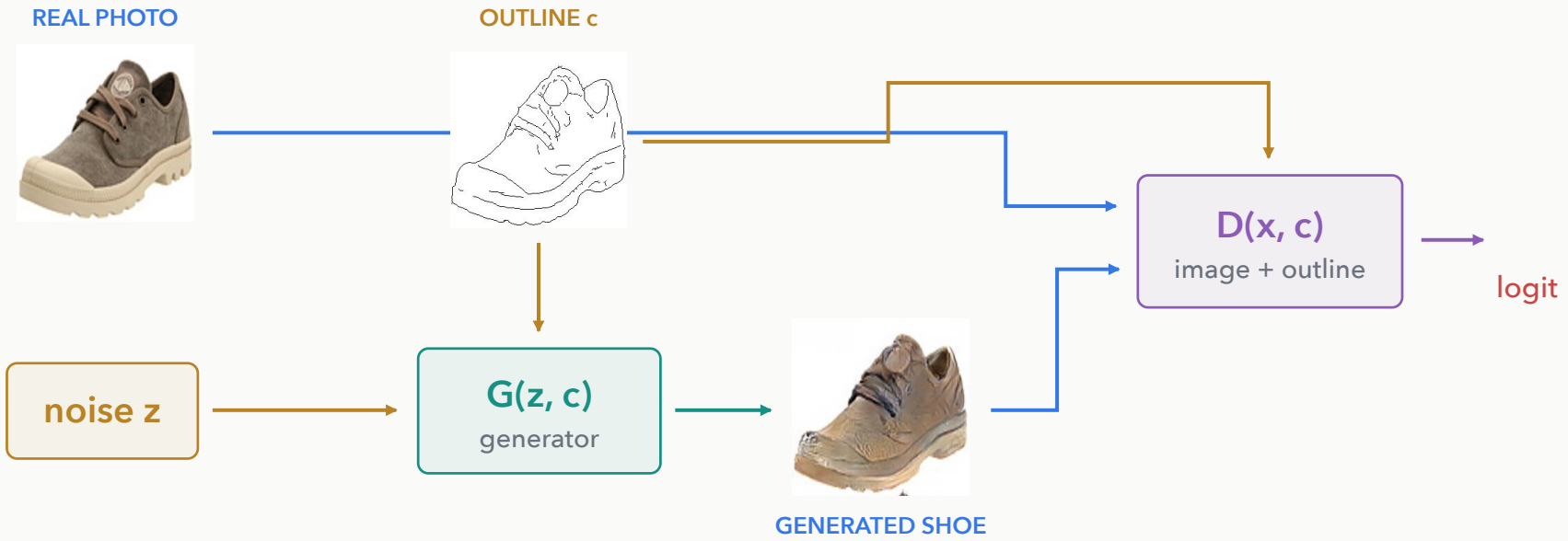
NOTICE Give the requested condition to the generator.

SAY An output can look like a shoe and still fail to match the outline.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

How can both players use a condition?



The same conditioning idea applies to semantic layouts.

Published pix2pix shoe example · conceptual conditioning paths

$$D(x, c)$$

TAKEAWAY D needs the condition too, so it can judge whether the output fits the request.

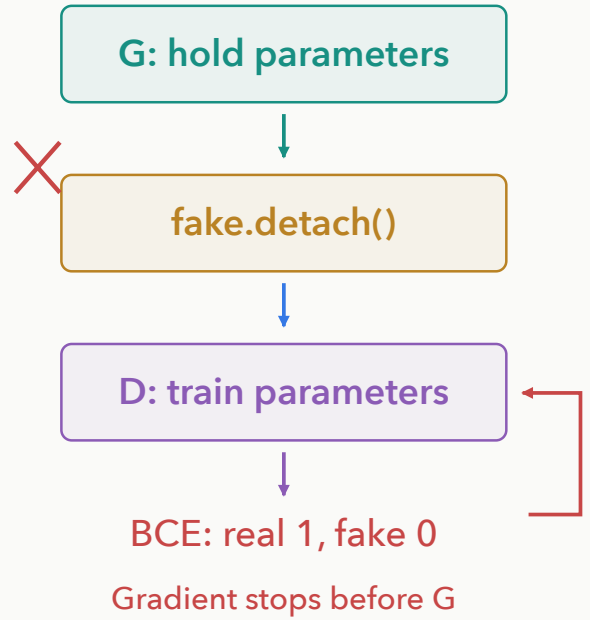
SAY An output can look like a shoe and still fail to match the outline.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Can you find the broken gradient path?

```
opt_D.zero_grad(set_to_none=True)
opt_G.zero_grad(set_to_none=True)
fake = G(torch.randn(B, 32))
real_logits = D(real)
fake_logits = D(fake.detach())
loss_D = BCE_logits(real_logits, ones)
loss_D += BCE_logits(fake_logits, zeros)
loss_D.backward()
opt_D.step()
# G has no new gradients
```



BCE_logits = binary_cross_entropy_with_logits · Complete runnable models

NOTICE Find the detach boundary in D's update and the open path in G's update.

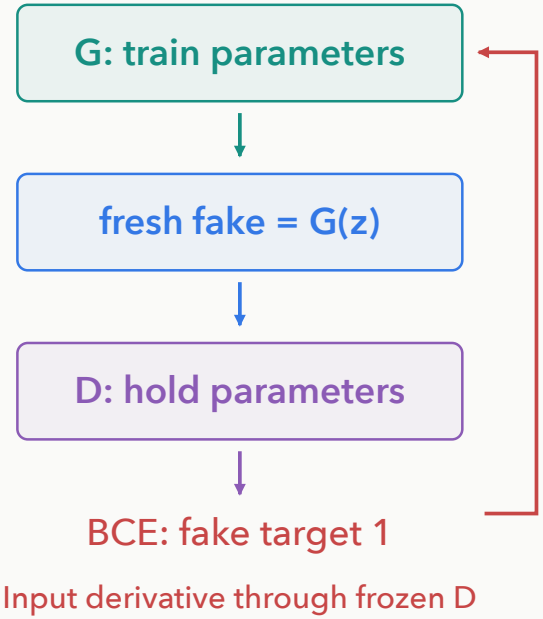
SAY Find the cut in the graph before changing the optimiser or the learning rate.

Goodfellow et al. · GANs · authored teaching diagram

G / GAN

Can you find the broken gradient path?

```
D.requires_grad_(False)
opt_G.zero_grad(set_to_none=True)
fake = G(torch.randn(B, 32))
fake_logits = D(fake)
loss_G = BCE_logits(fake_logits, ones)
# no detach; no no_grad around D
loss_G.backward()
opt_G.step()
D.requires_grad_(True)
```



BCE_logits = binary_cross_entropy_with_logits · Complete runnable models

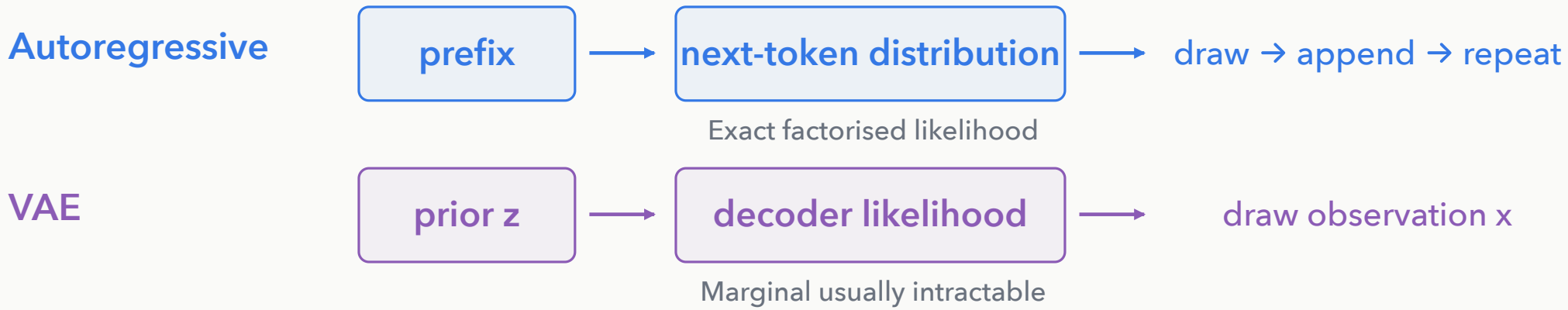
TAKEAWAY In the lab, compare a working GAN, a missing mode and a cut generator gradient.

SAY Find the cut in the graph before changing the optimiser or the learning rate.

Goodfellow et al. · GANs · authored teaching diagram

H / COMPARISON

How do the three recipes compare?

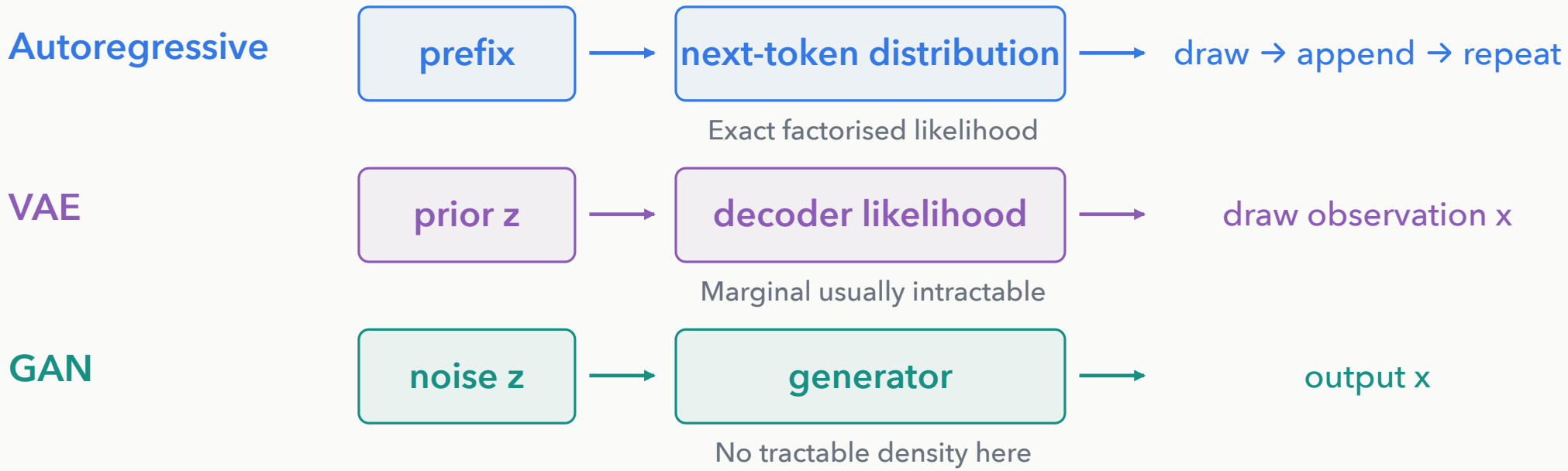


NOTICE Compare the learned distributions and the work needed to sample.

SAY Compare what each model lets us calculate and how it produces one new sample.

H / COMPARISON

How do the three recipes compare?



TAKEAWAY The sampling path and access to likelihood depend on the model we choose.

SAY Compare what each model lets us calculate and how it produces one new sample.

H / COMPARISON

What can you explain now?

1. What does the VAE encoder do?
2. Which two KLs appear?
3. What does “freeze D” mean?

NOTICE Explain prior sampling, the two KLs, and the gradient through frozen D.

SAY Explain the sampling path, the two KLs, and the gradient through frozen D in your own words.

H / COMPARISON

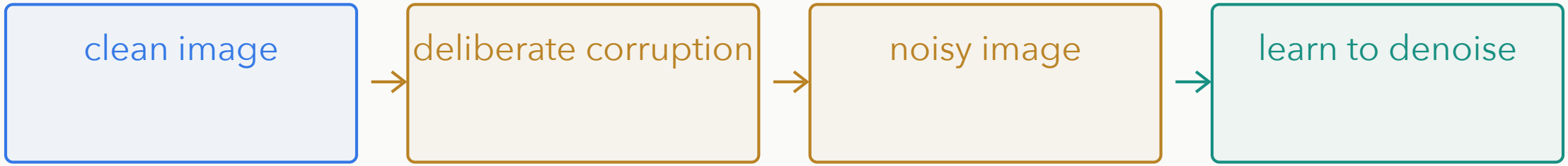
What can you explain now?

1. What does the VAE encoder do?
2. Which two KLs appear?
3. What does “freeze D” mean?

It approximates a posterior after observing x.

Prior KL in the objective; true-posterior KL in the gap.

Hold parameters fixed; differentiate through the input.



TAKEAWAY Next: can deliberately corrupting data give us targets for learning to generate?

SAY Explain the sampling path, the two KLs, and the gradient through frozen D in your own words.

OPTIONAL APPENDIX

Appendix · Derive the Gaussian KL

APPENDIX · DERIVATION

$$\log \frac{q(z)}{p(z)} = -\log \sigma - \frac{(z-\mu)^2}{2\sigma^2} + \frac{z^2}{2}$$

For diagonal covariance, sum the coordinate contributions.

NOTICE Write the Gaussian log-density ratio before taking its expectation.

SAY Take the expectation term by term, using the mean and variance of q.

OPTIONAL APPENDIX

Appendix · Derive the Gaussian KL

APPENDIX · DERIVATION

$$\log \frac{q(z)}{p(z)} = -\log \sigma - \frac{(z-\mu)^2}{2\sigma^2} + \frac{z^2}{2}$$

$$D_{\mathrm{KL}}(q\|p) = \tfrac{1}{2}(\mu^2 + \sigma^2 - 1 - \log \sigma^2)$$

For diagonal covariance, sum the coordinate contributions.

TAKEAWAY Use $E[(z-\mu)^2] = \sigma^2$ and $E[z^2] = \mu^2 + \sigma^2$.

SAY Take the expectation term by term, using the mean and variance of q.

Kingma & Welling · AEVB · authored teaching diagram

OPTIONAL APPENDIX

Appendix · Expand the posterior gap

APPENDIX · DERIVATION

$$\mathbb{E}_q \log \frac{q(z \mid x)}{p(z \mid x)} = \mathbb{E}_q[\log q - \log p(x, z)] + \log p(x)$$

NOTICE Bayes' rule turns posterior KL into a difference of objectives.

SAY Substitute Bayes' rule, then collect the terms that already form the ELBO.

Kingma & Welling · AEVB · authored teaching diagram

OPTIONAL APPENDIX

Appendix · Expand the posterior gap

APPENDIX · DERIVATION

$$\mathbb{E}_q \log \frac{q(z \mid x)}{p(z \mid x)} = \mathbb{E}_q[\log q - \log p(x, z)] + \log p(x)$$

$$\log p_\theta(x) = \mathcal{E} + D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x))$$

TAKEAWAY Rearrange, retaining the conditioning on x.

SAY Substitute Bayes' rule, then collect the terms that already form the ELBO.

Kingma & Welling · AEVB · authored teaching diagram

OPTIONAL APPENDIX

Appendix · The ideal discriminator

APPENDIX · DERIVATION

$$D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_g(x)}$$

NOTICE Assume densities under a common measure and optimise D pointwise where their sum is positive.

SAY This result assumes an ideal discriminator. A finite training run need not reach it.

OPTIONAL APPENDIX

Appendix · The ideal discriminator

APPENDIX · DERIVATION

$$D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_g(x)}$$

$$V(D^*, G) = -\log 4 + 2\,\mathrm{JS}(p_{\text{data}} \| p_g)$$

JS uses M = (p_data + p_g)/2. Zero-density regions do not constrain D*.

TAKEAWAY For the minimax value at D*, distribution matching minimises JS. This is idealised analysis.

SAY This result assumes an ideal discriminator. A finite training run need not reach it.